

Research Methodology and IPR (22MBA23)

Module-1

Introduction to Business Research:

Meaning, types, process of research- management problem, defining the research problem, formulating the research Hypothesis, developing the research proposals, research design formulation, sampling design, planning and collecting the data for research, data analysis and interpretation. Research Application in business decisions, Ethical issues in business research. Features of a good research study.

Module-2

Business Research Design:

Meaning, types and significance of research design, errors affecting research design.

Exploratory Research: Meaning, purpose, methods, Literature search, experience survey, focus groups and comprehensive case methods.

Conclusive Research Design: Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.

Experimental Research Design: Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

Module-3

Sampling: Concepts, Types of Sampling, **Probability Sampling:** simple random sampling, systematic sampling, stratified random sampling, cluster sampling.

Non Probability Sampling: convenience sampling- judgmental sampling, snowball sampling, quota sampling, Errors in sampling.

Module-4

Data Collection: Meaning, types, **Data collection methods:** Observations, survey and interview techniques, **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary Data Sources: advantages and disadvantages. **Measurement and Scaling Techniques:** Basic measurement Scales-Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale, **Multi-Dimensional Scaling:** Non comparative scaling techniques.

Module-5

Data Analysis and Report Writing: Editing, Coding, Classification, Tabulation, Validation. Analysis and Interpretation, Report writing and presentation of results, Importance of report writing, types of research reports, Report structure, Guidelines for effective documentation.

Module-6

Intellectual Property Rights:

Meaning and Concepts of Intellectual Property, Nature and Characteristics of Intellectual Property, Origin and Development of Intellectual Property, Kinds of Intellectual Property, Intellectual Property System in India, IPRs- Invention and Creativity- Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs)- **A brief summary of:** Patents, Copyrights, Trademarks, TRIPS and TRIMS , Industrial Designs- Integrated Circuits-Geographical Indications-Establishment of WIPO-Application and Procedures.

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Research Meaning:

- Research in common man's language refers to "search for knowledge"
- Research is an art of scientific investigation.
- The Advanced Learner's Dictionary of Current English lays down the meaning of research as,

a careful investigation or inquiry specially through search for new facts in any branch of knowledge".

It is also a systematic design, collection, analysis and reporting the findings and solutions for problems of a company.

- Research is a structured enquiry that utilizes acceptable scientific methodology to solve problems and create new knowledge that is generally applicable.
- Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested.
- Research is an art of scientific investigation. It is also a systematic design, collection, analysis and reporting the findings and solutions for the problems of a company.

Definition:

Research can be defined as an organized and systematic study of materials and sources in order to discover new things and establish facts and reach new conclusions. Essentially speaking, research involves a well-developed plan, a systematic approach to develop new theories or finding solutions to various problems.

According to Clifford Woody, “research comprises defining and refining problems, formulating hypothesis or suggested solutions, collecting, organizing and evaluating data, making decisions and reaching conclusions to determine whether they fit the formulating hypothesis”

Business Research

- A Systematic inquiry that provides information to guide managerial decisions.
- Systematic and objective process of gathering, recording and analyzing data for aid in making business decisions.

Scientific Method

Scientific research is one which yields the same results when repeated by different individuals. Techniques or procedures used to analyze empirical evidence in an attempt to confirm or disprove prior concepts

Empirical means verifiable by observation, experimentation or experience.

- A scientific undertaking which by means of logical and systematic techniques, aims to:
 1. Discover new facts or verify and test old facts.
 2. Analyze their sequences, interrelationships and causal explanations.
 3. Develop new scientific tools, concepts and theories, which would facilitate valid and reliable study of human behavior.

Characteristics of Scientific Method

- **Validity** – Validity is the ability of a measuring instrument to measure what it is supposed to.
- **Reliability** – Measurement by instruments such as thermometer can be repeated any number of times by different individuals, but the results will be the same.

Types of research

Research can be classified on the basis of purpose, time, methodology, setting etc. however one can broadly classify research as basic and applied research.

1. **Basic research:** research for the sake of enhancing knowledge is termed as basic research.
 - It is also called as theoretical research

- Basic or pure or fundamental research is done with the intention of empowering the unknown, Undertaken for the sake of knowledge without any intention to apply or practice.
- It is primarily concerned with developing and formulating theories and generalization.
- It addresses itself to more fundamental questions and not the problems with immediate commercial potential.
- Not directly involved with practical problems.
- Contributions:
 - i. By developing principle, pure research *offers solutions* to many *practical problems*.
 - ii. Helps to *find critical factors* in a practical problem.
 - iii. Develops many *alternative solutions*.
 - iv. How did the universe begin?
- What are protons, neutrons, and electrons composed of?
- What is the specific genetic code of the fruit fly?
- Example : Theory of Relativity (by Einstein)

2. Applied Research

- It is also termed as need based research having high practical relevance.
- Specific in nature, result oriented and is driven by clear aim.
- Undertaken to find solutions for real-life problem requiring an action or policy decision.
- It may incidentally contribute to the development of theoretical knowledge.
- Contributions:
 - Can contribute new facts about existing theories.
 - Puts theories to test.
 - May aid in conceptual clarification.

- Integrate previously existing theories. Example :
 - To develop a new market for the product.
 - improve the energy efficiency of homes, offices, or modes of transportation
- 3. **Qualitative Research** – An unstructured, exploratory research methodology based on small samples that provides insights and understanding of the problem setting.
 - The aim is a complete, detailed description.
 - Data is in the form of words, pictures or objects.
 - Case studies, group interviews
- 4. **Quantitative Research** – A research methodology that seeks to quantify the data and typically applies some form of statistical analysis.
 - The aim is to classify features, count them, and construct statistical models in an attempt to explain what is observed.
 - Data is in the form of numbers and statistics.
 - Experiments and surveys
- 5. **Historical Research**
 - Historical study is a study of past records and data in order to understand the future trends and development of the organization or market.
 - There is no direct observation.
 - Research has to depend on the conclusions or inferences drawn in the past.
 - Main objective is to derive explanation and generalization from the past trends in order to understand the present and anticipate the future.
 - Example : Investors in the share market study the past records or prices of shares which he/she intends to buy. Studying the share prices of a particular company enables the investor to take decision whether to invest in the shares of a company.
- 6. **Exploratory Research:** this kind of research is carried out at the very beginning when the problem is not clear or vague.

- Provides insights into and comprehension of an issue or situation.
- Often relies on secondary research such as reviewing available literature and/or data, or qualitative approaches such as informal discussions with consumers, employees, management or competitors and more formal approaches through in-depth interviews, focus groups, case studies or pilot studies.
- Example : Exploring the reasons for sales decline of a product in a company.

7. Descriptive Research

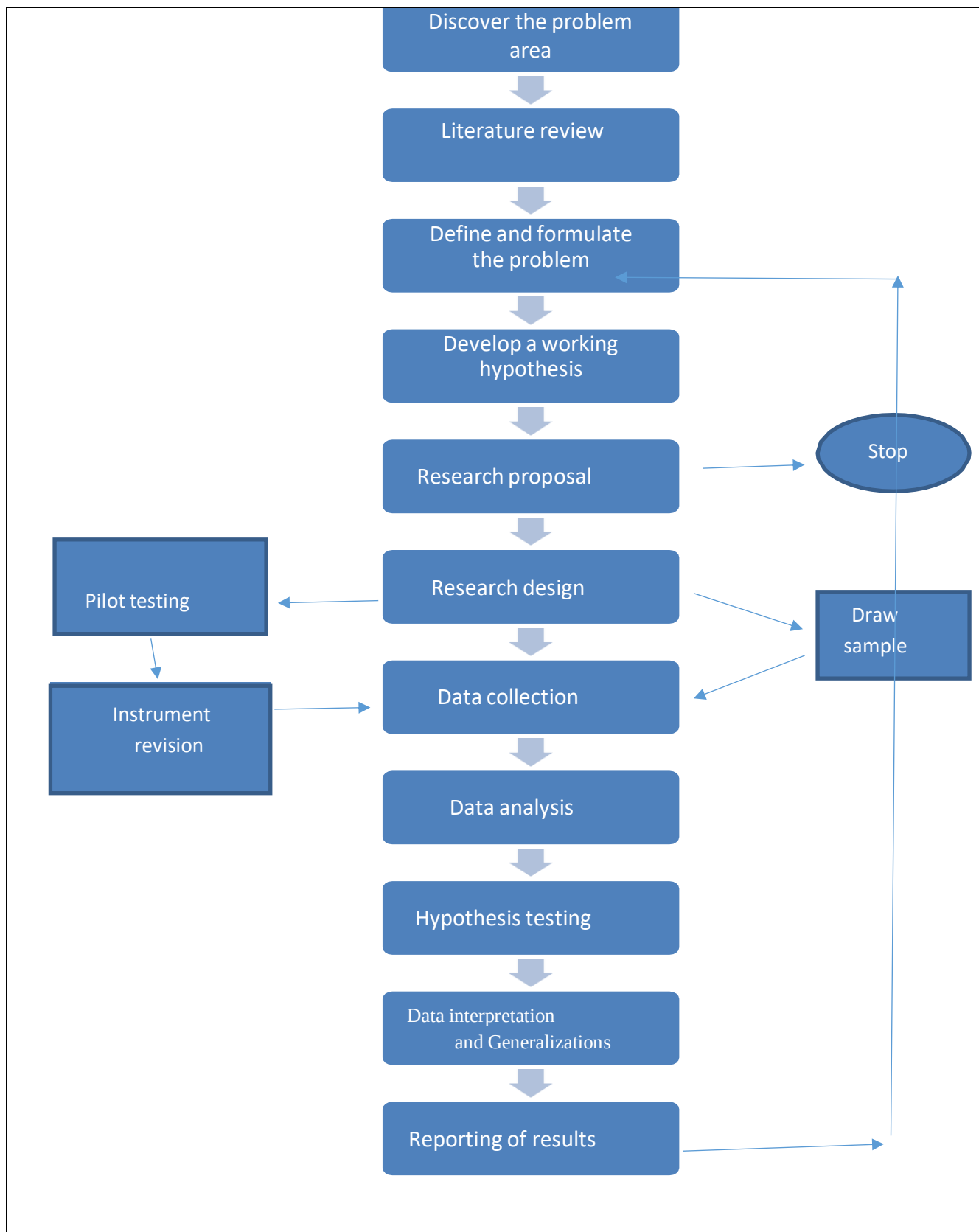
- Describes data and characteristics about the population or phenomenon being studied.
- Answers the questions who, what, where, when and how.
- Describes the characteristics of the respondent in relation to a particular product. Deals with demographic characteristics of the consumer.
- Example : Degree of viewing TV Channels, its variation with age, income level, profession of respondent as well as time of viewing.

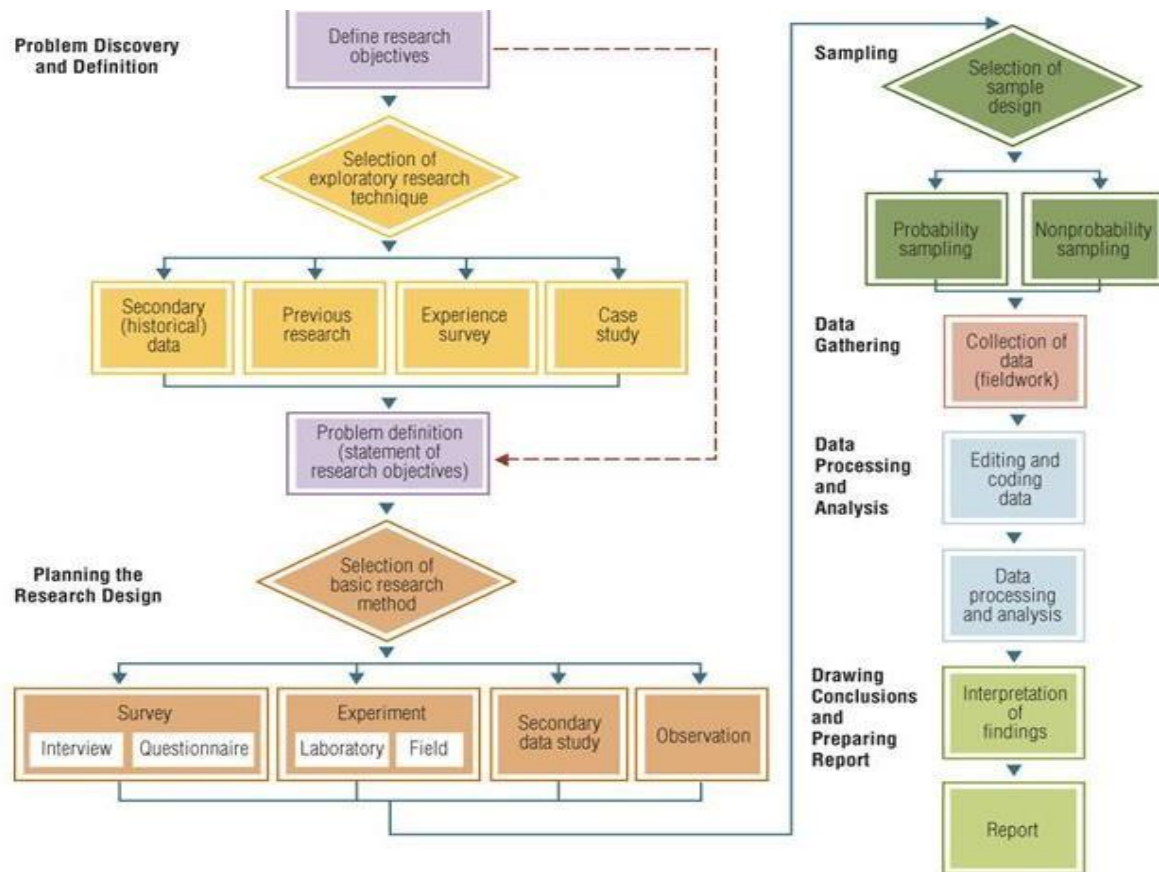
8. Causal Research

- Conducted to determine the cause and effect relationship between the two variables.
- Way of seeing how actions now will affect a business in the future.
- Example:
 - Training and productivity of an employees
 - Advertisement and sales

9. Simulation research

- This research involves the creation of an artificial environment which is very similar to the real environment. Therefore with in the artificial environment the variables are manipulated and studied.
- Ex: study of consumer buying behavior by giving specific amount of time and money in a mall.

Research process



Note: Diamond-shaped boxes indicate stages in the research process in which a choice of one or more techniques must be made. The dotted line indicates an alternative path that skips exploratory research.

- 1. DISCOVERING THE PROBLEM AREA:** to start with a research, one needs to discover the problem which needs solution. Any researcher is pretty sure of the subject on which to conduct his research, but the exact problem area may remain exclusive.

The best way to identify the problem would be to look for an unresolved query, a gap in the existing knowledge or an unfulfilled need within the chosen subject. The researcher should take care that the problem should be the one which can be clearly defined and formulated. Further while choosing the research area, the researcher should look into the availability of information relevant to the topic. Mere availability is not enough. It must be accessible.

- 2. LITERATURE REVIEW:** it involves a comprehensive review of published and unpublished work from the secondary sources of data available in the relevant area of study.

Literature review helps the researcher in 2 ways

- Specifying his research in a meaningful context
- It would provide him with an insight into the methods and techniques adopted for handling such problems

3. **PROBLEM DEFINITION:** initially the focus of the problem is ambiguously defined. However after the literature review, the researcher is in a position to formulate problem clearly. A problem well formulated alone yield fruitful results. It is a clear, precise and concise statement of the matter that is to be investigated.

4. **DEVELOPING A WORKING HYPOTHESIS:** hypothesis is a tentative hypothesis regarding a solution to the problem under study.

Hypothesis is a predictive statement which is made in the light of available facts relating to the problem under study.

The type of data to be collected and the tools of analysis are influenced by the hypothesis.
Ex: a study conducted to find out the amount of research investment being done by the companies will have the following hypothesis.

H0: companies invest 1% of their sales revenue in research activities(null hypothesis)

H1: companies invest less than 1% of their sales revenue in research activities(alternative hypothesis)

5. **RESEARCH PROPOSAL:** it is a brief summary outlining the objectives of study and the modus operandi of conducting the research.

In case of thesis, the research proposal are in the form of synopsis stating

- Research objectives
- Proposed methodology of research
- Benefits of study
- Detailed bibliography

In case of government or business organizations, in addition to the above information there are other details like

- Researcher's qualification
- Time and cost schedule
- Resources and special facilities required during the research.

Essentially they contain an executive summary in the beginning stating the objectives of study and benefits derived thereon.

6. **RESEARCH DESIGN:** it outlines the conditions for collection and analysis of data. The what, when, where, how much and the method of data collection are detailed in the research

design. It will specially contain information about:

- **Sampling design**
- **Data collection design**
- **Instrument design**

Sampling design: the plan devised to draw a sample is termed as sampling design. A sampling design will include a decision on the

- Sampling unit
- Sampling technique
- Sample size

Data collection design: the method of gathering data is planned in data collection design. Briefly speaking, some of the data collection techniques available to researcher are

- Questionnaire
- Interview
- Observation

Instrument design: instrument designing schedule or the questionnaire the researcher would use to collect data. The researcher while designing the questionnaire must think in terms of

- Type of data
- Communication approach
- Question structure
- Question wording etc

7. DATA COLLECTION

This involves sending questionnaire to respondents, training field workers in interview and observation methods and making them to collect necessary data.

8. DATA ANALYSIS

All the data gathered in the previous step has no meaning till the time it is not subject to data analysis.

Processing of raw data will yield some kind of relevant information. Data analysis is concerned with reducing the bulk of accumulated data to a manageable size.

Generally speaking, it involves the following procedures:

- Editing

- Coding
- Tabulation
- Statistical analysis

9. HYPOTHESIS TESTING

There are various parametric and non-parametric tests like z-test, t-test, f-test etc

Hypothesis testing helps a researcher to establish if the difference between values is due to chance or actual.

10. DATA INTERPRETATION AND GENERALISATION(finding out meaning)

Researcher can use inductive or deductive approach. After the data is been analysed and tested repeatedly for arriving at conclusive results, generalizations are made to build new theory.

11. REPORTING OF RESULTS

The last step is concerned with bringing in public the results of the research so that the design can be put to some use.

Format of the report

Preliminary section

Title of the report, table of contents, list of tables, graphs, preface and an executive summary

Main text

Problem background, research objectives, research methodology, importance of conducting research and the conclusion arrived thereof.

End section

Appendices supporting the research items like questionnaire, schedule used etc and bibliography.

Management problem

When a problem has been clearly and accurately identified can a research project be conducted properly. It is a concise description of the issues that need to be addressed by a problem solving team and should be presented to them (or created by them) before they try to solve the problem. It asks what the decision maker needs to do. Mostly action oriented and Focuses on symptoms.

Defining the research problem

The research problem is a general statement of an issue meriting research. Its nature will suggest appropriate forms for its investigation

Problem definition involves stating the general marketing research problem and identifying its specific components. Only when research problem has been clearly defined can research be designed and conducted properly.

Management Problem vs Research Problem

Management Problem involves “what needs to be done?”

Research Problem involves “what information needs to be provided and how can the information is obtained”. It refers to some difficulty an organization faces and wishes to obtain solution for the same.

Basis of comparison	Management problem	Research problem
Nature	Asks what the Decision Makers needs to do	Asks what information needed and how it can be obtained
Orientation	Action Oriented	Information oriented
Focus	Focuses on Symptoms	Focuses on underlying causes
Common errors	Problem definition is too broad	Problem definition is too narrow
example	Should a new product be introduced	To determine consumer preferences and purchase intentions for the proposed new product
example	Should the advertising campaign be changed	To determine the effectiveness of the current advertising campaign
example	Should the price of the brand be increased	To determine the price elasticity of the demand and the impact on sales and profits of various levels of price changes

Formulating the research Hypothesis

A good hypothesis relates and explains the known facts. It should also predict new facts. It *must* be stated in such a way that we can test it by experimentation or further observation, or it is of no scientific value. Also, it must be stated in a way that would allow us to show if it is incorrect, i.e., it must be "falsifiable." A scientist *must* be willing to accept the possibility that his or her hypothesis is incorrect, and this point often separates true science from pseudoscience. (In fact, most scientists work hard to develop good hypotheses, and then spend a great deal of effort trying to *disprove* them. Pseudo scientists tend to settle on a hypothesis that suits their needs or expectations, and then spend a great deal of effort trying to prove that they are "true".

Developing the research proposals

Research proposal is outline, a draft plan of a research work proposed by the researcher. It is also termed as synopsis. It contains

- A statement of the problem
- The proposed methodology
- The benefits of the research
- The reasons required to conduct the research

Contents of research proposal

The research proposal contents differ depending on the type of proposal and intended audience.

1. Problem statement
2. Research objectives
3. Literature review
4. Importance/ benefits of study
5. Research design
6. Data analysis
7. Nature and form of results
8. Schedule
9. Facilities and special resources
10. Bibliography
11. Appendices
12. Measurement instrument

Research design formulation

A research design is a framework or blueprint for conducting the marketing research project. It details the procedures necessary for obtaining the required information, and its purpose is to design a study that will test the hypotheses of interest, determine possible answers to the research questions, and provide the information needed for decision making. Decisions are also made regarding what data should be obtained from the respondents (e.g., by conducting a survey or an experiment). A questionnaire and sampling plan also are designed in order to select the most appropriate respondents for the study. The following steps are involved in formulating a research design:

- Secondary data analysis (based on secondary research)
- Qualitative research
- Methods of collecting quantitative data (survey, observation, and experimentation)
- Definition of the information needed
- Measurement and scaling procedures
- Questionnaire design
- Sampling process and sample size
- Plan of data analysis

Sampling design

Sampling is a means of selecting a subset of units from a target population for the purpose of collecting information. This information is used to draw inferences about the population as a whole. The subset of units that are selected is called a sample. The sample design encompasses all aspects of how to group units on the frame, determine the sample size, allocate the sample to the various classifications of frame units, and finally, select the sample. Choices in sample design are influenced by many factors, including the desired level of precision and detail of the information to be produced, the availability of appropriate sampling frames, the availability of suitable auxiliary

variables for stratification and sample election, the estimation methods that will be used and the available budget in terms of time and resources.

Planning and collecting the data for research

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes. Generally there are three types of data collection and they are

1. Surveys: Standardized paper-and-pencil or phone questionnaires that ask predetermined questions.

2. Interviews: Structured or unstructured one-on-one directed conversations with key individuals or leaders in a community.

3. Focus groups: Structured interviews with small groups of like individuals using standardized questions, follow-up questions, and exploration of other topics that arise to better understand participants

Data analysis and interpretation

Analysis of data is a process of inspecting, cleaning, transforming, and modeling data with the goal of discovering useful information, suggesting conclusions, and supporting decision-making. Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, in different business, science, and social science domains.

Literature review

It involves a comprehensive review of published and unpublished work from the secondary sources of data available in the relevant area of study.

Importance of literature review:

1. Identification of problem for the study
2. Monitors the repetition
3. Helps avoiding mistakes
4. Creates appropriate theoretical structure
5. Facilitates interpretation of study

Research application in business decision

☐ Businesses use research to ascertain the success of their advertising. For instance, a dairy manufacturer may want to find out what percentage of the people saw its latest TV commercial. The dairy manufacturer may find that more individuals become aware of its advertising the longer the TV ad runs. The company may decide to run its TV ads at different times if few people have seen the ads.

☐ A business can make well-informed decisions due to research. In the research process, the business will be able to acquire details about key business areas, analyze it, create a strategy and distribute business information. Reports, presented to the top management, often contain details on consumer and employee preferences and all the available channels for sales, marketing, finance and production. Management makes use of these details to **determine the best strategy**.

☐ Research is a necessity at all stages and phases of business operations. Initial research is needed to evaluate whether getting into the given kind of business would be profitable and whether there exists demand for the proposed product.

☐ Regarding the staff, a correctly carried out research can uncover important details on their satisfaction quotient, the difficulties experienced by them and how the problems related to relationships at the workplace could be handled. An analysis into the results would allow the management to bring about changes for the all round effective functioning of the organization and its employees. The workers **can be trained and coached in line with the needs**. This would help personal as well as professional development improving overall organizational performance.

☐ Research is important for **managerial decision making**. All strategic business areas are analyzed and evaluated; then techniques for more efficient procedures are created. Through proper research, the organization will be able to pick the most effective, productive and profitable one.

☐ Research could possibly be applied to marketing, production, finance, IT and Human resources.

☐ Research can answer questions for **various problems**, from getting a grip on industry trends, identifying new products to produce and deliver to the market, or deciding on which site to locate an outlet, to better understanding what it needs to fulfill customer demands. Research can also help evaluate if a product is accepted in the market. Research aids expansion into new markets.

☐ Research helps in **testing the potential success of new products**. Businesses have to understand what kinds of products consumers would like before they market them. For instance, a restaurant may in the beginning, interview focus groups to test types of burgers. The focus groups will probably include small teams of consumers. The goal of the focus group may be to figure out which burger customers prefer. Ultimately, the company may test the burgers through surveys with larger groups of people. The above points state the importance of research in business decision making. Research is necessary to gather facts and statistics with regards to a company's customers, employees and competitors. Based on these numbers, businesses are able to make better managerial decisions.

☐ **New Knowledge:** The fascination and desire for new knowledge, new facts for business cycles, environment analysis and technological up gradation are the primary reasons of research.

☐ **Research assists** the company to find the right supplier at the right price and at the right time. An appropriate supplier choice makes it possible for the company to obtain or acquire top quality raw materials which result into production of good quality items which are consumed by the end user.

Ethical issues in business research.

1 . Study design and ethics approval

According to COPE, “good research should be well adjusted, well-planned, appropriately designed, and ethically approved. To conduct research to a lower standard may constitute misconduct.” This may appear to be a stringent criterion, but it highlights the basic requirement of a researcher is to conduct a

research responsibly. To achieve this, a research protocol should be developed and adhered to. It must be carefully agreed to by all contributors and collaborators, and the precise roles of each team member should be spelled out early, including matters of authorship and publications. Research should seek to answer specific questions, rather than just collect data.

It is essential to obtain approval from the Institutional Review Board, or Ethics Committee, of the respective organisations for studies involving people, medical records, and anonymised human tissues. The research proposal should discuss potential ethical issues pertaining to the research. The researchers should pay special attention to vulnerable subjects to avoid breach of ethical codes (e.g. children, prisoners, pregnant women, mentally challenged, educationally and economically disadvantaged). Patient information sheet should be given to the subjects during recruitment, detailing the objectives, procedures, potential benefits and harms, as well as rights to refuse participation in the research. Consent should be explained and obtained from the subjects or guardians, and steps should be taken to ensure confidentiality of information provided by the subjects.

2. Data analysis

It is the responsibility of the researcher to analyse the data appropriately. Although inappropriate analysis does not necessarily amount to misconduct, intentional omission of result may cause misinterpretation and mislead the readers. Fabrication and falsification of data do constitute misconduct. For example, in a clinical trial, if a drug is found to be ineffective, this study should be reported. There is a tendency for the researchers to under-report negative research findings, and this is partly contributed by pressure from the pharmaceutical industry which funds the clinical trial.

To ensure appropriate data analysis, all sources and methods used to obtain and analyse data should be fully disclosed. Failure to do so may lead the readers to misinterpret the results without considering possibility of the study being underpowered. The discussion section of a paper should mention any issues of bias, and explain how they have been dealt with in the design and interpretation of the study.

3. Authorship

There is no universally agreed definition of authorship. It is generally agreed that an author should have made substantial contribution to the intellectual content, including conceptualising and designing the study; acquiring, analysing and interpreting the data. The author should also take responsibility to certify that the manuscript represents valid work and take public responsibility for the work. Finally, an author is usually involved in drafting or revising the manuscript, as well as approving the submitted manuscript. Data collection, editing of grammar and language, and other routine works by itself, do not deserve an authorship.

It is crucial to decide early on in the planning of a research who will be credited as authors, as contributors, and who will be acknowledged. It is also advisable to read carefully the “Advice to Authors” of the target journal which may serve as a guide to the issue of authorship.

4. Conflicts of interest

This happens when researchers have interests that are not fully apparent and that may influence their judgments on what is published. These conflicts include personal, commercial, political, academic or financial interest. Financial interests may include employment, research funding, stock or share ownership, payment for lecture or travel, consultancies and company support for staff. This issue is especially pertinent in biomedical research where a substantial number of clinical trials are funded by pharmaceutical company.

Such interests, where relevant, should be discussed in the early stage of research. The researchers need to take extra effort to ensure that their conflicts of interest do not influence the methodology and outcome of the research. It would be useful to consult an independent researcher, or Ethics Committee, on this issue if in doubt. When publishing, these conflicts of interest should be declared to editors, and readers will judge for themselves whether the research findings are trustworthy.

5. Redundant publication and plagiarism

Redundant publication occurs when two or more papers, without full cross reference, share the same hypothesis, data, discussion points, or conclusions. However, previous publication of an abstract during the proceedings of meetings does not preclude subsequent submission for publication, but full disclosure should be made at the time of submission. This is also known as self-plagiarism. In the increasing competitive environment where appointments, promotions and grant applications are strongly influenced by publication record, researchers are under intense pressure to publish, and a growing minority is seeking to bump up their CV through dishonest means.

On the other hand, plagiarism ranges from unreferenced use of others’ published and unpublished ideas, including research grant applications to submission under “new” authorship of a complete paper, sometimes in different language.

Therefore, it is important to disclose all sources of information, and if large amount of other people’s written or illustrative materials is to be used, permission must be sought.

Features of a good research study.

- **Systematic-** research should be structured.
- **Logical-** there must be logical reasoning
- **Empirical-** factual investigation must be possible. Its validity can be checked through

reliable sources of evidences.

- **Replicable-** the research conducted can be repeated any number of times and yield the same result
- **Purpose clearly defined** – The purpose of the business research-the problem involved or the decision to be made-should be clearly defined and sharply delineated in terms as unambiguous as possible. The statement of the decision problem should include its scope, its limitations, and the precise meanings of all words and terms significant to the research.
- **Research process detailed** – The research procedures used should be described in sufficient detail to permit another researcher to repeat the research. Except when secrecy is imposed, research reports should reveal the sources of data and the means by which they were obtained.
- **Research design thoroughly planned** – The procedural design of the research should be carefully planned to yield results that are as objective as possible. A survey of opinions or recollections ought not to be used when more reliable evidence is available from documentary sources or by direct observation.
- **High ethical standards applied** – Ethical issues in research reflect important moral concerns about the practice of responsible behavior in society. A research design that includes safeguards against causing mental or physical harm to participants and makes data integrity a first priority should be highly valued.
- **Limitations frankly revealed** – The researcher should report, with complete frankness, flaws in procedural design and estimate their effect on the findings.
- **Analysis adequate for decision maker's needs-** Analysis of the data should be extensive enough to reveal its significance, what managers call “insights”. The methods of analysis used should be appropriate
- **Findings presented unambiguously** – Language that is restrained, clear, and precise; assertions that are carefully drawn and hedged with appropriate reservations; and an apparent effort to achieve maximum objectivity tend to leave a favorable impression of the researcher with the decision maker.
- **Conclusions justified** – Conclusions should be limited to those for which the data provide an adequate basis. Good researchers always specify the conditions under which their conclusions seem to be valid.

Module-2

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Exploratory Research: Meaning, purpose, methods, Literature search, experience survey, focus groups and comprehensive case methods.

Conclusive Research Design: Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.

Experimental Research Design: Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

Business Research Design

Meaning

The formidable problem that follows the task of defining the research problem is the preparation of the design of the research project, popularly known as the “research design”.

Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design. “A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.”¹ In fact, the research design is the conceptual structure within which research is conducted; it constitutes the blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data. More explicitly, the designing decisions happen to be in respect of:

- (i) What is the study about?
- (ii) Why is the study being made?
- (iii) Where will the study be carried out?
- (iv) What type of data is required?
- (v) Where can the required data be found?
- (vi) What periods of time will the study include?
- (vii) What will be the sample design?
- (viii) What techniques of data collection will be used?
- (ix) How will the data be analysed?
- (x) In what style will the report be prepared?

Keeping in view the above stated design decisions, one may split the overall research design into the following parts:

- (a) the sampling design which deals with the method of selecting items to be observed for the given study;
- (b) the observational design which relates to the conditions under which the observations are to be made;
- (c) the statistical design which concerns with the question of how many items are to be observed and how the information and data gathered are to be analysed; and
- (d) the operational design which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.

From what has been stated above, we can state the important features of a research design as under:

- (i) It is a plan that specifies the sources and types of information relevant to the research problem.
- (ii) It is a strategy specifying which approach will be used for gathering and analysing the data.
- (iii) It also includes the time and cost budgets since most studies are done under these two constraints.

In brief, research design must, at least, contain—(a) a clear statement of the research problem; (b) procedures and techniques to be used for gathering information; (c) the population to be studied; and (d) methods to be used in processing and analysing data.

Types of Research Design

There are different types of research design depend on the nature of the problem and objectives of the study. Following are the four types of research design.

1. Explanatory Research Design
2. Descriptive Research Design
3. Diagnostic Research Design
4. Experimental Research Design

1. Explanatory Research Design

In explanatory research design a researcher uses his own imaginations and ideas. It is based on the researcher personal judgment and obtaining information about something. He is looking for the unexplored situation and brings it to the eyes of the people. In this type of research there is no need of hypothesis formulation.

2. Descriptive Research Design

In descriptive research design a researcher is interested in describing a particular situation or phenomena under his study. It is a theoretical type of researcher design based on the collection

designing and presentation of the collected data. Descriptive research design covers the characteristics of people, materials, Socio-economics characteristics such as their age, education, marital status and income etc. The qualitative nature data is mostly collected like knowledge, attitude, beliefs and opinion of the people. Examples of such designs are the newspaper articles, films, dramas, and documentary etc.

3. Diagnostic Research Design

Here researcher wants to know about the root causes of the problem. He describes the factors responsible for the problematic situation. It is a problem solving research design that consists mainly:

1. Emergence of the problem
2. Diagnosis of the problem
3. Solution for the problem and
4. Suggestion for the problem solution

4. Experimental Research Design

In this type of research design is often used in natural science but it is different in social sciences. Human behavior cannot be measured through test-tubes and microscopes. The social researcher uses a method of experiment in that type of research design. One group is subjected to experiment called independent variables while other is considered as control group called dependent variable. The result obtained by the comparison of both the two groups. Both have the cause and effect relationship between each other.

Significance of Research Design

Research design offers the investigator an opportunity to carry out different research operations efficiently. This makes research as valuable as possible producing maximum information with minimum effort, time and money.

Researcher needs to consider all necessary precautions when preparing the design, as any error may upset the whole project. The reliability of result, which a researcher is looking, is proportional with design that constitutes a firm foundation of entire body of research work.

Significance of Research Design in Research Methodology

Research design is significant simply because it allows for the smooth sailing of the various research operations, thus making research as efficient as possible producing maximum information with nominal expenses of effort, time and money.

Just as for better, economical and attractive construction of a home, we require a blueprint (or what is typically known as the map of the home) well planned and prepared by an expert architect, in the same way we require a design or a plan in advance of data collection and analysis for our research study. It means advance planning of the techniques to be implemented for accumulating

the appropriate data and the strategies to be employed in their analysis, keeping in view the purpose of the research and the availability of staff, time and money.

Preparation of the design must be carried out meticulously as any error in it may upset the complete project. Research design, actually, has a great significance and impact on the reliability of the results achieved and as such constitutes the firm base of the entire edifice of the research work.

Even then the necessity for a well planned design is at times not realized by many people. The significance which this problem warrants is not given to it. Because of this many researches do not serve the purpose for which they are undertaken. The truth is, they may even provide misleading conclusions.

Thoughtlessness in developing the research project may lead to rendering the research exercise futile. It is, for that reason, crucial that an efficient and appropriate design should be prepared before beginning research operations.

The design assists the researcher to organize his ideas in a form whereby it will be possible for him to watch out for flaws and inadequacies. This type of design can also be given to others for their comments and critical evaluation. In the absence of such a strategy, it will likely be challenging for the critic to supply a comprehensive review of the offered study.

A research design isn't just a work plan. A work plan details what needs to be done to complete the project but the work plan will flow from the project's design. The **function of a research design** is to make certain that evidence obtained allows us to answer the initial question as unambiguously as possible. Acquiring relevant evidence involves specifying the kind of evidence required to answer the research question, to test a theory, to judge a programme or to precisely describe some phenomenon.

Errors Affecting Research Designs

The research design must attempt to reduce the 8 types of potential errors that can influence research results, viz.

1. Surrogate Information error
2. Measurement error.
3. Experimental error.
4. Population Specification error.
5. Frame Error.
6. Sampling error.
7. Selection error.
8. Non Response error.

However, it should also be kept in mind that research design must reduce total error, not just one or two aspects of total error.

1. Surrogate Information Error:

Surrogate information error is caused by a variation between the information required to solve the problem and the information sought by the researcher. The so-called price-

quality relationship, where a consumer uses the price of a brand to represent its quality level, is a common example of a measure that is subject to surrogate information error (because price level does not always reflect quality level). It has been argued that, in part, the earlier mentioned taste test research done by Coca-Cola India resulted in surrogate information. The company based its decision on taste preferences. The resultant consumer backlash was caused by surrogate information error, as consumers purchase Coke for reasons other than taste alone.

2. Measurement error:

Measurement error is caused by a difference between the information desired by the researcher and the information provided by the measurement process. In other words, not only is it possible to seek the wrong type of information (surrogate information error) but it is also possible to gather information that is different from what is being sought. This is one of the most common and serious errors. For example, respondents may exaggerate their income in order to impress an interviewer; the reported income will then reflect an unknown amount of measurement error, measurement error is particularly difficult to control because it can arise from many different sources.

3. Experimental Error:

Experiments are designed to measure the impact of one or more independent variables on a dependent variable. Experimental error occurs when the effect of the experimental situation itself is measured rather than the effect of the independent variable. For example, a retail chain may increase the price of selected items in four outlets and leave the price of the same items constant in four similar outlets, in an attempt to discover the best pricing strategy. However, unique weather patterns, traffic condition, or competitors' activities may affect the sales at one set of stores and not the other. Thus, the experimental results will reflect the impact of variables other than price. Like measurement error, experimental error can arise from a number of sources.

4. Population Specification Error:

Population specification error is caused by selecting an inappropriate universe or population from which to collect data. This is a potentially serious problem in both industrial and consumer research. A firm wishing to learn the criteria that are considered most important in the purchase of certain machine tools might conduct a survey among purchasing agents. Yet, in many firms, the purchasing agents do not determine or necessarily even know the criteria behind brand selections. These decisions may be made by the machine operators, by committee, or by higher Level executives. A study that focuses on the purchasing agent as the person who decides which brands to order may be subject to population specification error.

5. Frame Error:

The sampling frame is the list of population members from which the sample units are selected. An ideal frame identifies each member of the population once and only once. Frame error is caused by using an inaccurate or incomplete sampling frame. For example, using the telephone directory as a sampling frame for the population of a community contains a potential for frame error. Those families who do not have listed numbers, both voluntarily and non-voluntarily, are likely to differ from those with listed numbers in such respects as income, gender and mobility.

6. Sampling Error:

Sampling error is caused by the generation of a non-representative sample by means of a probability sampling method. For example, a random sample of one hundred university students could produce a sample composed of all females (or all seniors or all business majors). Such a

sample would not be representative of the overall student body. Yet it could occur using probability sampling techniques.

7. Selection Error:

Selection error occurs when a non-representative sample is obtained by non-probability sampling methods. For example, if an interviewer was afraid of dogs, it would inevitably happen that in those surveys that allowed any freedom of choice, this interviewer would have avoided homes with dogs present. Obviously, such a practice may introduce error into the survey results. Selection error is a major problem in non-probability samples.

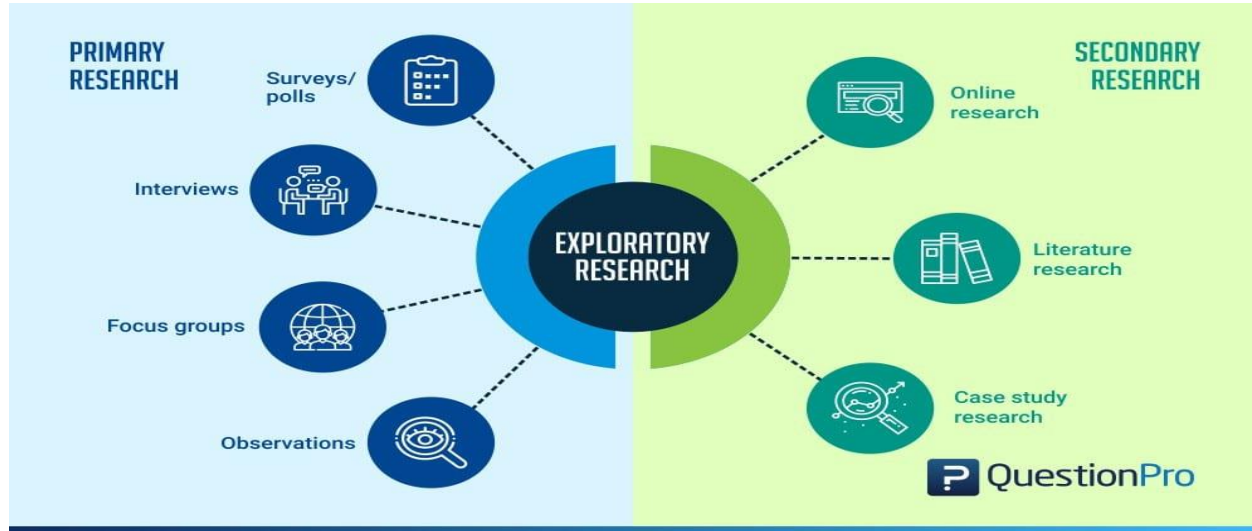
8. Non-response Error:

Non-response error is caused by i) A failure to contact all members of a sample and/or ii) The failure of some contacted members of the sample to respond to all or specific parts of the measurement instrument. Individuals who are difficult to contact or who are reluctant to co-operate will differ, on at least some characteristics, from those who are relatively easy to contact or who readily co-operate. If these differences include the variable of interest, non-response error has occurred. For example, people are more likely to respond to a survey on a topic that interests them. If a firm were to conduct a mail survey to estimate the Incidence of athlete foot among adults, non-response error would be of major concern. Why? Those most likely to be interested in athlete's foot, and thus most likely to respond to the survey, are current or recent sufferers of the problem. If the firm were to use the percentage of those responding who report having athlete's foot as an estimate of the total population having athlete's foot, the company would probably greatly over estimate the extent of the problem.

Exploratory research: Definition

Exploratory research is defined as a research used to investigate a problem which is not clearly defined. It is conducted to have a better understanding of the existing research problem, but will not provide conclusive results. For such a research, a researcher starts with a general idea and uses this research as a medium to identify issues, that can be the focus for future research. An important aspect here is that the researcher should be willing to change his/her direction subject to the revelation of new data or insight. Such a research is usually carried out when the problem is at a preliminary stage. It is often referred to as grounded theory approach or interpretive research as it used to answer questions like what, why and how.

Types and methodologies of Exploratory research



While it may sound difficult to research something that has very little information about it, there are several methods which can help a researcher figure out the best research design, data collection methods and choice of subjects. There are two ways in which research can be conducted namely primary and secondary.. Under these two types, there are multiple methods which can be used by a researcher. The data gathered from these research can be qualitative or quantitative. Some of the most widely used research designs include the following:

Primary research methods

Primary research is information gathered directly from the subject. It can be through a group of people or even an individual. Such a research can be carried out directly by the researcher himself or can employ a third party to conduct it on their behalf. Primary research is specifically carried out to explore a certain problem which requires an in-depth study.

- **Surveys/polls:** Surveys/polls are used to gather information from a predefined group of respondents. It is one of the most important quantitative method. Various types of surveys or polls can be used to explore opinions, trends, etc. With the advancement in technology, surveys can now be sent online and can be very easy to access. For instance, use of a survey app through tablets, laptops or even mobile phones. This information is also available to the researcher in real time as well. Nowadays, most organizations offer short length surveys and rewards to respondents, in order to achieve higher response rates.

For example: A survey is sent to a given set of audience to understand their opinions about the size of mobile phones when they purchase one. Based on such information organization can dig deeper into the topic and make business related decision.

- **Interviews:** While you may get a lot of information from public sources, but sometimes an in person interview can give in-depth information on the subject being studied. Such a research is a qualitative research method. An interview with a subject matter expert can give you meaningful insights that a generalized public source won't be able to provide. Interviews are carried out in person or on telephone which have open-ended questions to get meaningful information about the topic.

For example: An interview with an employee can give you more insights to find out the degree of job satisfaction, or an interview with a subject matter expert of quantum theory can give you in-depth information on that topic.

- **Focus groups:** Focus group is yet another widely used method in exploratory research. In such a method a group of people is chosen and are allowed to express their insights on the topic that is being studied. Although, it is important to make sure that while choosing the individuals in a focus group they should have a common background and have comparable experiences.

For example: A focus group helps a research identify the opinions of consumers if they were to buy a phone. Such a research can help the researcher understand what the consumer value while buying a phone. It may be screen size, brand value or even the dimensions. Based on which the organization can understand what are consumer buying attitudes, consumer opinions, etc.

- **Observations:** Observation research can be qualitative observation or quantitative observation. Such a research is done to observe a person and draw the finding from their reaction to certain parameters. In such a research, there is no direct interaction with the subject.

For example: An FMCG company wants to know how it's consumer react to the new shape of their product. The researcher observes the customers first reaction and collects the data, which is then used to draw inferences from the collective information.

Secondary research methods

Secondary research is gathering information from previously published primary research. In such a research you gather information from sources likes case studies, magazines, newspapers, books, etc.

- **Online research:** In today's world, this is one of the fastest way to gather information on any topic. A lot of data is readily available on the internet and the researcher can download it whenever he needs it. An important aspect to be noted for such a research is the

genuineness and authenticity of the source websites that the researcher is gathering the information from.

For example: A researcher needs to find out what is the percentage of people that prefer a specific brand phone. The researcher just enters the information he needs in a search engine and gets multiple links with related information and statistics.

- **Literature research:** Literature research is one of the most inexpensive method used for discovering a hypothesis. There is tremendous amount of information available in libraries, online sources, or even commercial databases. Sources can include newspapers, magazines, books from library, documents from government agencies, specific topic related articles, literature, Annual reports, published statistics from research organizations and so on.

However, a few things have to be kept in mind while researching from these sources. Government agencies have authentic information but sometimes may come with a nominal cost. Also, research from educational institutions is generally overlooked, but in fact educational institutions carry out more number of research than any other entities.

Furthermore, commercial sources provide information on major topics like political agendas, demographics, financial information, market trends and information, etc.

For example: A company has low sales. It can be easily explored from available statistics and market literature if the problem is market related or organization related or if the topic being studied is regarding financial situation of the country, then research data can be accessed through government documents or commercial sources.

- **Case study research:** Case study research can help a researcher with finding more information through carefully analyzing existing cases which have gone through a similar problem. Such analysis are very important and critical especially in today's business world. The researcher just needs to make sure he analyses the case carefully in regards to all the variables present in the previous case against his own case. It is very commonly used by business organizations or social sciences sector or even in the health sector.

For example: A particular orthopedic surgeon has the highest success rate for performing knee surgeries. A lot of other hospitals or doctors have taken up this case to understand and benchmark the method in which this surgeon does the procedure to increase their success rate.

Exploratory research: Steps to conduct a research

- **Identify the problem:** A researcher identifies the subject of research and the problem is addressed by carrying out multiple methods to answer the questions.
- **Create the hypothesis:** When the researcher has found out that there are no prior studies and the problem is not precisely resolved, the researcher will create a hypothesis based on the questions obtained while identifying the problem.
- **Further research:** Once the data has been obtained, the researcher will continue his study through descriptive investigation. Qualitative methods are used to further study the subject in detail and find out if the information is true or not.

Characteristics of Exploratory research

- They are not structured studies
- It is usually low cost, interactive and open ended.
- It will enable a researcher answer questions like what is the problem? What is the purpose of the study? And what topics could be studied?
- To carry out exploratory research, generally there is no prior research done or the existing ones do not answer the problem precisely enough.
- It is a time consuming research and it needs patience and has risks associated with it.
- The researcher will have to go through all the information available for the particular study he is doing.
- There are no set of rules to carry out the research per se, as they are flexible, broad and scattered.
- The research needs to have importance or value. If the problem is not important in the industry the research carried out is ineffective.
- The research should also have a few theories which can support its findings as that will make it easier for the researcher to assess it and move ahead in his study
- Such a research usually produces qualitative data, however in certain cases quantitative data can be generalized for a larger sample through use of surveys and experiments.

Advantages of Exploratory research

- The researcher has a lot of flexibility and can adapt to changes as the research progresses.
- It is usually low cost.
- It helps lay the foundation of a research, which can lead to further research.
- It enables the researcher understand at an early stage, if the topic is worth investing the time and resources and if it is worth pursuing.
- It can assist other researchers to find out possible causes for the problem, which can be further studied in detail to find out, which of them is the most likely cause for the problem.

Disadvantages of Exploratory research

- Even though it can point you in the right direction towards what is the answer, it is usually inconclusive.
- The main disadvantage of exploratory research is that they provide qualitative data. Interpretation of such information can be judgmental and biased.
- Most of the times, exploratory research involves a smaller sample, hence the results cannot be accurately interpreted for a generalized population.
- Many a times, if the data is being collected through secondary research, then there is a chance of that data being old and is not updated.

Importance of Exploratory research

Exploratory research is carried out when a topic needs to be understood in depth, especially if it hasn't been done before. The goal of such a research is to explore the problem and around it and not actually derive a conclusion from it. Such kind of research will enable a researcher to set a strong foundation for exploring his ideas, choosing the right research design and finding variables that actually are important for the analysis. Most importantly, such a research can help organizations or researchers save up a lot of time and resources, as it will enable the researcher to know if it worth pursuing.

Conclusive Research Design

Conclusive research design, as the name implies, is applied to generate findings that are practically useful in reaching conclusions or decision-making. In this type of studies research objectives and data requirements need to be clearly defined. Findings of conclusive studies usually have specific uses. Conclusive research design provides a way to verify and quantify findings of exploratory studies.

Conclusive research design usually involves the application of quantitative methods of data collection and data analysis. Moreover, conclusive studies tend to be deductive in nature and research objectives in these types of studies are achieved via testing hypotheses.

Factor	Conclusive	Exploratory
Objectives	To test hypothesis and relationships	To get insights and understanding
Characteristics	Information needs a clearly defined	Information needs are loosely defined
	Research process is formal and structured	Research process is unstructured and flexible
	Large representative sample	Small, non-representative sample
	Data analysis is quantitative	Primary data analysis is qualitative
Findings	Conclusive	Only tentative
Outcome	Findings used as input to decision making	Generally followed by further exploratory conclusive research

The table below illustrates the main differences between conclusive and exploratory research design:

Main differences between conclusive and exploratory research design

It has to be noted that “conclusive research is more likely to use statistical tests, advanced analytical techniques, and larger sample sizes, compared with exploratory studies. Conclusive research is more likely to use quantitative, rather than qualitative techniques”[1]. Conclusive research is helpful in providing a reliable or representative picture of the population through the application of valid research instrument.

Conclusive research design can be divided into two categories: descriptive research and causal research.

Descriptive research is used to describe some functions or characteristics of phenomenon and can be further divided into the following groups:

1. Case study;
2. Case series study;
3. Cross-sectional study;
4. Longitudinal study;
5. Retrospective study.

Causal research, on the other hand, is used to research cause and affect relationships. Two popular research methods for causal studies are experimental and quasi-experimental studies.

Descriptive Research

Meaning

The descriptive research design involves observing and collecting data on a given topic without attempting to infer cause-and-effect relationships. The goal of descriptive research is to provide a comprehensive and accurate picture of the population or phenomenon being studied and to describe the relationships, patterns, and trends that exist within the data.

Descriptive research methods can include surveys, observational studies, and case studies, and the data collected can be qualitative or quantitative. The findings from descriptive research provide valuable insights and inform future research, but do not establish cause-and-effect relationships.

Types of Descriptive Research

1. Survey Research

Surveys are a type of descriptive research that involves collecting data through self-administered or interviewer-administered questionnaires. Additionally, they can be administered in-person, by mail, or online, and can collect both qualitative and quantitative data.

2. Observational Research

Observational research involves observing and collecting data on a particular population or phenomenon without manipulating variables or controlling conditions. It can be conducted in naturalistic settings or controlled laboratory settings.

3. Case Study Research

Case study research is a type of descriptive research that focuses on a single individual, group, or event. It involves collecting detailed information on the subject through a variety of methods, including interviews, observations, and examination of documents.

4. Focus Group Research

Focus group research involves bringing together a small group of people to discuss a particular topic or product. Furthermore, the group is usually moderated by a researcher and the discussion is recorded for later analysis.

5. Ethnographic Research

Ethnographic research involves conducting detailed observations of a particular culture or community. It is often used to gain a deep understanding of the beliefs, behaviors, and practices of a particular group.

Cross sectional studies and longitudinal studies.

Cross-sectional study

Both the cross-sectional and the longitudinal studies are observational studies. This means that researchers record information about their subjects without manipulating the study environment. In our study, we would simply measure the cholesterol levels of daily walkers and non-walkers along with any other characteristics that might be of interest to us. We would not influence non-walkers to take up that activity, or advise daily walkers to modify their behaviour. In short, we'd try not to interfere.

The defining feature of a cross-sectional study is that it can compare different population groups at a single point in time. Think of it in terms of taking a snapshot. Findings are drawn from whatever fits into the frame.

To return to our example, we might choose to measure cholesterol levels in daily walkers across two age groups, over 40 and under 40, and compare these to cholesterol levels among non-walkers in the same age groups. We might even create subgroups for gender. However, we would not consider past or future cholesterol levels, for these would fall outside the frame. We would look only at cholesterol levels at one point in time.

The benefit of a cross-sectional study design is that it allows researchers to compare many different variables at the same time. We could, for example, look at age, gender, income and educational level in relation to walking and cholesterol levels, with little or no additional cost.

However, cross-sectional studies may not provide definite information about cause-and-effect relationships. This is because such studies offer a snapshot of a single moment in time; they do not consider what happens before or after the snapshot is taken. Therefore, we can't know for sure if our daily walkers had low cholesterol levels before taking up their exercise regimes, or if the behaviour of daily walking helped to reduce cholesterol levels that previously were high.

Longitudinal Study

A longitudinal study, like a cross-sectional one, is observational. So, once again, researchers do not interfere with their subjects. However, in a longitudinal study, researchers conduct several observations of the same subjects over a period of time, sometimes lasting many years.

The benefit of a longitudinal study is that researchers are able to detect developments or changes in the characteristics of the target population at both the group and the individual level. The key here is that longitudinal studies extend beyond a single moment in time. As a result, they can establish sequences of events.

To return to our example, we might choose to look at the change in cholesterol levels among women over 40 who walk daily for a period of 20 years. The longitudinal study design would account for cholesterol levels at the onset of a walking regime and as the walking behaviour continued over time. Therefore, a longitudinal study is more likely to suggest cause-and-effect relationships than a cross-sectional study by virtue of its scope.

In general, the research should drive the design. But sometimes, the progression of the research helps determine which design is most appropriate. Cross-sectional studies can be done more quickly than longitudinal studies. That's why researchers might start with a cross-sectional study to first establish whether there are links or associations between certain variables. Then they would set up a longitudinal study to study cause and effect.

Experimental Design Definition

In Statistics, the experimental design or the design of experiment (DOE) is defined as the design of an information-gathering experiment in which a variation is present or not, and it should be performed under the full control of the researcher. This term is generally used for controlled experiments. These experiments minimise the effects of the variable to increase the reliability of the results. In this design, the process of an experimental unit may include a group of people, plants, animals, etc.

Classification/Types of Experimental Designs

There are different types of experimental designs of research. They are:

1. Formal & Informal Research Design
2. Pre-experimental Research Design
3. True-experimental Research Design
4. Quasi-Experimental Research Design
5. Statistical experimental design.
6. Randomised Block Design

In this article, we are going to discuss these different experimental designs for research with examples.

Formal Research

Formal research is a type of research study conducted using a systematic approach and scientific methods. A formal research study typically involves several components: abstract, introduction, literature review, research design and method, results and analysis, conclusion, bibliography. All in all, formal research is more scientific in nature than informal research and aims to discover new information or solve a problem. It always uses up to date, reliable sources. Moreover, the sources of information used in research are also complied using a reference system.

Researchers usually begin a formal research study with a hypothesis; then, they test this hypothesis rigorously. They also explore and analyze the literature already available on their research subject. This allows them to study the research subject from multiple perspectives, acknowledging different problems that need to be solved. In comparison to informal research, formal research tends to be objective or unbiased. This mainly happens because formal research uses scientific methods which involve objective information. If the research involves participants, for example, in surveys or interviews, the researcher has to obtain their written consent first. Furthermore, we can apply the findings from formal research to a larger audience.

Informal Research

Informal research is the use of nonscientific methods to gather and analyze data. Since the research uses nonscientific methods, its finding cannot be applied to a wider audience. Most of us conduct informal research on a regular basis. For example, conducting a simple survey, informal interviews with a target audience, seeking opinions of colleagues, searching through company files, etc. are all nonscientific methods of gathering data and are therefore forms of informal research.

Informal research does not take much time or effort. This usually involves just looking through some information or gathering some data. For example, suppose you are going to buy a new laptop. You will ask for the opinions of your colleagues and may look through some websites and reviews to know more information about laptop brands. Your decision here will be based on informal research. But it's important to know this information may be subjective. Moreover, as this is an informal research, you might also not worry about the source of information.

Difference Between Formal and Informal Research

Definition

Formal research is a type of research study conducted using a systematic approach and scientific methods while informal research is the use of nonscientific methods to gather and analyze data.

Methods

While formal research uses scientific methods, informal research uses nonscientific methods

Time and Effort

Informal research takes less time and effort than formal research.

Objectivity

Formal research tends to be more objective or unbiased than informal research.

Sources

It's necessary to list the sources of information in a formal research, but informal research does not follow this practice.

Application of Findings

Findings of formal research can be applied to a larger group, whereas findings of informal research can be applied only to a smaller group.

Pre-experimental Research Design

The simplest form of experimental research design in Statistics is the pre-experimental research design. In this method, a group or various groups are kept under observation, after some factors are recognised for the cause and effect. This method is usually conducted in order to understand whether further investigations are needed for the targeted group. That is why this process is considered to be cost-effective. This method is classified into three types, namely,

- Static Group Comparison
- One-group Pretest-posttest Experimental Research Design
- One-shot Case Study Experimental Research Design

True-experimental Research Design

This is the most accurate form of experimental research design as it relies on the statistical hypothesis to prove or disprove the hypothesis. This is the most commonly used method implemented in Physical Science. True experimental research design is the only method that establishes the cause and effect relationship within the groups. The factors which need to be satisfied in this method are:

- Random variable
- Variable can be manipulated by the researcher
- Control Groups (A group of participants are familiar with the experimental group, but the experimental rules do not apply to them)
- Experimental Group (Research participants where experimental rules are applied)

Quasi-Experimental Design

A quasi-experimental design is similar to a true experimental design, but there is a difference between the two.

In a true experiment design, the participants of the group are randomly assigned. So, every unit has an equal chance of getting into the experimental group.

In a quasi-experimental design, the participants of the groups are not randomly assigned. So, the researcher cannot make a cause or effect conclusion. Thus, it is not possible to assign the participants to the group.

Apart from these types of experimental design research in statistics, there are other two methods used in the research process such as randomized block design and completely randomized design.

Statistical experimental design.

The (statistical) design of experiments (DOE) is an efficient procedure for planning experiments so that the data obtained can be analyzed to yield valid and objective conclusions. DOE begins with determining the objectives of an experiment and selecting the process factors for the study.

4 types of experimental design statistics

Four major design types with relevance to user research are experimental, quasi-experimental, correlational and single subject. These research designs proceed from a level of high validity and generalizability to ones with lower validity and generalizability.

the statistical design process?

The Statistical Process has five steps:

- Design the study
- Collect the data
- Describe the data
- Make inferences
- Take action.

In a designed experiment, researchers control the conditions of the study. In an observational study, researchers don't control the conditions but only observe what happens.

Randomised Block Design

The randomised block design is preferred in the case when the researcher is clear about the distinct difference among the group of objects. In this design, the experimental units are classified into subgroups of similar categories. Those groups are randomly assigned to the group of treatment. The blocks are classified in such a way in the variability within each block should be less than the variability among the blocks. This block design is quite efficient as it reduces the variability and produces a better estimation.

Example:

In a drug testing experiment, the researcher believes that age is the most significant factor. So he divides the units according to the age groups such as

- Under 15 years old
- 15 – 35 years old
- 36 – 55 years old
- Over 55 years old

Completely Randomised Design

Of all the types, the simplest type of experimental design is the completely randomized design, in which the participants are randomly assigned to the treatment groups. The main advantage of using this method is that it avoids bias and controls the role of chance. This method provides a solid foundation for Statistical analysis as it allows the use of probability theory.

Module-3

Sampling: Concepts, Types of Sampling, **Probability Sampling:** simple random sampling, systematic sampling, stratified random sampling, cluster sampling,

Non Probability Sampling: convenience sampling- judgmental sampling, snowball sampling, quota sampling, Errors in sampling.

Sampling: Concepts

What is Sampling?

Sampling is a process in which the fixed numbers of observations are taken randomly from a larger population. A technique which is fundamental for behavioral research is known as sampling and without using it, research work is not possible. It is impracticable and Impossible to study the whole population due to practical limitations of cost, time and other factors that are indispensable and operative in studying the whole population. For the sole reason of making the research findings economical and accurate, the concept of sampling has been introduced.

For example, for taking decision about to purchase or not to purchase the fruits, a fruit merchant inspects randomly only a few of them rather than examining each and every fruit. The important objective of sampling is to obtain maximum information of the population under study using minimum of money, labour and time.

Sampling in Research:

A definite plan to obtain a sample from the sampling frame is sample design. The method or technique which is adopted by the researcher in selecting the units of sampling from the population is called sampling design.

The basis for selecting a sample survey is the framework or road-map which is called sample design and it affects other important aspects of the survey. For obtaining some type of relevant information using survey, researchers execute it for some population, or universe. The sampling frame must be defined in such a way that it represents the population of interest out of which a sample is to be drawn at random. This sampling frame should be either identical to the population or may be a part of it. The sampling frame can be indirectly related to the population subject to some under coverage (e.g., the population is preschool children and the frame is a listing of preschools). The design of the sample may be simple or complex which provides the fundamental plan and methodology for the selection of the sample.

Definition of Sampling

According to Davis S. Fox :

"In the social science, it is not possible to collect data from every respondent relevant to our study but only from some fractional part is called sampling".

According to Cochran :

"In every branch of science we lack the resources, to study more than a fragment of the phenomena that might advance our knowledge". In this definition a fragment is the sample and phenomena is the population.

Basic Concepts of Sampling

Following are the basic concepts of sampling in research methodology :

1) Universe/Population :

The entire group of units, being the focus of study, is represented by the population or universe consisting of all the persons in the country, in a particular geographical location, or a special domestic or economic group based on the purpose and scope of study. Non-human entities like farms, houses or business establishments also comprise population.

For example, if a study is to be conducted on the marks obtained by the students of Statistics then the 'universe' will be all the students of that subject in that class. If the class is consisting of 50 students then the 50 students is the "universe".

"Population or universe" is the sum total of the objects under study, both animated and in animated. Thus, numerical details of the collection of the individuals or of their qualities, or of results of the operations are known as universe. Universe can be classified as follows :

i) Finite Universe :

A 'finite universe' comprises of finite number of entities or members. For example, the universe of the heights of students in a particular class.

ii) Infinite Universe :

If the infinite numbers of members are included in a universe then it is called "infinite universe". For example, the universe of pressures at various points in the atmosphere.

2) Statistical Population :

To draw out statistical inferences from the set of entities of statistical population is based on random sampling of the population. For example, to derive generalizations about the crows, to only the set of crows should be the interest of study. However, if one chooses a population such as 'all crows', then the limitations will be in observing crows both the existing as well as existing in the future. There will be additional limitations of geography where the resources are limited for studying crows.

The total of quantities that are measurable or set of numbers is 'statistical population'. Thus, when only a single character symbolized every element of such a set then it is defined as 'single population', c.g., income of individuals. Depending on the finite and infinite number of elements, statistical population can be infinite as well as finite. A random set may not be a statistical population.

For example, a set of crows at a point of time will not represent a statistical population.

3) Sample :

A part of the population which is examined to estimate the characteristics of the population is called a sample.

For example,

- A portion of rice is examined to assess the quality of rice in the bag where the portion of rice examined is the sample and the bag of rice is the population.
- From a large consignment, proportion of defective articles is estimated only from a portion of the consignment. The selected portion is called sample.

4) Census:

When in a research every item or entity of the universe is included, that approach is called "census". In this method, detailed information regarding every population unit is collected. It is also called complete enumeration. This method provides more accuracy in the results in comparison to the sampling method.

For example, population survey about various issues, such as, literacy, education, income, etc., include all the items of the population.

5) Sampling Frame:

The actual set of units and a list containing every member of the population from which a sample is drawn at random is called sample frame.

For example, if one wants to know the factors that are responsible for the patients being admitted in the hospital after an acute attack of asthma in a specified area, then it becomes necessary to know the names of all those people of that particular area being admitted into the hospital for that reason.

6) Sample Unit :

Every sample is made up of several members or components. These members or components are called "sampling units". A sampling unit may or may not be a sampling element. Generally sampling unit is referred as sampling element. But in case of sampling units consisting of several population units, each sampling unit is called a cluster of units. The individual population units which come under the cluster of units, are called sampling elements. Therefore in this case a sampling element is an individual object on which the research is done, and it is a component of sampling unit.

For example, in case of inspecting the quality of oranges, only few oranges are selected to examine and each selected orange is a sampling unit. But in case of finding, the literacy level of a particular colony, several houses are selected in that colony for research. These houses are sampling units. However the individual members of each house are approached for examining the literacy level, thus these individual members of several houses are sampling elements.

7) Sample Design :

Designing a plan for effectively drawing out a sample from the sampling frame is called 'sample design'. In this different tools or techniques are used by researchers for drawing sampling units from the sampling frame.

Characteristics of a Good Sample

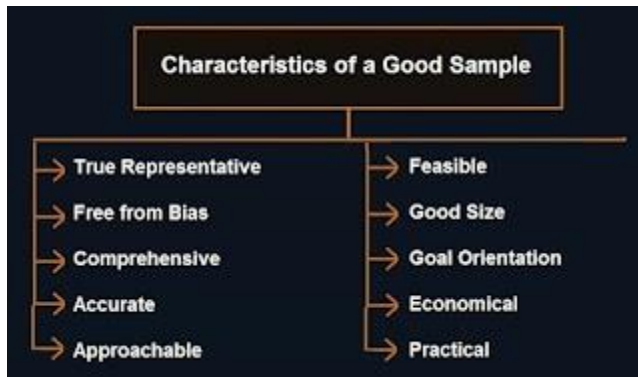
The various features of a good sample design are as follows :

A good sampling frame should be :

- Relevant : Research topic must be directly linked with relevant things.
- Complete : Coverage of all relevant items.
- Precise : It should exclude all the irrelevant items.
- Up-to-Date : Incorporating recent changes and additions, and deleting redundant items.

Characteristics of a Good Sample

The various features of a good sample design are as follows :



1) True Representative : The true representative of the population and matching its properties is termed as good sample where aggregate of certain properties is the population and sample is the sub-total of the universe.

2) Free from Bias : A good sample does not allow prejudices, pre-conceptions, and imaginations which affect its choice and it is unbiased.

3) Accurate : A sample is called good when it yields accurate estimates or statistics and free from errors.

4) Comprehensive : A sample that is true representative of the population is also comprehensive in nature which is controlled by definite purpose of investigation. A sample's characteristic may be comprehensive but it may not be a good representative of the population.

5) Approachable : The subjects of good sample are easily accessible where the tools of research are easily conducted and easy collection of data possible.

6) Good Size : The size of good sample is such that it yields an accurate result and the error due to probability can be estimated.

7) Feasible : A good sample creates the research work more feasible.

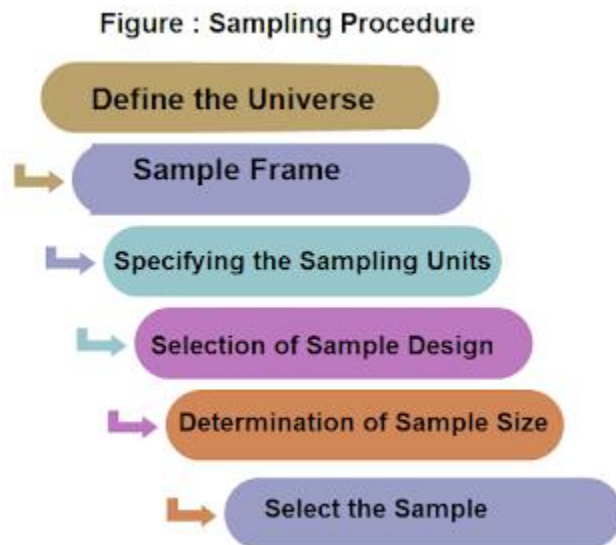
8) Goal Orientation : Any sample which is selected by the researcher should be able to satisfy the objectives of the research. The sample should be taken in proper number. It should be customized to fit the environment under which the research is going to be conducted. If the sample is changed as per the requirement of the survey design then it can come out with better results and outcomes.

9) Practical : It means that the concepts of sample selection should be applied properly while conducting the research. The researcher should be well experienced and well instructed. The instructions which are passed to observer should be clear, complete and correct in all terms so as to avoid errors and biasness on their part. The sample should be selected on basis of the sample design. The sampling units should be representative. The sample design should be practical and feasible in nature.

10) Economical : It refers that the research should not incur huge cost, time or efforts. One of the objectives of any research is to complete the research with minimum effort, time, money and resources. The researcher calculates per unit cost for each respondent. The researcher should choose that sampling design which gives minimum per respondent cost and maximum accuracy. The funding agency is highly concerned with per unit cost and the corresponding accuracy.

Procedure of Sampling

A brief description of which measurements will be taken at what times, on which material, in what manner, and by whom, is termed as 'sampling plan'. The designing of the sampling plans should incorporate that data to be evaluated which is a representative sample of the parameters of interest and be answerable to all the questions as stated in the goals. Following are the steps involved in sampling procedure in research or developing a sampling plan :



1) Define the Universe : The universe can be restricted to some geographical limits, a particular type of product or some other constraints. Defining the universe is the initial problem procedures of sampling. The collection of elements or objects possessing information desired by the researcher regarding what inferences are to be made is termed as the target population or the universe. Precise definition of the target population is must as ambiguous definition will result in ineffective research and deceptive in meaning. Proper defining of the target population requires translating the problem definition precisely as what should and should not be included in the sample. The description of the target population should be on the basis of elements, sampling units, the period and time. Thus, in survey research element is like an object from which or about information is required. In which survey research, the elements are usually the respondents.

For example, consider a marketing research project in which researcher wants to take response of the consumer for a new brand of men's perfume. It has to decide that who should be included in the target population. It is the important question all the men included in the research or the men who used the perfume during the last month or men who are 17 years of age, or older. Should females be included, because some women buy colognes for their husbands?. Appropriate description of target population requires prior resolution of such types of questions.

2) Sample Frame : The frame is enlisted by some population that already exists or it is constructed for some study to be conducted by the researcher. After the specification of the population to be studied, the frame of the population is developed. Sampling frame consists of all the sampling units of the population which is built by the researcher for the subject of his study or

it consists of already existing list of the population. This frame does not consist of names only but it may include a permanent location, a boundary, an address, or the set of certain rules that describes a sampling unit.

A set of boundaries which is restricting the universe is termed as a frame that may be in the form of lists, indices, maps, directories, population records, electoral rolls, city tax rolls, enrollment of the students in an university, etc. This frame is very mandatory in marketing studies and every element of the population enlisted appears only once which constitutes a sample frame. For a good sampling frame it should be accurate, free from duplication, and conveniently available. For the better performance of sampling procedure, sampling should be essential for marketing research. The representation of the elements of the target populations sampling frame which consists of a list to identify the target population.

For example, firms in industry that are enlisted in association directory, telephone book, a mailing list purchased from a commercial organisation, a city directory, or a map.

3) Specifying the Sampling Units : A sampling unit is decided on the basis of the sampling frame which is the primary unit containing the sampling elements of the population like, city blocks, households, a business organisation, etc. The design of the project is the partial basis of the selection of the sampling unit. The units that are served as initial sampling are called as primary sampling units. The objective of the inquiry is the basis of the composition of one or more units of the population.

For example, for the assessment of the consumer response to a new line of lipsticks, Lakme wants to sample females who are over 18 years of age where it can be the possibility of sampling such females directly and the sampling unit would be the same as an element. Otherwise, households might be the sampling units which are sampled. All the females above 18 in the selected households would be interviewed where the sampling unit differs from the population element. The time period under consideration is the extent of the geographical boundaries and the time factor.

4) Selection of Sample Design : The procedure of selecting units in the sample involves two primary methods of sampling namely, probability and non-probability methods that can be further sub divided into specific methods of selection. The probability sample being one and the procedure involved in selecting the units has some specific chance of its inclusion in the sample. An arbitrary method is adopted in selecting the units which is not depending upon the chance for the non probability sample. The purpose of inquiry and the attitude or convenience of the investigators is the basis of this method.

Two decisions are involved in the selection of the sample design:

- To use probability or non-probability method of selection.
- Specific sample design to use in collecting the data.

The following considerations affect the choice of the researcher :

- Usage of the probability sampling for the evaluation of the sampling error.
- Probability sample should be used for ensuring randomness in the selection of the units.
- Non-probability sampling should be used in the absence of proper sample frame.

- Non-probability sampling should be used when the considerations of time and money are important.

To achieve the objectives of investigation in the best possible way sample design should be selected after the decision of the method of selection whether probability or non-probability method. Unconcerned about the design being chosen finally the researcher has to defend his design at the time of the final presentation of the study results.

5) Determination of Sample Size : The desired degree of accuracy in investigation is directly proportional to the size the sample which also depends upon the nature of the population and the method of selection. The ideal samples size in marketing research investigations depends upon the the type of series and the population size. There will be more degree of heterogeneity and larger units s will be drawn in the sample due to larger size of the population. Hence the size of the sample should be larger for being fully representative.

6) Select the Sample : Execution of the actual sampling process is called as selection of the sample and the real selection of the elements of sample requiring considerable amount of office and field work especially when the personal interviews are involved. Detailed specification of the sampling design is the require is requirement for the execution of the sampling process with respect to the implementation of the size of population, sampling frame, sampling unit, sampling technique and sample size. Sampling units and sample size. being the household's requirement is the operational definition of a household, procedure for those houses that are vacant and if nobody is at home then the execution of the call backs is mandatory. For all the sampling design, decisions information must be provided in detail.

Advantages of Sampling

Following are the importance of sampling in research :

- 1) Saves Time, Money and Effort :** The subjects involved in sampling are small in number which gives him less time to calculate, tabulate, present, analyse, and interpret and hence the researcher saves time, money and effort.
- 2) More Effective :** Due to the size of the sample being small than that of the population, the tiredness is reduced in collecting the information and investigator works more effectively.
- 3) Faster and Cheaper :** Due to small sample size the data collection, tabulation, presentation, analysis, and interpretation takes less time and it involves nominal expenses.
- 4) More Accurate :** Small errors are in sampling because small data are involved in collection, tabulation, presentation, analysis and interpretation.
- 5) Gives More Comprehensive Information :** Thorough investigation of the study is the result of small sample which provides more complete information as all the members of the population have equal chance to be included in the sample.

Disadvantages of Sampling

Following are the disadvantages of sampling :

- 1) Biased Selection :** Biased selection of the respondents may be included in sampling by the research worker.

2) Difficulty in Selection : It is very difficult to select a truly representative sample because a large number of factors obstructing the method of selecting good samples.

3) Specialized Knowledge Needed : Specialized knowledge is required in the sampling method where the investigator may commit serious mistakes due to its absence.

4) Problem of Cooperation : Due to scattered sample the subjects are uncooperative with the researcher.

5) Less Accuracy : When the higher standard of accuracy is expected then the sampling is not suitable.

6) Limited Nature : Due to small or dissimilar universe it is impossible to derive a representative sample where census study is the best possible substitute.

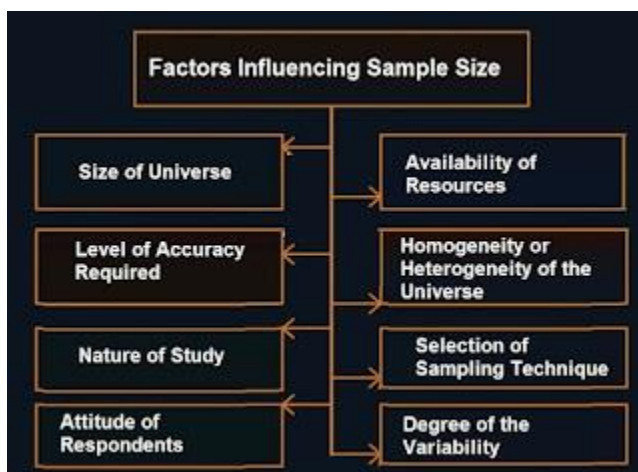
Sample Size

The sample size for a research refers to the total number of elements of the population to be included in the sample for conducting the research study. Both qualitative and quantitative points are involved in specifying the size of the sample for the research study. The accuracy of research depends on the size of the sample. It has been observed that a larger sample gives more accuracy and estimate levels. Apart from it, the availability of money, resources and efforts also specify the size of a sample.

The sample size of any research study is represented by 'n'. If the population is highly heterogeneous, then the sample will be divided into various groups with an indication that the sample size would be larger. If the population is homogeneous then the sample will be small. Thus, at pre-assigned level there is a fixation of the sampling error and the standard error of the statistic, thus having sample size as 'n' is chosen.

Factors Influencing Sample Size/ Sample Size Constraints

The following points should be taken care of while deciding the sample size :



1) Size of Universe : It has been observed and statistically proven that the sample size should be large so that it can represent the whole target population. If the universe size is large and heterogeneous then the researcher should take large sample size and vice versa.

2) Availability of Resources : The researcher needs a lot of resources to complete the task of research. These resources bind the researcher. It means that the researcher has to complete the research within those allotted resources. Resources can be money, time, experts or any other variable. If the resources are easily available then the sample size can be large otherwise smaller sample would be appropriate.

3) Level of Accuracy Required : For any research, the level of the accuracy affects sample size as bigger sample represents the whole population properly than a smaller sample. However, it is not always mandatory that a large sample will always give more accurate results. The selection of the sampling technique and research tools also impact the accuracy level of the research.

4) Homogeneity or Heterogeneity of the Universe : Homogeneous universe is that universe in which all the elements are similar, on the other hand heterogeneous universe is that universe where the elements are different from each other. A homogeneous universe requires a smaller sample size, while researcher needs to take bigger sample size in case of heterogeneous universe because a small sample may not represent the whole population.

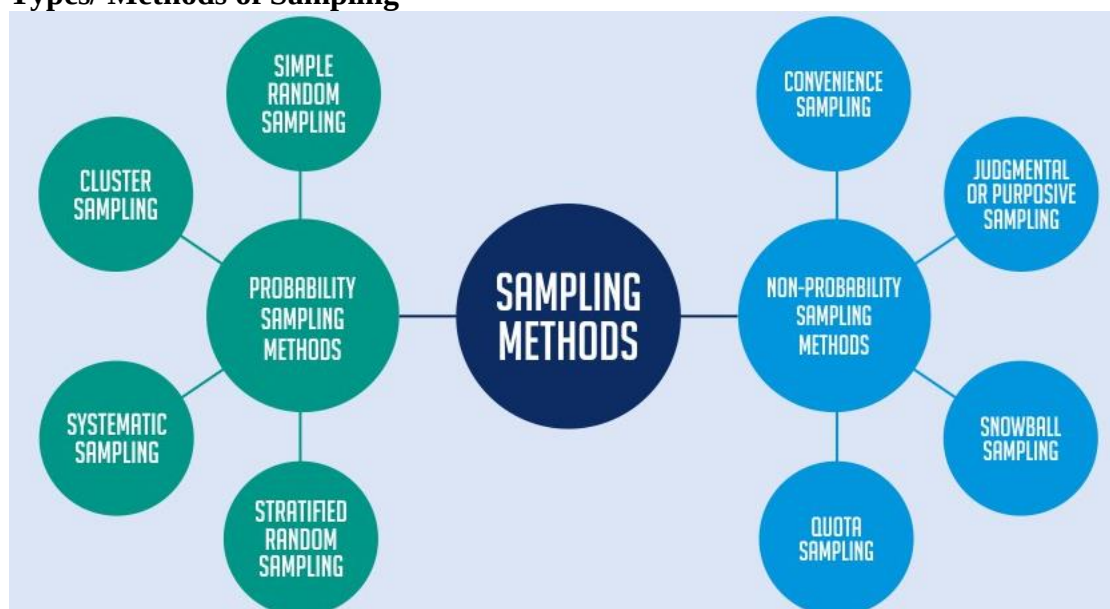
5) Nature of Study : The researcher may choose a small sample, if the research is intensive and fundamental (continuous) in nature. But in those cases, where the research is extensive and applied (researches which are not going to be repeated soon), then the researcher is bound to select larger sample size. So, it is quite clear that the nature of study also affects the sample size.

6) Selection of Sampling Technique : The sample size is also impacted by the kind of sampling technique selected by the researcher for conducting the research. For example, when a researcher adopts the simple random sampling technique then he must choose large sample size, but on the other hand in stratified sampling technique the researcher needs a comparatively smaller sample size for accurate and effective results.

7) Attitude of Respondents : The researcher is an experienced individual, so he will always choose a larger sample size in which he thinks that the non-response rate will be higher. In simple words, if there are chances that the respondents will not reply and return the questionnaire, then large sample size is chosen.

8) Degree of the Variability : Degree of variability specifies the extent to which the scores in any distribution are spread in the population. Larger numbers always indicate greater variability of scores in a population. Sometimes dispersion is substituted for variability, and this term is widely used in statistics. If the population is heterogeneous, then the researcher uses larger sample size so as to receive a specified precision level. On the other hand, if the population is more homogeneous then the smaller sample size is selected by the researcher.

Types/ Methods of Sampling



Probability Sampling:

Probability sampling refers to the selection of a sample from a population, when this selection is based on the principle of randomization, that is, random selection or chance. Probability sampling is more complex, more time-consuming and usually more costly than non-probability sampling.

Simple random sampling

Simple random sampling is a statistical method in which everyone in a population has an equal chance of being selected into a sample. The sample represents a smaller and more manageable portion of the people that can be studied and analyzed. It's a fundamental technique to gather data and make inferences about a population.

Simple random sampling is considered a fair and unbiased sample selection method. This type of sampling is the most straightforward sample selection method.

What is Simple Random Sampling?

Simple random sampling is a technique where every item in the population has an even chance and likelihood of being selected. Here, the selection of items entirely depends on luck or probability. Therefore, this sampling technique is also a method of chance.

Simple random sampling is a fundamental method and can easily be a component of a more complex method. The main attribute of this sampling method is that every sample has the same probability of being chosen.

The sample size in simple random sampling method should ideally be more than a few hundred so that it can be applied appropriately. This method is theoretically simple to understand but difficult to implement practically. Working with a large sample size isn't an easy task, and it can sometimes be challenging to find a realistic sampling bias frame.

Simple Random Sampling Methods

Researchers follow these methods to select a simple random sample:

1. They prepare a list of all the population members initially, and each member is marked with a specific number (for example, if there are nth members, then they will be numbered from 1 to N).
2. Researchers from this population choose random samples using random number tables and random number generator software. Researchers prefer random number generator software, as no human interference is necessary to generate samples.

Two approaches aim to minimize any biases in the process of this method:

- **Method of lottery**

Using the lottery method is one of the oldest ways and is a mechanical example of random sample. Researchers draw numbers from the box randomly to choose samples. In this method, the researcher gives each member of the population a number.

- **Use of random numbers**

Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Random Number Table									
1	69	24	40	68	29	39	95	60	30
97	23	70	59	79	4	47	19	38	20
13	44	5	71	12	99	78	34	9	96
34	55	83	21	72	3	37	85	61	2
22	80	18	82	54	32	84	16	46	88
7	43	6	48	11	92	63	53	86	28
56	90	36	91	64	45	15	73	10	87
49	65	50	14	51	33	89	52	74	57
98	17	100	58	5	8	77	25	62	31
27	76	66	81	26	93	41	94	67	42

Simple Random Sampling Formula

Consider that a hospital has 1000 staff members and must allocate a night shift to 100 members. All their names will be put in a bucket to be randomly selected. Since each person has an equal chance of being selected. Since we know the population size (N) and sample size (n), the calculation can be as follows:

$$P=1- \frac{N-1}{N} \cdot \frac{N-2}{N-1} \dots \frac{N-n}{N-(n-1)}$$

$$\text{Cancelling} = 1 - \frac{N-n}{N}$$

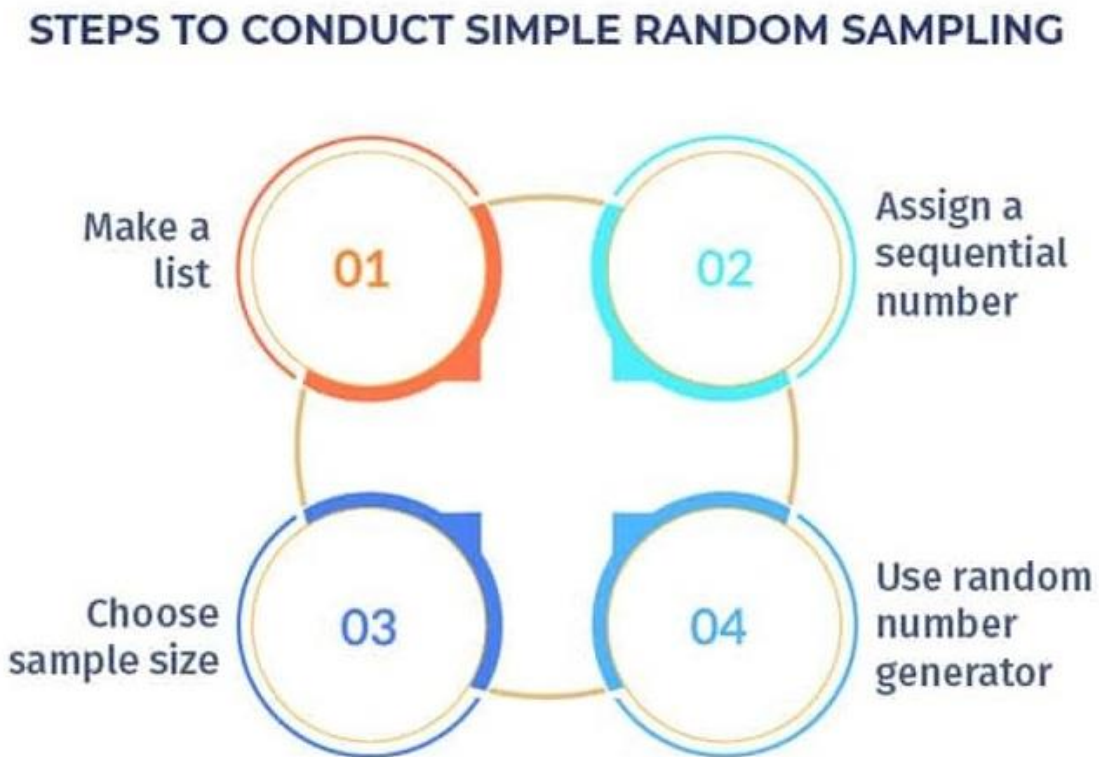
$$= \frac{n}{N}$$

$$= \frac{100}{1000}$$

$$= 10\%$$

Simple Random Sampling Steps

Follow these steps to extract a simple random sample of 100 employees out of 500.



1. **Make a list** of all the employees working in the organization. (as mentioned above, there are 500 employees in the organization, so the record must contain 500 names).
2. **Assign a sequential number** to each employee (1,2,3...n). This is your sampling frame (the list from which you draw your sample).
3. **Figure out what your sample size is going to be.** (In this case, the sample size is 100).
4. **Use a random number generator** to select the sample, using your frame (population size) from Step 2 and your sample size from Step 3. For example, if your sample size is 100 and your population is 500, generate 100 random numbers between 1 and 500.

It is important to note that Simple Random Sampling is just one of many sampling methods available, and it may not always be the best option for your specific research needs.

Simple Random Sampling in Research

Today's market research projects are much larger and involve an indefinite number of items. It is practically impossible to study every member of the population's thought process and derive inference from the study.

If, as a researcher, you want to save your time and money, simple random sampling is one of the best probability sampling methods that you can use. Getting data from a sample is more advisable and practical.

Using a census or a sample depends on several factors, such as the type of census, the degree of homogeneity/heterogeneity, costs, time, feasibility to study, the degree of accuracy needed, etc.

Advantages of Simple Random Sampling

Simple random sampling has several advantages, including:

1. It is a fair sampling method, and if applied appropriately, it helps reduce any bias involved compared to any other sampling method.
2. Since it involves a large sample frame, it is usually easy to pick a smaller sample size from the existing larger population.
3. The person conducting the research doesn't need to have prior knowledge of the data he/she is collecting. One can ask a question to gather the researcher need not be a subject expert.
4. This sampling method is a fundamental method of collecting the data. You don't need any technical knowledge. You only require essential listening and recording skills.
5. Since the population size is vast in this type of sampling method, there is no restriction on the sample size that the researcher needs to create. From a larger population, you can get a small sample quite quickly.

6. The data collected through this sampling method is well informative; the more samples better is the quality of the data.

Overall, this is a valuable and versatile method for gathering data and making inferences about populations.

Systematic sampling

Systematic sampling is a statistical method used to select a sample from a larger population systematically and randomly. It is a widespread technique for researchers and analysts who want to gather data from a large population without surveying every individual. This method is beneficial when the population is large, diverse, or hard to reach.

This article will delve into the details of systematic sampling, including its definition and advantages. We will also provide examples of how it can be used in various research and data analysis contexts and tips on implementing it effectively. Whether you are a researcher, analyst, or simply interested in understanding this sampling method, this article will give you an understanding of what it is and how to use it for better decision-making.

What is systematic sampling?

Systematic sampling is a statistical method that researchers use to zero down on the desired population they want to research. Researchers calculate the sampling interval by dividing the entire population size by the desired sample size determination. It is an extended implementation of probability sampling in which each group member is selected regularly to form a sample.

Systematic sampling definition

Systematic sampling is a probability sampling method where the researcher chooses elements from a target population by selecting a random starting point and selecting sample members after a fixed 'sampling interval.'

For example, in school, while selecting the captain of a sports team, most of our coaches asked us to call out numbers such as 1-5 (1-n) and the students with a random number decided by the coach. For instance, three would be called out as team captains. It is a non-stressful selection process for both the coach and the players. There's an equal opportunity for every member of a population to be selected using this sampling bias technique.

What are the steps to form a sample using the systematic sampling technique?

Here are the steps to form a systematic sample:

Step one: Develop a defined structural audience to start working on the sampling aspect.

Step two: As a researcher, figure out the ideal size of the sample, i.e., how many people from the entire population to choose to be a part of the sample.

Step three: Once you decide the sample size, assign a number to every member of the sample.

Step four: Define the interval of this sample. This will be the standard distance between the elements.

For example, the sample interval should be 10, which is the result of the division of 5000 (N= size of the population) and 500 (n=size of the sample).

$$\text{Systematic Sampling Formula for interval (i)} = N/n = 5000/500 = 10$$

Step five: Select the members who fit the criteria which in this case will be 1 in 10 individuals.

Step six: Randomly choose the starting member (r) of the sample and add the interval to the random number to keep adding members in the sample. r, r+i, r+2i, etc. will be the elements of the sample.

How systematic sampling works

When you are sampling, ensure you represent the population fairly. Systematic sampling is a symmetrical process where the researcher chooses the samples after a specifically defined interval. Sampling like this leaves the researcher no room for bias regarding choosing the sample. To understand how it exactly works, take the example of the gym class where the instructor asks the students to line up and asks every third person to step out of the line. Here, the instructor has no influence over choosing the samples and can accurately represent the class.

Systematic sampling example

For instance, if a local NGO is seeking to form a systematic sample of 500 volunteers from a population of 5000, they can select every 10th person in the population to build a sample systematically.

Systematic Sampling Types

Here are the types of systematic sampling:

1. Systematic random sampling
2. Linear systematic sampling
3. Circular systematic sampling

Let's take a closer look at these sampling techniques.

Systematic random sampling:

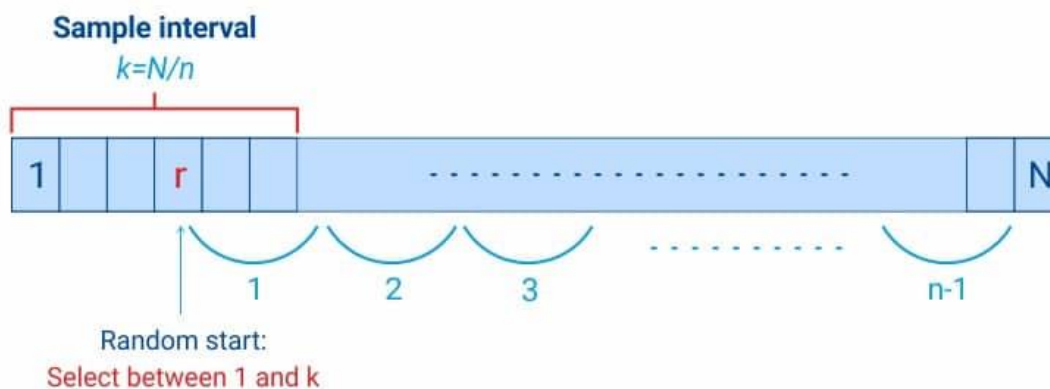
Systematic random sampling is a method to select samples at a particular preset interval. As a researcher, select a random starting point between 1 and the sampling interval. Below are the example steps to set up a systematic random sample:

1. First, calculate and fix the sampling interval. (The number of elements in the population divided by the number of elements needed for the sample.)
2. Choose a random starting point between 1 and the sampling interval.
3. Lastly, repeat the sampling interval to choose subsequent elements.

Linear systematic sampling:

Linear systematic sampling is a method where samples aren't repeated at the end and 'n' units are selected to be a part of a sample having 'N' population units. Rather than selecting these 'n' units of a sample randomly, a researcher can apply a skip logic to select these. It follows a linear path and then stops at the end of a particular population.

This sampling or skip interval (k) = N (total population units)/ n (sample size)

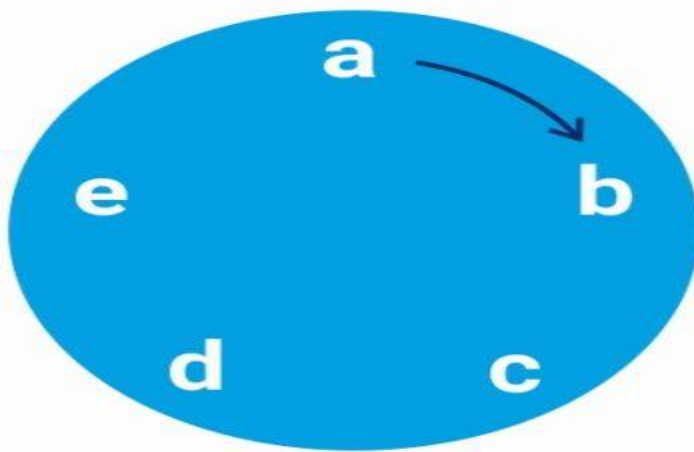


How is a Linear systematic sample selected?

- Arrange the entire population in a classified sequence.
- Select the sample size (n)
- Calculate sampling interval (k) = N/n
- Select a random number between 1 to k (including k)
- Add the sampling interval (k) to the chosen random number to add the next member to a sample and repeat this procedure to add remaining members of the sample.
- In case k isn't an integer, you can select the closest integer to N/n .

Circular systematic sampling:

In circular systematic sampling, a sample starts again from the same point once again after ending; thus, the name. For example, if $N = 7$ and $n = 2$, $k=3.5$. There are two probable ways to form sample:



1. If we consider $k=3$, the samples will be – ad, be, ca, db and ec.
2. If we consider $k=4$, the samples will be – ae, ba, cb, dc and ed.

How is a circular systematic sample selected?

- Calculate sampling interval (k) = N/n . (If $N = 11$ and $n = 2$, then k is taken as 5 and not 6)
- Start randomly between 1 to N
- Create samples by skipping through k units every time until you select members of the entire population.
- In the case of this method, there will be N number of samples, unlike k samples in the linear systematic sampling method.

Difference between linear systematic sampling and circular systematic sampling:

Here is the difference between linear and circular.

Linear	Circular
Create samples = k (sampling interval)	Create samples = N (total population)
The start and endpoints of this sample are distinct.	It restarts from the start point once the entire population is considered.
All sample units should be arranged in a linear manner prior to selection.	Elements will be arranged in a circular manner.

What are the advantages of systematic sampling?

Here are the advantages:

- It's extremely simple and convenient for the researchers to create, conduct, analyze samples.
- As there's no need to number each member of a sample, it is better for representing a population in a faster and simpler manner.
- The samples created are based on precision in member selection and free from favoritism.
- In the other methods of probability sampling methods such as cluster sampling and stratified sampling or non-probability methods such as convenience sampling, there are chances of the clusters created to be highly biased which is avoided in systematic sampling as the members are at a fixed distance from one another.
- The factor of risk involved in this sampling method is extremely minimal.
- In case there are diverse members of a population, this sampling technique can be beneficial because of the even distribution of members to form a sample.

Other probability sampling techniques like cluster sampling and stratified random sampling can be very unorganized and challenging due to which researchers and statisticians have turned to methods like systematic sampling or simple random sampling for better sampling results. It consumes the least time as it requires a selection of sample size and identification of the starting point for this sample, which needs to be continued at regular intervals to form a sample.

When to use systematic sampling?

Let's take an example where you want to form a sample of 500 individuals out of a population of 5000; you'd have to number every person in the population.

Once the numbering is done, the researcher can select a number randomly, for instance, 5. The 5th individual will be the first to be a part of the systematic sample. After that, the 10th member will be added into the sample, so on and so forth (15th, 25th, 35, 45th, and members till 4995).

Here are 4 other situations of when to use Systematic Sampling:

1. **Budget restrictions:** In comparison to other sampling methods like simple random sampling, this sampling technique is more suitable for conditions where there are budget restrictions and also the extremely uncomplicated accomplishment of the study.
2. **Uncomplicated implementation:** As systematic sampling depends on the defined sampling intervals to decide the sample, it becomes simple for the researchers and statisticians to manage samples with more respondents. This is because the time invested in creating samples is minimal, and the cost spent is also restricted due to the periodic nature of systematic sampling.
3. **Absence of data pattern:** There are specific data that don't have an arrangement in place. This data can be analyzed in an unbiased manner, using systematic sampling.
4. **Low risk of data manipulation in research:** It is highly productive while researching a broad subject, especially when there's a negligible risk of data manipulation.

Stratified random sampling

Stratified random sampling is a widely used statistical technique in which a population is divided into different subgroups, or strata, based on some shared characteristics. The purpose of stratification is to ensure that each stratum in the sample and to make inferences about specific population subgroups.

This technique is beneficial in cases where the population is heterogeneous, and a simple random sample may not provide accurate results. By dividing the population into strata, researchers can ensure that their sample is representative of the population and avoid sampling biases.

What is Stratified Random Sampling?

Stratified random sampling is a type of probability method using which a research organization can branch off the entire population into multiple non-overlapping, homogeneous groups (strata) and randomly choose final members from the various strata for research which reduces cost and improves efficiency. Members in each of these groups should be distinct so that every member of all groups gets an equal opportunity to be selected using simple probability. This sampling method is also called "random quota sampling."

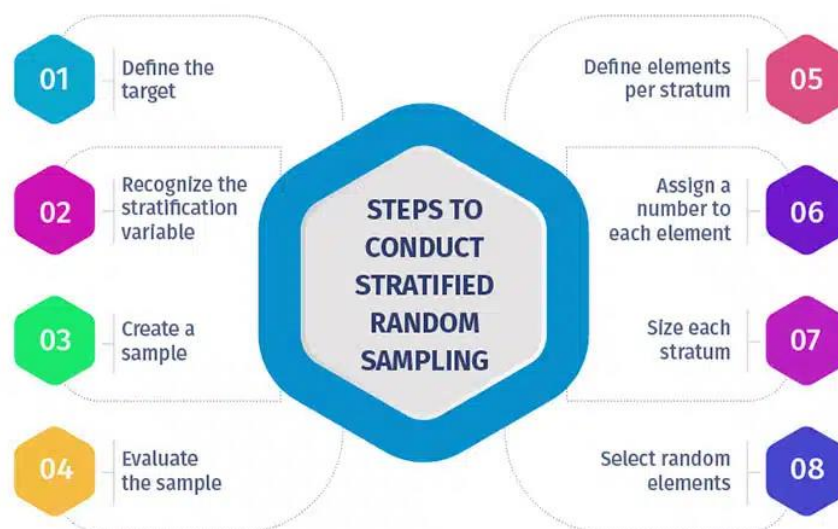
Let's consider a situation where a research team seeks opinions about religion among various age groups. Instead of collecting feedback from 326,044,985 U.S citizens, random samples of around 10000 can be selected for research. These 10000 citizens can be divided into groups according to age, i.e., 18-29, 30-39, 40-49, 50-59, and 60 and above. Each stratum will have distinct members

and the number of members—age, socioeconomic divisions, nationality, religion, educational achievements, and other classifications.

8 Steps to Conduct Stratified Random Sampling

The following are the steps to select a stratified random sample:

The following are the steps to select a stratified random sample:



1. Define the target audience.
2. Recognize the stratification variable or variables and figure out the number of strata to be used. These stratification variables should be in line with the objective of the research. Every additional information decides the stratification variables. For instance, if the objective of the research is to understand all the subgroups, the variables will be related to the subgroups.
3. Use an already-existent sampling frame or create a frame that's inclusive of all the information of the stratification variable for all the elements in the target audience.
4. Make changes after evaluating the sampling frame on the basis of lack of coverage, over-coverage, or grouping.
5. Considering the entire population, each stratum should be unique and should cover each and every member of the population. Within the stratum, the differences should be minimum, whereas each stratum should be extremely different from one another. Each element of the population should belong to just one stratum.
6. Assign a random, unique number to each element.

7. Figure out the size of each stratum according to your requirement. The numerical distribution amongst all the elements in all the strata will determine the type of sampling to be implemented. It can either be proportional or disproportional stratified sampling.
8. The researcher can select random elements from each stratum to form the sample. A minimum of one piece must be chosen from each stratum so that there's representation from every stratum, but if two elements from each stratum are selected, quickly calculate the error margins of the calculation of collected data.

Using this method helps ensure that the sample is representative of the population and reduces sampling error, leading to more accurate results.

Types of Stratified Random Sampling

- **Proportionate Sampling:**

In this approach, each stratum sample size is directly proportional to the population size of the entire population of strata. That means each strata sample has the same sampling fraction.

Proportionate Stratified Random Sampling Formula: $n_h = (N_h / N) * n$

n_h = Sample size for h^{th} stratum

N_h = Population size for h^{th} stratum

N = Size of entire population

n = Size of entire sample

Stratum	A	B	C	D
Population Size	500	1000	1500	2000
Sampling Fraction	1/2	1/2	1/2	1/2
Final Sampling Size Results	250	500	750	1000

If you have four strata with 500, 1000, 1500, and 2000 respective sizes, the research organization selects $\frac{1}{2}$ as the sampling fraction. A researcher must choose 250, 500, 750, and 1000 members from the separate stratum.

Irrespective of the sample size of the population, the sampling fraction will remain uniform across all the strata.

- **Disproportionate Sampling:**

The sampling fraction is the primary differentiating factor between proportionate and disproportionate stratified random sampling. In excessive sampling, each stratum will have a different sampling fraction.

The success of this sampling method depends on the researcher's precision at fraction allocation. If the allotted fractions aren't accurate, the results may be biased due to the overrepresented or underrepresented strata.

Stratum	A	B	C	D
Population Size	500	1000	1500	2000
Sampling Fraction	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$
Final Sampling Size Results	250	333	375	400

Stratified Random Sampling Examples

Researchers and statisticians use stratified random sampling to analyze relationships between two or more strata. As stratified random sampling involves multiple layers or strata, it's crucial to calculate the strata before calculating the sample value.

Following is a classic stratified random sampling example:

Let's say 100 (N_h) students of a school having 1000 (N) students were asked questions about their favorite subject. It's a fact that the students of the 8th grade will have different subject preferences

than the students of the 9th grade. For the survey to deliver precise results, the ideal manner is to divide each step into various strata.

Here's a table of the number of students in each grade:

Grade	Number of students (n)	Calculate the sample of each grade using the formula:
5	150	Stratified Sample (n_5) = $100 / 1000 * 150 = 15$
6	250	Stratified Sample (n_6) = $100 / 1000 * 250 = 25$
7	300	Stratified Sample (n_7) = $100 / 1000 * 300 = 30$
8	200	Stratified Sample (n_8) = $100 / 1000 * 200 = 20$
9	100	Stratified Sample (n_9) = $100 / 1000 * 100 = 10$

Advantages of Stratified Random Sampling

Stratified random sampling is a statistical technique that offers several advantages over other methods. Some of the critical benefits include the following:

- Better accuracy in results in comparison to other probability sampling methods such as cluster sampling, simple random sampling, and systematic sampling or non-probability methods such as convenience sampling. This accuracy will be dependent on the distinction of various strata.
- Convenient to train a team to stratify a sample due to the exactness of the nature of this sampling technique.
- Due to statistical accuracy of this method, smaller sample sizes can also retrieve highly useful results for a researcher.
- This sampling technique covers maximum population as the researchers have complete charge over the strata division.
- Health surveys can benefit from this method to represent the population concerning health status, disease prevalence, or risk factors.

Stratified random sampling is a valuable and efficient method for a population with distinct subgroups. Its ability to increase precision, represent each subgroup proportionately, control for bias, save costs, and improve data quality make it a valuable tool for researchers in many fields.

When to use Stratified Random Sampling?

Stratified random sampling should be used when the population under study has distinct subgroups, and the goal is to make inferences about each subgroup or the population as a whole. This technique is beneficial when:

1. Stratified random sampling is a highly productive method sampling in situations where the researcher intends to focus only on specific strata from the available population data. This way, the desired characteristics of the strata can be found in the survey sample.
2. Researchers rely on this sampling method in cases where they intend to establish a relationship between two or more different strata. If this comparison is conducted using simple random sampling, there is a higher likelihood of the target groups not being equally represented.
3. Samples with a population which are difficult to access or contact can be quickly involved in the research process using the stratified random sampling technique.
4. The accuracy of statistical results is higher than simple random sampling since the elements of the sample are chosen from relevant strata. The diversification within the strata will be much lesser than the diversification in the target population. Due to the accuracy involved, the required sample size will be much lesser, which will help researchers save time and effort.
5. Public opinion polls often use stratified random sampling to represent the population concerning demographic characteristics such as region, political affiliation, or age.

Stratified random sampling is practical when the population has distinct subgroups, and the goal is to make inferences about each subgroup or the population as a whole.

Cluster sampling

Cluster sampling is defined as a sampling method where the researcher creates multiple clusters of people from a population where they are indicative of homogeneous characteristics and have an equal chance of being a part of the sample.

Consider a scenario where a data organization is looking to survey the performance of smartphones across Germany. They can divide the entire country's population into cities (clusters), select further towns with the highest population, and filter those using mobile devices.

What is Cluster Sampling?

Cluster sampling is a probability sampling technique where researchers divide the population into multiple groups (clusters) for research. So researchers then select random groups with a simple random or systematic random sampling technique for data collection and unit of analysis.

Example: A researcher wants to conduct a study to judge the performance of sophomores in business education across the U.S. It is impossible to conduct a research study that involves a student in every university. Instead, cluster sampling allows the researcher to club the universities from each city into one cluster. These clusters then define the sophomore student population in the U.S. Next, either using simple random sampling or systematic random sampling and randomly picking clusters for the research study. Subsequently, by using simple or systematic sampling, the sophomore's from each of these selected clusters can be chosen on whom to conduct the research study.

In this sampling technique, researchers analyze a sample that consists of multiple sample parameters such as demographics, habits, background – or any other population attribute, which may be the focus of conducted research. This method is usually conducted when groups that are similar yet internally diverse form a statistical population. Instead of selecting the entire population, cluster sampling allows the researchers to collect data by bifurcating the data into small, more productive groups.

Types of Cluster Sampling

There are two ways to classify this sampling technique. The first way is based on the number of stages followed to obtain the cluster sample, and the second way is the representation of the groups in the entire cluster analysis. In most cases, sampling by clusters happens over multiple stages. A stage is considered to be the step taken to get to the desired sample. We can divide this technique into single-stage, two-stage, and multiple stages.

Single-stage Cluster Sampling:

As the name suggests, sampling is done just once. An example of single-stage cluster sampling – An NGO wants to create a sample of girls across five neighboring towns to provide education. Using single-stage sampling, the NGO randomly selects towns (clusters) to form a sample and extend help to the girls deprived of education in those towns.

Two-stage Cluster Sampling:

Here, instead of selecting all the elements of a cluster, only a handful of members are chosen from each group by implementing systematic or simple random sampling. An example of two-stage cluster sampling – A business owner wants to explore the performance of his/her plants that are spread across various parts of the U.S. The owner creates clusters of plants. So he/she then select random samples from these clusters to conduct research.

Multiple-stage Cluster Sampling:

Multiple-stage cluster sampling takes a step or a few steps further than two-stage sampling.

For conducting effective research across multiple geographies, one needs to form complicated clusters that can be achieved only using the multiple-stage sampling technique. An example of Multiple stage sampling by clusters – An organization intends to survey to analyze the performance of smartphones across Germany. They can divide the entire country's population into cities (clusters) and select cities with the highest population and also filter those using mobile devices.

Steps to conduct Cluster Sampling

Here are the steps to perform cluster sampling:



1. **Sample:** Decide the target audience and also the sample size.
2. **Create and evaluate sampling frames:** Create a sampling frame by using either an existing framework or creating a new one for the target audience. Evaluate frameworks based on coverage and clustering and make adjustments accordingly. These groups will be varied, considering the population, which can be exclusive and comprehensive. Members of a sample are selected individually.
3. **Determine groups:** Determine the number of groups by including the same average members in each group. So make sure each of these groups is distinct from one another.
4. **Select clusters:** Choose clusters by applying a random selection.

5. **Create sub-types:** It is bifurcated into two-stage and multi-stage subtypes based on the number of steps followed by researchers to form clusters.

Applications of Cluster Sampling

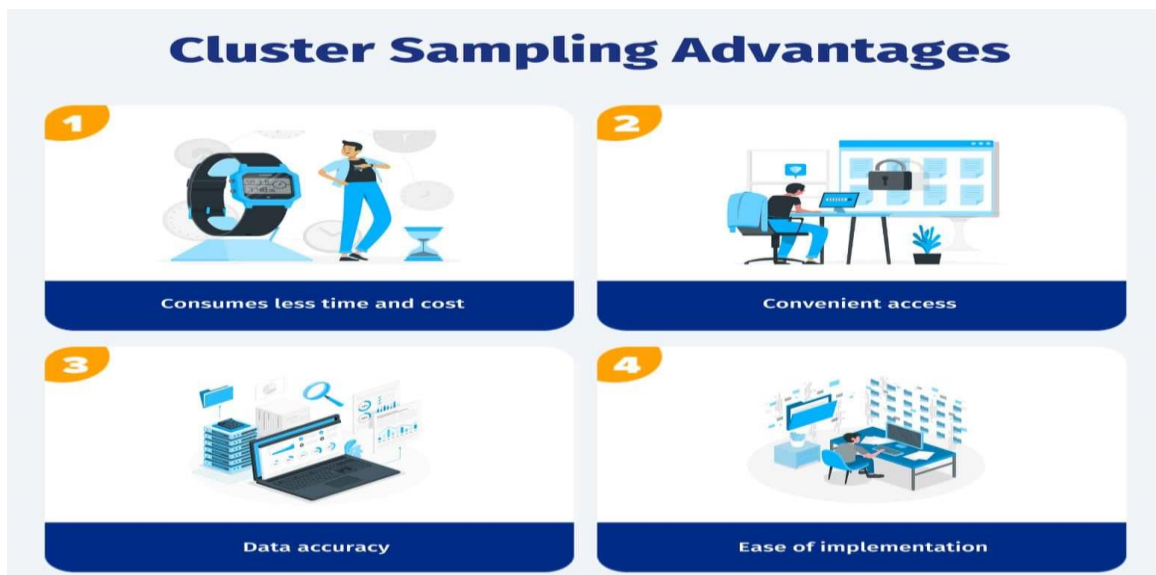
This sampling technique is used in an area or geographical cluster sampling for market research. A broad geographic location can be expensive to survey compared to surveys sent to clusters divided based on region. The sample numbers must be increased to achieve accurate results, but the cost savings make this process of rising clusters attainable.

Cluster Sampling in statistics

The technique is widely used in statistics where the researcher can't collect data from the entire population. So, it is the most economical and practical solution for research statisticians. Take the example of a researcher looking to understand smartphone usage in Germany. In this case, the cities of Germany will form clusters. This sampling method is also used in wars and natural calamities to draw inferences about a population, where collecting data from every individual residing is impossible.

Cluster Sampling Advantages

There are multiple advantages to using cluster sampling. Here they are:



- **Consumes less time and cost:** Sampling of geographic segmentation divided groups requires less work, time, and cost. It's a highly economical method to observe clusters instead of randomly doing them throughout a particular region by allocating a limited number of resources to those selected clusters.
- **Convenient access:** Researchers can choose large samples with this convenience sampling technique, and that'll increase accessibility to various clusters.
- **Data accuracy:** Since there can be large samples in each cluster, loss of data accuracy in information per individual can be compensated.
- **Ease of implementation:** Cluster sampling facilitates information from various areas and groups. Researchers can quickly implement it in practical situations compared to other probability sampling methods.

In comparison to simple random sampling, this technique can be useful in deciding the characteristics of a group such as population, and researchers can implement it without having a sampling frame for all the elements for the entire population.

Cluster Sampling vs Stratified Sampling

Since cluster sampling and stratified random sampling are pretty similar, there could be issues with understanding their finer nuances. So let's talk about the significant differences between cluster sampling and stratified sampling:

Cluster Sampling	Stratified Sampling
Elements of a population are randomly selected to be a part of groups (clusters).	The researcher divides the entire population into even segments (strata).
Members from randomly selected clusters are a part of this sample.	Researchers consider individual components of the strata randomly to be a part of sampling units.
Researchers maintain homogeneity between clusters.	Researchers maintain homogeneity within the strata.
Researchers divide the clusters naturally.	The researchers or statisticians primarily decide the strata division.
The key objective is to minimize the cost involved and enhance competence.	The key objective is to conduct accurate sampling, along with a properly represented population.

Non Probability Sampling:

Non-probability sampling is a method of selecting units from a population using a subjective (i.e. non-random) method. Since non-probability sampling does not require a complete survey frame, it is a fast, easy and inexpensive way of obtaining data.

Convenience sampling

Convenience sampling is a simple and easy way to get information compared to other sampling methods. Most of the time, simple and easy go well together. But you need to know what it is so you know when to use it and when not to.

It is a type of sampling that doesn't depend on chance and is often used in research studies. This sampling technique involves choosing people who are easy for the researcher to reach and get in touch with.

Instead of picking people at random from a certain population, convenience sampling involves picking the people who are easiest for the researcher to get information from.

Convenience sampling is often used when other types of sampling methods are hard or impossible to use because of time, cost, or other issues. Even though it can be a quick and easy way to get data, it can also have biases and limitations that can affect how well the results can be used in the real world and how reliable they are.

What is convenience sampling?

Convenience sampling is defined as a method adopted by researchers where they collect market research data from a conveniently available pool of respondents.

It is the most commonly used sampling as it's incredibly prompt, uncomplicated, and economical. Members are often readily approachable to be a part of the sample.

Researchers use various sampling techniques in situations where there are large populations. In most cases, testing the entire community is practically impossible because they are not easy to reach. Researchers use convenience sampling in situations where additional inputs are not necessary for the principal research. There are no criteria required to be a part of this sample.

Thus, it becomes incredibly simplified to include elements in this sample. All components of the population are eligible and dependent on the researcher's proximity to get involved in the sample.

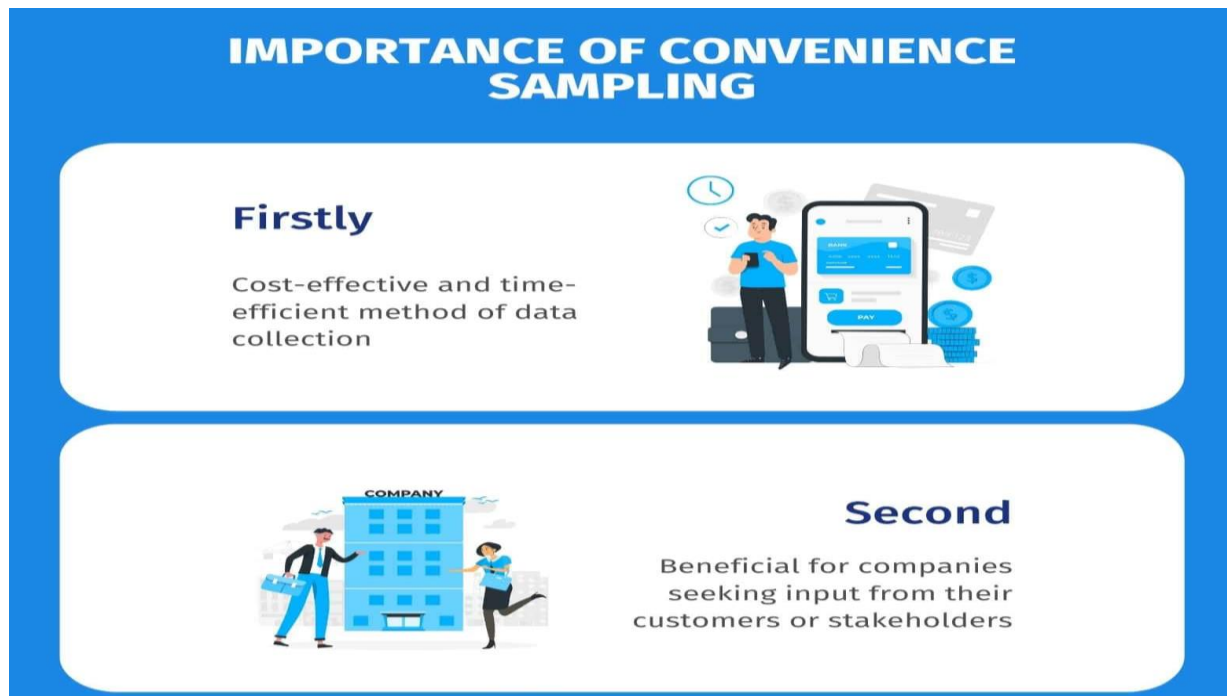
The researcher chooses members merely based on proximity and doesn't consider whether they represent the entire population or not.

Using this technique, they can observe habits, opinions, and viewpoints in the easiest possible manner.

A good example of convenience sampling is A new NGO that wants to establish itself in 20 cities. It selects the top 20 cities to serve based on the proximity to where they're based.

Importance of convenience sampling

Convenience sampling is beneficial to businesses in a variety of ways.



- Firstly, it can be a cost-effective and time-efficient method of data collection. Businesses can save time and money by picking readily available participants instead of employing more elaborate sampling procedures.
- Secondly, it can be beneficial for companies seeking input from their customers or stakeholders. Businesses can gain useful insights and feedback by selecting individuals who have interacted with the firm or used its products/services.

It is important to highlight that convenience sampling has limits and biases that can affect the reliability and generalizability of the findings.

As a result, before depending on this sampling technique to make crucial decisions, organizations must carefully analyze the potential biases and limits.

Applications of convenience sampling:

Convenience sampling is applied by brands and organizations to measure their perception of their image in the market. Data is collected from potential customers to understand specific issues or manage opinions of a newly launched product. In some cases, it is the only available option.

For example, a university student working on a project and wants to understand the average consumption of soda on campus on a Friday night will most possibly call his/her classmates and friends and ask how many cans of soda they consume. Or may go to a party nearby and conduct an easy online survey.

There is always a chance that the randomly selected population may not accurately represent the population of interest, thus increasing the chances of research bias.

Example of convenience sampling

A basic example of a convenience sampling method is when companies distribute their promotional pamphlets and ask questions at a mall or on a crowded street with randomly selected participants.

Businesses use this sampling technique to gather information to address critical issues arising from the market. They also use it when collecting feedback about a particular feature or a newly launched product from the sample created.

During the initial stages of survey research, researchers usually prefer using this as it's quick and easy to deliver results. Even if many statisticians avoid implementing this technique, it is vital in situations where you intend to get insights in a shorter period or without investing too much money.

For instance, a marketing student needs to get feedback on the “scope of content marketing in 2020.” The student may quickly create an online survey, send a link to all the contacts on your phone, share a link on social media, and talk to people you meet daily, face-to-face.

What is convenience sampling in qualitative research?

Convenience sampling is a qualitative research sampling strategy that involves selecting participants based on their accessibility and availability to the researcher. Rather than being drawn at random from a bigger population, participants in this strategy are picked because they are easily available to the researcher.

It is frequently used in qualitative research because it can be a cost-effective and time-efficient way to collect data from individuals who are difficult to reach or may not be interested in participating in the study.

For example, a researcher conducting a study on cancer survivors' experiences may choose to recruit volunteers using social media groups or cancer support networks because they are easily accessible and likely to be interested in the study.

Convenience sampling in qualitative research can also contribute to data biases because the sample may not be representative of the greater community or may be prejudiced towards participants with certain qualities or experiences.

While evaluating the findings of their study, researchers must carefully evaluate the various limits and biases of this sampling strategy.

Advantages of using convenience sampling

Here are the advantages of adopting a convenience sampling approach:



1. Collect data quickly

In situations where time is a constraint, many researchers choose this method for quick data collection. The rules to gather elements for the sample are the least complicated in comparison to techniques such as simple random sampling, stratified sampling, and systematic sampling. Due to this simplicity, data collection takes minimal time.

2. Inexpensive to create samples
The money and time invested in other probability sampling methods are quite large compared to convenience sampling. It allows researchers to generate more samples with less or no investment and in a brief period.
3. Easy to do research
The name of this surveying technique clarifies how samples are formed. Elements are easily accessible by the researchers and so, collecting members for the sample becomes easy.
4. Low cost
Low cost is one of the main reasons why researchers adopt this technique. When on a small budget, researchers – especially students, can use the budget in other areas of the project.
5. Readily available sample
Data collection is easy and accessible. Most convenience sampling considers the population at hand. Samples are readily available to the researcher. They do not have to move around too much for data collection. Quotas are met quickly, and the data collection can commence even within a few hours.
6. Fewer rules to follow
It doesn't require going through a checklist to filter members of an audience. Here, gathering critical information and data becomes uncomplicated. For instance, if an NGO wants to survey women's empowerment, they can go to schools, colleges, offices, etc., in their proximity and gather quick responses.

Disadvantages of convenience sampling

Convenience sampling includes a number of drawbacks that can reduce the reliability and validity of research findings. The following are some of the major drawbacks:

1. Bias in sampling
When participants are not chosen at random from a larger population, this might result in sampling bias. This indicates that the sample may not be typical of the greater population, and the findings may not apply to other groups.
2. Lack of variety
Because the researcher may choose participants who are easily accessible and have comparable features, this may result in a lack of variety in the sample. This can potentially narrow the spectrum of opinions and experiences represented in the sample.
3. External validity is limited
It may have poor external validity due to the possibility of sampling bias and a lack of diversity. As a result, the findings may not apply to different locations or populations.
4. Unknown errors
Convenience sampling can lead to unknown errors since the researcher may be unaware of how skewed or unrepresentative the sample is of the population.
5. Possibility of researcher bias
Researchers may be more prone to select participants who they believe would offer data that supports their theory, perhaps contributing to researcher bias.

While it can be a valuable and cost-effective method of data collection in some contexts, it is crucial to understand its limitations and inherent biases. When selecting to utilize it in their research, researchers should carefully assess its advantages and disadvantages.

How to reduce bias in convenience sampling?

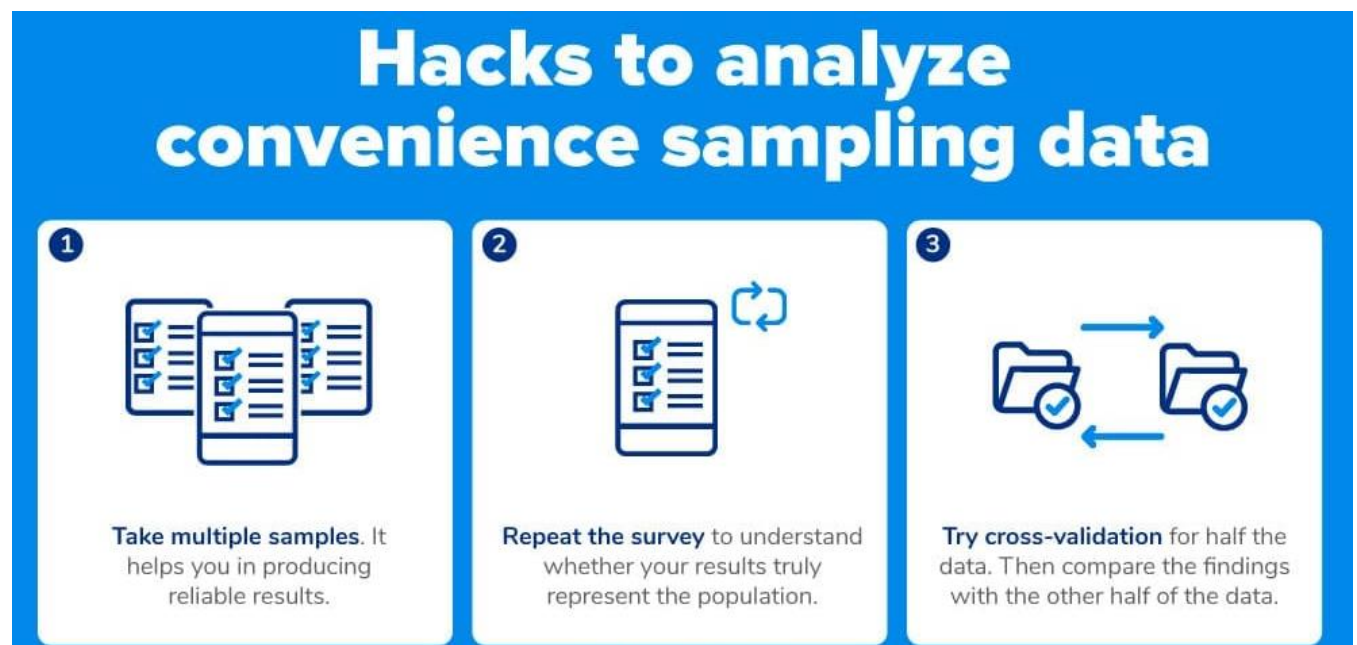
The best way of reducing bias in convenience sampling is to use it along with probability sampling. Since it is usually biased, probability sampling gets the measurement parameter with it to keep this sampling bias under check.

After receiving a fair idea about this bias using probability sampling, the researcher can use both convenience sampling and probability sampling techniques to draw a more accurate estimation. The probability aspect used, along with the convenience sample, will have to be powerful enough to overcome it.

Bias can make the entire sample futile, which is the last thing a researcher needs. This bias can be reduced or eliminated by including probability sampling.

How to efficiently analyze convenience sampling data?

Here are three quick hacks to efficiently analyze convenience sampling data. It is best to use probability sampling, but when that is not possible, here are three hacks you should keep in mind.



- Take multiple samples. It helps you in producing reliable results.
- Repeat the survey to understand whether your results truly represent the population.

- For a big sample size, try cross-validation for half the data. Then compare the findings with the other half of the data.

Judgmental sampling

Definition

Judgmental sampling, also called purposive sampling or authoritative sampling, is a non-probability sampling technique in which the sample members are chosen only on the basis of the researcher's knowledge and judgment. As the researcher's knowledge is instrumental in creating a sample in this sampling technique, there are chances that the results obtained will be highly accurate with a minimum margin of error.

The process of selecting a sample using judgmental sampling involves the researchers carefully picking and choosing each individual to be a part of the sample. The researcher's knowledge is primary in this sampling process as the members of the sample are not randomly chosen.



When to execute Judgmental Sampling?

Judgmental sampling is most effective in situations where there are only a restricted number of people in a population who own qualities that a researcher expects from the target population. Researchers prefer to implement Judgmental sampling when they feel that other sampling techniques will consume more time and that they have confidence in their knowledge to select a sample for conducting research.

Judgmental or Expert sampling is usually used in situations where the target population comprises of highly intellectual individuals who cannot be chosen by using any other probability or non-

probability sampling technique. It is also used in situations where the sample selected using other sampling methods need to be approved or filtered. For instance, in situations where a researcher conducts convenience sampling to gather feedback from professors about their university but the fact that there are high chances of the results to be skewed, researchers prefer judgmental sampling to select those professors who will provide 100% feedback about the university.

Selecting each individual of the sample is a critical challenge that an intellectual researcher will undertake. It is a tedious task to handpick members of a sample while ensuring there is no bias involved.

The authority involved in the selection process may not necessarily be “experts” in the field but they have to comply with certain characteristics expected from a Judgmental sampling authority and sampling errors. Education or work experience is not considered while appointing authorities for the selection process.

Purposive sampling is used where there is time-constraint for sample creation and the authorities involved would prefer relying on their knowledge and not on other sampling methods. But, one must keep in mind, the fact that a researcher may or may not have the appropriate proficiency to conduct an effective sampling process. This is the only disadvantage of purposive sampling. Each researcher who takes up the responsibility of creating a sample using expert sampling will have to be extremely confident in their own skills and understanding of the subject.

Examples of Judgmental Sampling

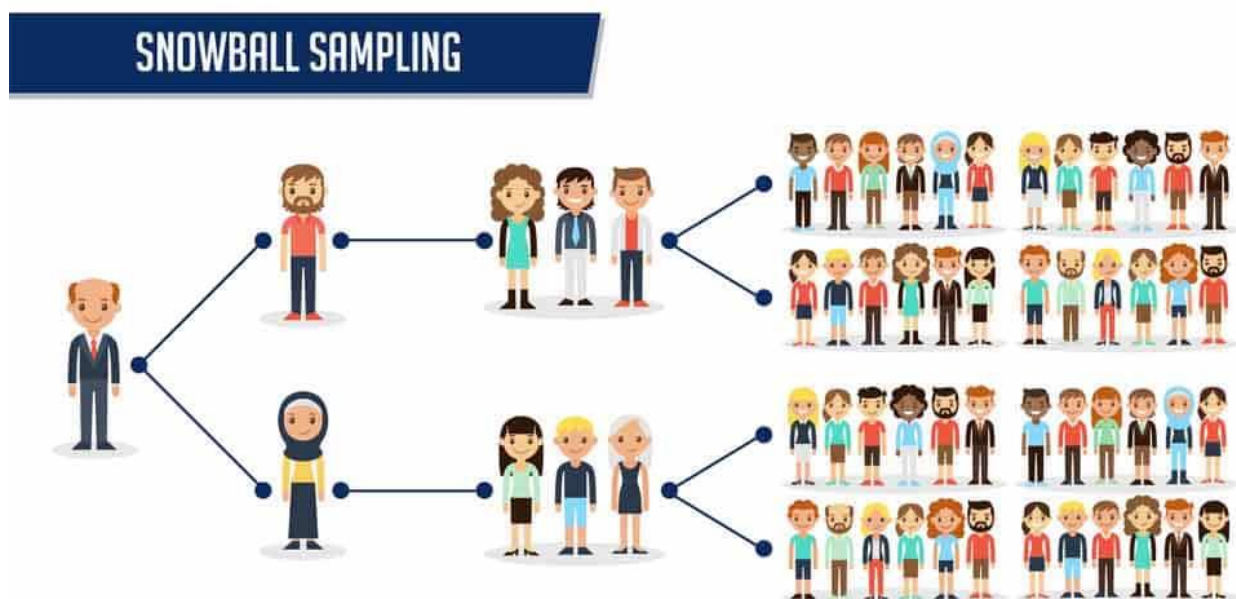
Here are two distinct Judgmental Sampling examples:

- Consider a scenario where a panel decides to understand what are the factors which lead a person to select ethical hacking as a profession. Ethical hacking is a skill which has been recently attracting youth. More and more people are selecting it as a profession. The researchers who understand what ethical hacking is will be able to decide who should form the sample to learn about it as a profession. That is when judgmental sampling is implemented. Researchers can easily filter out those participants who can be eligible to be a part of the research sample.
- There are many tribes in the world which have their own religious beliefs, for instance, the Balinese people follow syncretism, which is considered to be a mixture of Hinduism and Buddhism. For researchers who plan to study the culture of Southeast Asian countries, it is advised that they select strata using judgmental sampling as religious beliefs are considered to be highly sensitive in this part of the world. Due to the sensitivity of the topic, if samples of those who have appropriate knowledge are created and research is conducted with those samples, results will be highly accurate. Probability sampling techniques often produce altered results in such cases.

Judgmental Sampling Advantages

- **Consumes minimum time for execution:** In this sampling bias approach, researcher expertise is important and there are no other barriers involved due to which selecting a sample becomes extremely convenient.
- **Allows researchers to approach their target market directly:** There are no criteria involved in selecting a sample except for the researcher's preferences. Due to this, he/she can communicate directly with the target audience of their choice and produce desired results.
- **Almost real-time results:** A quick poll or survey can be conducted with the sample using judgmental sampling since the members of the sample will possess appropriate knowledge and understanding of the subject.

Snowball Sampling: Definition



Snowball sampling or chain-referral sampling is defined as a non-probability sampling technique in which the samples have rare traits. This is a sampling technique, in which existing subjects provide referrals to recruit samples required for a research study.

For example, if you are studying the level of customer satisfaction among the members of an elite country club, you will find it extremely difficult to collect primary data sources unless a member of the club agrees to have a direct conversation with you and provides the contact details of the other members of the club.

This sampling method involves a primary data source nominating other potential data sources that will be able to participate in the research studies. Snowball sampling method is purely based on referrals and that is how a researcher is able to generate a sample. Therefore this method is also called the chain-referral sampling method.

Snowball sampling is a popular business study method. The snowball sampling method is extensively used where a population is unknown and rare and it is tough to choose subjects to assemble them as samples for research.

This sampling technique can go on and on, just like a snowball increasing in size (in this case the sample size) till the time a researcher has enough data to analyze, to draw conclusive results that can help an organization make informed decisions.

Types of Snowball Sampling

1. **Linear Snowball Sampling:** The formation of a sample group starts with one individual subject providing information about just one other subject and then the chain continues with only one referral from one subject. This pattern is continued until enough number of subjects are available for the sample.
2. **Exponential Non-Discriminative Snowball Sampling:** In this type, the first subject is recruited and then he/she provides multiple referrals. Each new referral then provides with more data for referral and so on, until there is enough number of subjects for the sample.
3. **Exponential Discriminative Snowball Sampling:** In this technique, each subject gives multiple referrals, however, only one subject is recruited from each referral. The choice of a new subject depends on the nature of the research study.

Snowball Sampling Method

The nature of snowball sampling is such, that it cannot be considered for a representative sample or in that case for statistical studies. However, this sampling technique can be extensively used for conducting qualitative research, with a population that is hard to locate. Let us now explore how snowball sampling can be carried out:

Consider hypothetically, you as a researcher are studying the homeless in Texas City. It is obviously difficult to find a list of all the details of the number of homeless there. However, you are able to identify one or two homeless individuals who are willing to participate in your research studies.

Now, these homeless individuals provide you with the details of other homeless individuals they know. The first homeless individual that you found for your research is the primary data. You can collect the information and tabulate data from the primary data source and move on to other individuals who the primary data source has referred to. You as a researcher can continue to tap as many homeless you can find through the reference provided till you know you have collected enough data for your research.

The same strategy can be followed to conduct research or study individuals belonging to certain underground subculture, or individuals who have a hidden identity or are members of a cult etc. who don't want to be identified easily. Trust is an important part of any researcher.

An individual, who is ready to share information, needs to know that the information will be used discreetly and this kind of trust is especially important in snowball sampling. For a participant to agree to identify themselves or their group, researchers first need to develop that kind of rapport with the participants. Please know that this sampling technique may consume more time than anticipated because of its nature.

Snowball Sampling Applications

Snowball sampling is usually used in cases where there is no precalculated list of target population details (homeless people), there is immense pain involved in contacting members of the target population (victims of rare diseases), members of the target population are not inclined towards contributing due to a social stigma attached to them (hate-crime, rape or sexual abuse victims, sexuality, etc.) or the confidentiality of the organization respondents work for (CIA, FBI or terrorist organization).

Thus, this type of sampling is preferred in the following applications:

- **Medical Practices:** There are many less-researched diseases. There may be a restricted number of individuals suffering from diseases such as progeria, porphyria, Alice in Wonderland syndrome etc. Using snowball sampling, researchers can get in touch with these hard to contact sufferers and convince them to participate in the survey research.
- **Social research:** Social research is a field which requires as many participants as possible as it is a process where scientists learn about their target sample. When social research is to be conducted in domains where participants might not necessarily be willing to contribute such as homeless or the less-fortunate people.
- **Cases of discord:** In case of disputes such as an act of terrorism, violation of civil rights and other similar situations, the individuals involved may oppose giving their statements for evidential purposes. The researchers or management can use snowball sampling, to filter out those people from a population who are most likely to have caused the situation or are witness to the event to gather proof around the event.

Snowball Sampling Examples

For some population, snowball sampling is the only way of collecting data and meaningful information. Following are the instances, where snowball sampling can be used:

1. **No official list of names of the members:** This sampling technique can be used for a population, where there is no easily available data like their demographic information. For example, homeless or list of members of an elite club, whose personal details cannot be obtained easily.

2. **Difficulty to locate people:** People with rare diseases are quite difficult to locate. However, if a researcher is carrying out a research study similar in nature, finding the primary data source can be a challenge. Once he/she is identified, they usually have information about more such similar individuals.
3. **People who are not willing to be identified:** If a researcher is carrying out a study which involves collecting information/data from sex workers or victims of sexual assault or individuals who don't want to disclose their sexual orientations, these individuals will fall under this category.
4. **Secretiveness about their identity:** People who belong to a cult or are religious extremists or hackers usually fall under this category. A researcher will have to use snowball sampling to identify these individuals and extract information from them.

Advantages of Snowball Sampling

1. **It's quicker to find samples:** Referrals make it easy and quick to find subjects as they come from reliable sources. An additional task is saved for a researcher, this time can be used in conducting the study.
2. **Cost effective:** This method is cost effective as the referrals are obtained from a primary data source. It's is convenient and not so expensive as compared to other methods.
3. **Sample hesitant subjects:** Some people do not want to come forward and participate in research studies, because they don't want their identity to be exposed. Snowball sampling helps for this situation as they ask for a reference from people known to each other. There are some sections of the target population which are hard to contact. For example, if a researcher intends to understand the difficulties faced by HIV patients, other sampling methods will not be able to provide these sensitive samples. In snowball sampling, researchers can closely examine and filter members of a population infected by HIV and conduct a research by talking to them, making them understand the research objective, and eventually, analyzing the received feedback.

Disadvantages of Snowball Sampling

1. **Sampling bias and margin of error:** Since people refer those whom they know and have similar traits, this sampling method can have a potential sampling errors and margin of error. This means a researcher might only be able to reach out to a small group of people and may not be able to complete the study with conclusive results.
2. **Lack of cooperation:** There are fair chances even after referrals, people might not be cooperative and refuse to participate in the research studies.

Quota sampling:

Definition

Quota sampling is defined as a non-probability sampling method in which researchers create a convenience sample involving individuals that represent a population. Researchers choose these individuals according to specific traits or qualities. They decide and create quotas so that the market research samples can be useful in collecting data. These samples can be generalized to the entire population. The final subset will be decided only according to the interviewer's or researcher's knowledge of the population.

For example, a cigarette company wants to find out what age group prefers what brand of cigarettes in a particular city. They apply survey quota on the age groups of 21-30, 31-40, 41-50, and 51+. From this information, the researcher gauges the smoking trend among the population of the city.

Types of quota sampling:

Quota sampling can be of two kinds – controlled quota sampling and uncontrolled quota sampling. Here's what they mean:

Controlled quota sampling:

Controlled quota sampling imposes restrictions on the researcher's choice of samples. Here, the researcher is limited to the selection of samples.

Uncontrolled quota sampling:

Uncontrolled quota sampling does not impose any restrictions on the researcher's choice of samples. Here, the researcher chooses sample members at will.

Quota sampling example:

Let's look at a basic example of quota sampling:

A researcher wants to survey individuals about what smartphone brand they prefer to use. He/she considers a sample size of 500 respondents. Also, he/she is only interested in surveying ten states in the US. Here's how the researcher can divide the population by quotas:

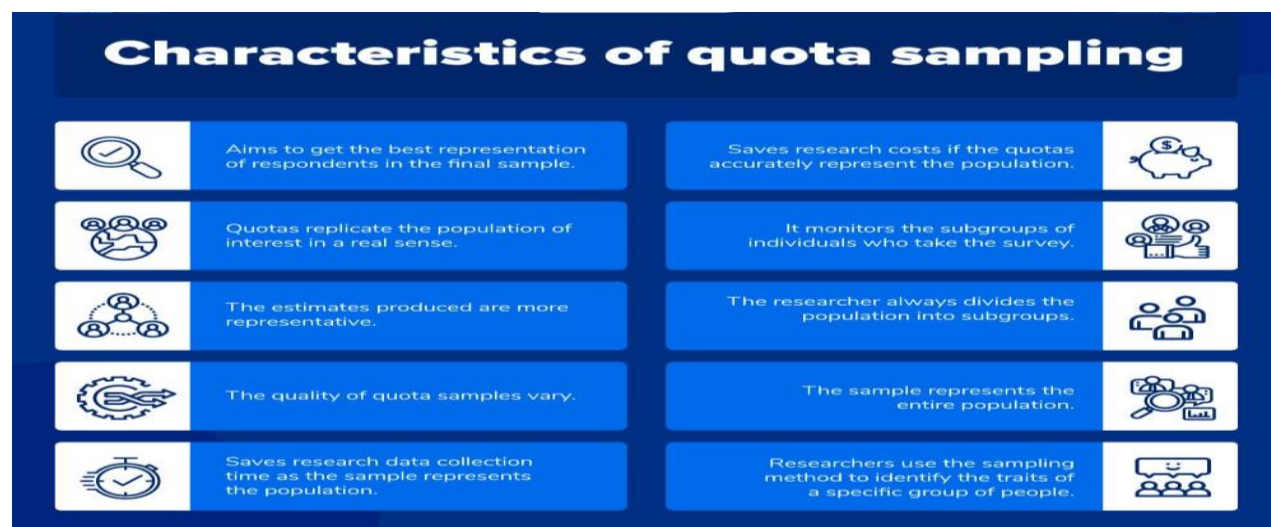
- Gender: 250 males and 250 females
- Age: 100 respondents each between the ages of 16-20, 21-30, 31-40, 41-50, and 51+
- Employment status: 350 employed and 150 unemployed people.
 - (Researchers apply further nested quotas . For eg, out of the 150 unemployed people, 100 must be students.)
- Location: 50 responses per state

Depending on the type of research, the researcher can apply quotas based on the sampling frame. It is not necessary for the researcher to divide the quotas equally. He/she divides the quotas as per his/her need (as shown in the example where the researcher interviews 350 employed and only 150 unemployed individuals). Random sampling can be conducted to reach out to the respondents.

How to perform quota sampling:

Probability sampling techniques involve a significant amount of rules that the researcher needs to follow to form samples. But, since quota sampling is a non-probability sampling technique, there are no rules for formally creating samples. Usually, there are four steps to form a quota sample. Here are the steps:

1. **Divide the sample population into subgroups:** With stratified random sampling, the researcher bifurcates the entire population into mutually exhaustive subgroups, i.e., the elements of each of the subgroups becomes a part of only one of those subgroups. Here, the researcher applies random selection.
2. **Figure out the weightage of subgroups:** The researcher evaluates the proportion in which the subgroups exist in the population. He/she maintains this proportion in the sample selected using this type of sampling method.
3. For example, if 58% of the people who are interested in purchasing your Bluetooth headphones are between the age group of 25-35 years, your subgroups also should have the same percentages of people belonging to the respective age group.
4. **Select an appropriate sample size:** In the third step, the researcher should select the sample size while maintaining the proportion evaluated in the previous step. If the population size is 500, the researcher can pick a sample of 50 elements. The sample chosen after following the first three steps should represent the target population.
5. **Conduct surveys according to the quotas defined:** Make sure to stick to the predefined quotas to achieve actual actionable results. Don't survey quotas that are full and focus on completing surveys for each quota.



Characteristics of quota sampling:

Here are the top ten characteristics of quota sampling

1. Aims to get the best representation of respondents in the final sample.
2. Quotas replicate the population of interest in a real sense.
3. The estimates produced are more representative.
4. The quality of quota samples vary.
5. Saves research data collection time as the sample represents the population.
6. Saves research costs if the quotas accurately represent the population.
7. It monitors the number of types of individuals who take the survey.
8. The researcher always divides the population into subgroups.
9. The sample represents the entire population.
10. Researchers use the sampling method to identify the traits of a specific group of people.

Advantages of quota sampling

Here are the top four advantages of quota sampling

1. **Saves time:** Because of the involvement of a quota for sample creation, this sampling process is quick and straightforward.
2. **Research convenience:** By using quota sampling and appropriate research questions, interpreting information and responses to the survey is a much convenient process for a researcher.
3. **Accurate representation of the population of interest:** Researchers effectively represent a population using this sampling technique. There is no room for over-representation as this sampling technique helps researchers to study the population using specific quotas.
4. **Saves money:** The budget required for executing this sampling method is minimalistic.

Applications of quota sampling:

Below are the instances where quota sampling is applied and used.

- In situations where researchers have specific criteria for conducting research, it allows the selection of subgroups, due to which it becomes extremely convenient for researchers to obtain desired results. A trait or characteristic can be the filter for subgroup formation.
- The researcher uses this method when he/she has time constraints. Applying quotas gives the researcher an idea of the whole population of interest in very little time.
- Quotas are applied when the researcher is on a tight budget. Instead of researching a large population, the researcher saves money by using a few quotas to get the whole picture of the population.
- Some research studies do not require pinpoint accuracy due to the nature of the research project. It is ideal for applying to quota sampling for these studies.

Errors in sampling.

Assume you are a market researcher of a company looking to introduce a new product to the market. You must collect data from a sample of potential customers as part of your research to determine their preferences and purchasing behavior. But how can you be sure that the information you get from your sample is accurate for all of the people who might buy your product? The idea of sampling error comes into play here.

It is the difference between what a sample has and what the entire population has. It can significantly affect how accurate and reliable market research data is.

We will cover ways to reduce sampling error in this article to get more accurate and reliable results. Now, grab your favorite coffee and get ready to explore what sampling error is.

What is a sampling error?

A sampling error occurs when the sample used in the study does not represent the entire population. Although sampling errors occur frequently, researchers always include a margin of error in their conclusions as a matter of statistical practice.

The margin of error is the amount allowed for a miscalculation to represent the difference between the sample and the actual population.

Sampling is a type of analysis where a small sample of observations is chosen from a larger population. The selection process can produce both sampling errors and non-sampling errors.

Recognizing sampling error

Sampling errors are differences between a sample's values and the actual population's values. This is because a sample does not accurately represent the whole population of data.

Since there was a mistake in collecting the data, the results from sampling are no longer valid. Also, when a sample is chosen randomly or because of bias, it doesn't represent the whole population, and sampling errors are likely to happen.

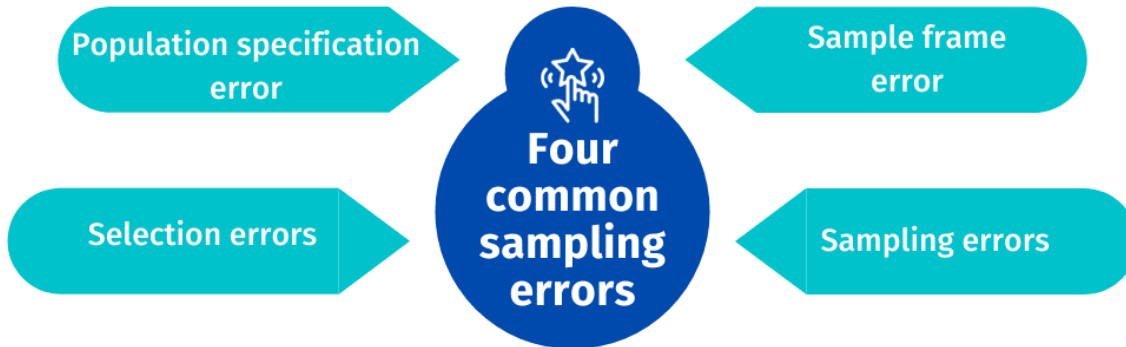
It is possible to avoid them if analysts carefully choose representative subsets of data from which to draw conclusions about the entire population. Factors including sample size and design, diversity in the population, and sampling percentage all contribute to sampling errors.

Diversity in the population increases the error in the estimations since it causes sampling to produce mixed results. Increasing the size of the samples allows them to represent the population more accurately, reducing the impact of population variation.

It is important to consider sampling errors before reporting survey results to establish confidence in the reliability of the estimates and their conclusions.

Types of common sampling error in market research

Here are the top four market research errors while sampling:



- **Population specification error**

A population specification error occurs when researchers don't know precisely who to survey.

For example, imagine a research study about kid's apparel. Who is the right person to survey? It can be both parents, only the mother, or the child. The parents make purchase decisions, but the kids may influence their choice.

- **Sample frame error**

Sampling frame error occurs when researchers target the sub-population wrongly while selecting the sample.

For example, picking a sampling frame from the telephone white pages book may have erroneous inclusions because people shift their cities. Erroneous exclusions occur when people prefer to un-list their numbers. Wealthy households may have more than one connection, thus leading to multiple inclusions.

- **Selection error**

Selection error occurs when respondents self-select themselves to participate in the study. You can control selection errors by going the extra step to request responses from the entire sample. Only interested ones respond.

Pre-survey planning, follow-ups, and a neat and clean survey design will boost respondents' participation rate. Also, try sampling methods like CATI surveys and in-person interviews to maximize responses.

- **Sampling errors**

Sampling errors occur due to a disparity in the representativeness of the respondents. It majorly happens when the researcher does not plan his sample carefully.

These sampling errors can be controlled and eliminated by creating a careful sample design, having a large enough sample to reflect the entire population, or using an online sample or survey audiences to collect responses.

Example of sampling error

Let's look more closely at this example.

Let's say a political party conducts a survey to find out how well-liked their candidate is before a major election. Instead of picking a random sample of the whole population to survey, they only ask their own members.

The sample would be skewed since party members may have extremely different opinions and tastes from the rest of the population. Party members may care more about their candidate's ideals or be more loyal to them. This may cause the survey to suggest more support than the general population has.

Suppose the survey results are used to make campaign decisions, such as where to allocate money or which issues to prioritize. In that case, they may not accurately reflect the candidate's support among the people. This could result in a poor campaign plan, affecting their election chances.

To avoid these kinds of sampling errors, it is essential to use a sampling method that is representative of the population being studied, such as random sampling or stratified random sampling, and to make sure that the sample size is big enough to give accurate results.

Controlling your sampling error

Statistical theories help researchers measure the probability of sampling errors in sample size and population. The size of the sample considered from the population primarily determines the size of the sampling error. Larger sample sizes tend to encounter a lower rate of errors.

Researchers use a metric known as the margin of error to understand and evaluate the margin of error. Usually, a confidence level of 95% is considered to be the desired confidence level.

Sampling error vs. Non-sampling error

Surveys can have sampling and non-sampling errors. Survey findings can be affected by sampling and non-sampling mistakes.

Sampling error arises when a survey sample does not accurately represent the population being researched due to random sampling. Sampling and non-response bias, measurement error, and sampling variability can cause this.

Non-sampling error includes all survey errors other than sampling errors. This includes questionnaire design, coding, data input, data collection, processing, and analysis errors. Improper interviewer training, insufficient or inaccurate data, or data analysis or reporting errors can create non-sampling errors.

Non-sampling error can be decreased by applying quality control measures and ensuring that all components of the survey process are properly designed, implemented, and monitored, whereas sampling error can be reduced by utilizing appropriate sampling procedures and increasing the sample size.

Sampling error vs. Sampling bias

Sampling in statistics means choosing the research group. Sampling error and sampling bias affect sample accuracy and representativeness in statistics.

Sampling error occurs because a sample is a subset of the population and may not precisely represent it. Instead, sampling bias occurs when the sample is not representative of the population. This may happen if the method used to choose the sample favors or excludes particular sorts of people, resulting in an overrepresentation or underrepresentation of particular groups.

Using stratified or random sampling and unbiased, population-representative sample selection can reduce sampling bias. On the other hand, sampling error can be reduced by using the proper sampling methods and making the sample size bigger.

What are the steps to reduce sampling errors?

Sampling errors are easy to identify. Here are a few simple steps to reduce sampling error:



- **Increase sample size**

A larger sample size is more accurate because the study gets closer to the actual population size.

- **Divide the population into groups**

Test groups according to their size in the population instead of a random sample. For example, if people of a specific demographic make up 20% of the population, make sure that your study is made up of this variable to reduce sampling bias.

- **Know your population**

Study your population and understand its demographic mix. Know what demographics use your product and service and ensure you only target the sample that matters.

We have also created a tool to help you determine your sample size easily: [Sample Size Calculator](#).

A sampling error is measurable, and researchers can use it to their advantage to estimate their findings' accuracy and variance.

Module-4 (9 Hours)

Data Collection: Meaning, types, **Data collection methods:** Observations, survey and interview techniques, **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary data Sources: advantages and disadvantages.

Measurement and Scaling Techniques: Basic measurement scales-Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale, **Multi-Dimensional Scaling:** Non comparative scaling techniques.

Data Collection:

Meaning

Data is a collection of facts, figures, objects, symbols, and events gathered from different sources. **Organizations collect data with various data collection methods to make better decisions.** Without data, it would be difficult for organizations to make appropriate decisions, so data is collected from different audiences at various points in time.

For instance, an organization must collect data on product demand, customer preferences, and competitors before launching a new product. If data is not collected beforehand, the organization's newly launched product may fail for many reasons, such as less demand and inability to meet customer needs.

Although data is a valuable asset for every organization, it does not serve any purpose until analyzed or processed to get the desired results.

What are Data Collection Methods?

Data collection methods are techniques and procedures used to gather information for research purposes. These methods can range from simple self-reported surveys to more complex experiments and can involve either quantitative or qualitative approaches to data gathering.

Some common data collection methods include surveys, interviews, observations, focus groups, experiments, and secondary data analysis. The data collected through these methods can then be analyzed and used to support or refute research hypotheses and draw conclusions about the study's subject matter.

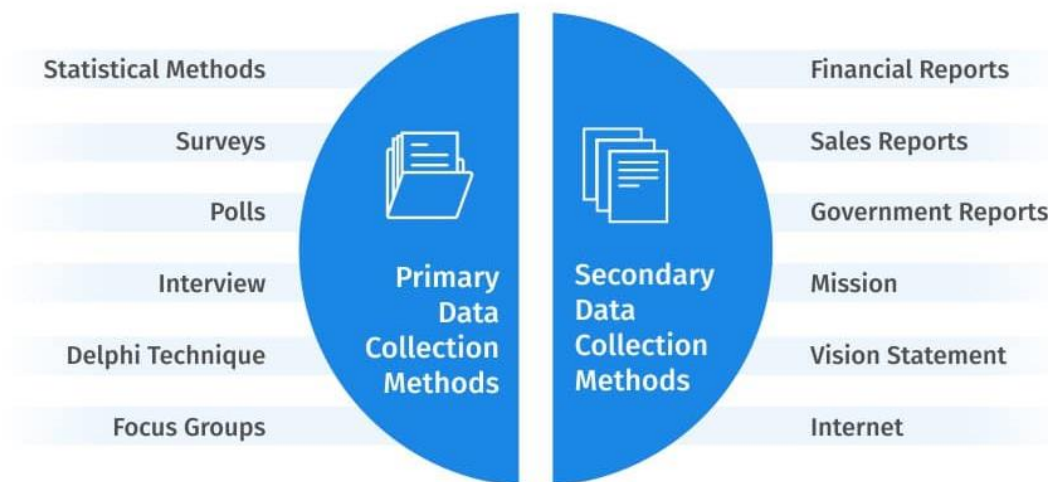
Importance of Data Collection Methods

Data collection methods play a crucial role in the research process as they determine the quality and accuracy of the data collected. Here are some major importance of data collection methods.

- Determines the quality and accuracy of collected data.
- Ensures that the data is relevant, valid, and reliable.
- Helps reduce bias and increase the representativeness of the sample.
- Essential for making informed decisions and accurate conclusions.
- Facilitates achievement of research objectives by providing accurate data.
- Supports the validity and reliability of research findings.

The importance of data collection methods cannot be overstated, as it plays a key role in the overall success and internal validity of the research study.

Types of Data Collection Methods



The choice of data collection method depends on the research question being addressed, the type of data needed, and the resources and time available. You can categorize data collection methods into primary methods of data collection and secondary methods of data collection.

Primary Data Collection Methods

Primary data is collected from first-hand experience and is not used in the past. The data gathered by primary data collection methods are specific to the research's motive and highly accurate.

Primary data collection methods can be divided into two categories: quantitative methods and qualitative methods.

Quantitative Methods:

Quantitative techniques for market research and demand forecasting usually use statistical tools. In these techniques, demand is forecasted based on historical data. These methods of primary data collection are generally used to make long-term forecasts. Statistical methods are highly reliable as subjectivity is minimal in these methods.



Time Series Analysis: The term time series refers to a sequential order of values of a variable, known as a trend, at equal time intervals. Using patterns, an organization can predict the demand for its products and services for the projected time.

Smoothing Techniques: In cases where the time series lacks significant trends, smoothing techniques can be used. They eliminate a random variation from the historical demand. It helps in identifying patterns and demand levels to estimate future demand. The most common methods used in smoothing demand forecasting techniques are the simple moving average method and the weighted moving average method.

Barometric Method: Also known as the leading indicators approach, researchers use this method to speculate future trends based on current developments. When the past events are considered to predict future events, they act as leading indicators.

Qualitative Methods:

Qualitative data collection methods are especially useful in situations when historical data is not available. Or there is no need of numbers or mathematical calculations. Qualitative research is closely associated with words, sounds, feeling, emotions, colors, and other elements that are non-quantifiable. These techniques are based on experience, judgment, intuition, conjecture, emotion, etc.

Quantitative methods do not provide the motive behind participants' responses, often don't reach underrepresented populations, and span long periods to collect the data. Hence, it is best to combine quantitative methods with qualitative methods.

Surveys: Surveys are used to collect data from the target audience and gather insights into their preferences, opinions, choices, and feedback related to their products and services. Most survey software often has a wide range of question types to select.

You can also use a ready-made survey template to save time and effort. Online surveys can be customized as per the business's brand by changing the theme, logo, etc. They can be distributed through several distribution channels such as email, website, offline app, QR code, social media, etc. Depending on the type and source of your audience, you can select the channel.

Once the data is collected, survey software can generate various reports and run analytics algorithms to discover hidden insights. A survey dashboard can give you statistics related to response rate, completion rate, filters based on demographics, export and sharing options, etc. Integrating survey builder with third-party apps can maximize the effort spent on online real-time data collection.

Polls: Polls comprise one single or multiple-choice question. You can go for polls when it is required to have a quick pulse of the audience's sentiments. Because they are short in length, it is easier to get responses from people.

Like surveys, online polls can also be embedded into various platforms. Once the respondents answer the question, they can also be shown how they stand compared to others' responses.

Interviews: In this method, the interviewer asks the respondents face-to-face or by telephone. In face-to-face interviews, the interviewer asks a series of questions to the interviewee in person and notes down responses. If it is not feasible to meet the person, the interviewer can go for a telephone interview. This form of data collection is suitable for only a few respondents. It is too time-consuming and tedious to repeat the same process if there are many participants.

Qualitative Methods



Delphi Technique: In delphi method, market experts are provided with the estimates and assumptions of forecasts made by other experts in the industry. Experts may reconsider and revise their estimates and assumptions based on the information provided by other experts. The consensus of all experts on demand forecasts constitutes the final demand forecast.

Focus Groups: In a focus group, a small group of people, around 8-10 members, discuss the common areas of the research problem. Each individual provides his insights on the issue concerned. A moderator regulates the discussion among the group members. At the end of the discussion, the group reaches a consensus.

Questionnaire: A questionnaire is a printed set of questions, either open-ended or closed-ended. The respondents must answer based on their knowledge and experience with the issue. The questionnaire is a part of the survey, whereas the questionnaire's end goal may or may not be a survey.

Secondary Data Collection Methods

Secondary data is the data that has been used in the past. The researcher can obtain data from the data sources, both internal and external, to the organizational data.

Internal sources of secondary data:

- Organization's health and safety records
- Mission and vision statements
- Financial Statements
- Magazines
- Sales Report
- CRM Software
- Executive summaries

External sources of secondary data:

- Government reports
- Press releases
- Business journals
- Libraries
- Internet

The secondary data collection methods can also involve quantitative and qualitative techniques. Secondary data is easily available and hence, less time-consuming and expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified.

The secondary data collection methods can also involve quantitative and qualitative observation techniques. Secondary data is easily available, less time-consuming, and more expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified.

Regardless of the data collection method of your choice, there must be direct communication with decision-makers so that they understand and commit to acting according to the results.

For this reason, we must pay special attention to the analysis and presentation of the information obtained. Remember that these data must be useful and functional to us, so the data collection method used has much to do with it.

Advantages of Secondary Research

Secondary research offers a number of advantages to researchers, including efficiency, the ability to build upon existing knowledge, and the ability to conduct research in situations where primary research may not be possible or ethical. By carefully selecting their sources and being thoughtful in their approach, researchers can leverage secondary research to drive impact and advance the field. Some key advantages are the following:

- 1. Most information in this research is readily available.** There are many sources from which relevant data can be collected and used, unlike primary research, where data needs to collect from scratch.
- 2. This is a less expensive and less time-consuming process** as data required is easily available and doesn't cost much if extracted from authentic sources. A minimum expenditure is associated to obtain data.
- 3. The data that is collected through secondary research gives organizations or businesses an idea about the effectiveness of primary research.** Hence, organizations or businesses can form a hypothesis and evaluate cost of conducting primary research.
- 4. Secondary research is quicker to conduct because of the availability of data.** It can be completed within a few weeks depending on the objective of businesses or scale of data needed.

As we can see, this research is the process of analyzing data already collected by someone else, and it can offer a number of benefits to researchers.

Disadvantages of Secondary Research

On the other hand, we have some disadvantages that come with doing secondary research. Some of the most notorious are the following:

1. Although data is readily available, credibility evaluation must be performed to understand the authenticity of the information available.
2. Not all secondary data resources offer the latest reports and statistics. Even when the data is accurate, it may not be updated enough to accommodate recent timelines.

3. Secondary research derives its conclusion from collective primary research data. The success of your research will depend, to a greater extent, on the quality of research already conducted by primary research.

Observation Method

Definition

The observation method is a process that involves human or mechanical observation to observe and describe the behavior of a subject. As the name suggests observational research is a way of collecting relevant information and data by involves observing people's behavior. The observational Research method is also referred to as a participatory study because the researcher has to establish a link with the respondent and for this has to immerse himself in the same setting as theirs. Only then can he use the observational research method to record and take notes.

Meaning and examples

Observational research is used in cases where you want to avoid an error that can be a result of bias during evaluation and interpretation processes. Observational research is one of the many other research methods to obtain objective data by watching a participant and recording it for analysis at a later stage.

A researcher can use observational research, for example in a school, and record the behavior of the children at a young age. Are the children comfortable sharing their tiffin at such an early age will make a good study for the researcher? In this example, the researcher can observe and record the details objectively. The observational research data collection method is associated with observational research methods but with a few ethical issues as it needs the full consent of a research participant.

The observation method in data collection can be

- **Structured observation method** – Structured Observation is a systematic observation method where data is collected as per a pre-defined schedule. The specific variable is used in this observation data collection method.
- **Unstructured observation method** – The unstructured observation method is different from structured observation because it is conducted in a free and open manner without using any pre-determined objectives, schedules, or variables.

Sampling Methods for Observational Data



While conducting observational research, the researcher has a vital role to play in collecting data, as he will have to collect, record and classify the data appropriately. The research methods primary sampling methods are

- **Event sampling** – In the event sampling observation method the researcher decides beforehand what events or behavior he will record and which ones he is going to ignore
- **Time sampling** – In the time sampling observation method, the researcher chooses the time when he will observe. He makes a record of the occurrence only in the specified and pre-determined period
- **Target-time or instantaneous sampling** – In the target-time or instantaneous sampling observation method, the researcher decides beforehand the moments when the observation will happen and will be recorded at that moment. Everything that happens before or after that moment is of no consequence and hence is typically ignored

Types of Observational Research Methods

the different types of observation methods in observational research are as follows-

1) Controlled observations

The controlled observation is carried out in a closed space. It is the researcher who has the authority to decide the place and the time where and when the observation will take place. He also decides who the participants will be and in what circumstances will he use the standardized process.

The participants are chosen for a variable group randomly. The researcher observes and records a detailed and descriptive quantitative data of behavior and divides it into a distinct category. Sometimes the researcher codes the action as per an agreed scale by using a behavior schedule. The coding can include letters or numbers or a range to measure behavior intensity and describe its characteristics.

The collected data from observational research methods is often turned into statistics for laboratory research. In a controlled observation method, the participants are informed by the researcher about the aim of the research. This makes them aware of being observed. The researcher avoids direct contact during the observation method and generally uses a two-way mirror to observe and record details.

2) Naturalistic observations

Social scientists and psychologists generally use the naturalistic observation method in observational research studies. The process of naturalistic observation involves observing and studying the spontaneous behavior of the participants in open or natural surroundings. The role of the researcher is to find and record whatever he can see and observe in natural habitat.

3) Participant observation

The participant observation method is often considered a variant of the naturalistic observation method because it has some similarities with it. The point of difference called disguised naturalistic observation, is that the researcher is not a distant observer anymore because he has joined the participants and become a part of their group. He does this to get a more in-depth and greater insight into their lives.

In participant observation, The researcher interacts with other members of the group freely, participates in their activities, studies their behavior and acquires a different way of life. Participant observation can be overt or covert.

1. **Overt participant observation:** When the researcher asks permission from a group to mingle the observation method is known as overt. He does so by revealing his true purpose and real identity to the group with whom he wants to mingle
2. **Covert participant observation:** When the researcher does not show either his true identity or real meaning to the group he wants to join then the observation is known as covert. He keeps both concealed and takes on a false role and identity to enter and mingle in the group. He generally acts as if he is a genuine member of that group

Advantages of the Observation Method

- Provides direct access to research phenomena
- By observing firsthand, the researcher can collect, check and record accurate data
- Greater flexibility in terms of application
- Generate a permanent record of phenomena and the researcher or others can refer with it later
- The organization method is one of the simplest methods of data collection. It does not require too much technical knowledge
- The observation method is one of the best ways to formulate a hypothesis. The researcher can observe and come to know about the activities, perceptions, likes, and dislikes to form a theory on his subject
- Observational methods are one of the most common methods used in all sciences and is very easy to follow and accept
- In some instances observation is the only available tool to collect essential data and information
- The observation method does not require the willingness of the participant to record. The researcher can observe from a distance and record his findings

Disadvantages of the Observation Method

- Observational methods Face a severe disadvantage because it takes a longer time frame compared to other data collection methods
- There is a chance of higher observer bias in the observation method
- Several personal behaviors are not open for observation and this proves a limitation in case of observation method
- There is a higher chance of the observer influencing the behavior of a sample group elements
- Uncertainties of the event cannot determine the actual time when the event will take place, and this is why every occurrence that is open to observation cannot be observed
- Many of the incidents are abstract like love, affection and the researcher can't gain an exact and correct account of those
- The social phenomena generalization made by observation are not considered reliable as it cannot be used for lab experiments
- In some cases, it is seen that two persons observing the same phenomena come at different results and this can lead to faulty perceptions
- Observation method is considered an expensive affair as it requires hard effort, plenty of time and high cost

Data collection methods: Survey



Survey Data: Definition

Survey data is defined as the resultant data that is collected from a sample of respondents that took a survey. This data is comprehensive information gathered from a target audience about a specific topic to conduct research. There are many methods used for survey data collection and statistical analysis.

Various mediums are used to collect feedback and opinions from the desired sample of individuals. While conducting survey research, researchers prefer multiple sources to gather data such as online surveys, telephonic surveys, face-to-face surveys, etc. The medium of collecting survey data decides the sample of people that are to be reached out to, to reach the requisite number of survey responses.

Factors of collecting survey data such as how the interviewer will contact the respondent (online or offline), how the information is communicated to the respondents etc. decide the effectiveness of gathered data.

Survey Data Collection Methods with Examples

The methods used to collect survey data have evolved with the change in technology. From face-to-face surveys, telephonic surveys to now online and email surveys, the world of survey data collection has changed with time. Each survey data collection method has its pros and cons, and every researcher has a preference for gathering accurate information from the target sample.

The survey response rates for each of these data collection methods will differ as their reach and impact are always different. Different ways are chosen according to specific target population characteristics and the intent to examine human nature under various situations.

Types of survey data based on deployment methods:

There are four main survey data collection methods – Telephonic Surveys, Face-to-face Surveys, and Online Surveys.

- **Online Surveys**

Online surveys are the most cost-effective and can reach the maximum number of people in comparison to the other mediums. The performance of these surveys is much more widespread than the other data collection methods. In situations where there is more than one question to be asked to the target sample, certain researchers prefer conducting online surveys over the traditional face-to-face or telephone surveys.

Online surveys are effective and therefore require computational logic and branching technologies for exponentially more accurate survey data collection vs any other traditional means of surveying. They are straightforward in their implementation and take a minimum time of the respondents. The investment required for survey data collection using online surveys is also negligible in comparison to the other methods. The results are collected in real-time for researchers to analyze and decide corrective measures.

A very good example of an online survey is a hotel chain using an online survey to collect guest satisfaction metrics after a stay or an event at the property.

Online surveys are safe and secure to conduct. As there is no in-person interaction or any direct form of communication, they are quite useful in times of global crisis. For instance, many organizations moved to contactless surveys during the pandemic. It helped them ensure that the employees are not experiencing any COVID-19 symptoms before they come to the office.

- **Face-to-face Surveys**

Gaining information from respondents via face-to-face mediums is much more effective than the other mediums because respondents usually tend to trust the surveyors and provide honest and clear feedback about the subject in-hand.

Researchers can easily identify whether their respondents are uncomfortable with the asked questions and can be extremely productive in case there are sensitive topics involved in the discussion. This online data collection method demands more cost-investment than in comparison to the other methods. According to the geographic segmentation or psychographic segmentation, researchers must be trained to gain accurate information.

For example, a job evaluation survey is conducted in person between an HR or a manager with the employee. This method works best face-to-face as the data collection can collect as accurate information as possible.

- **Telephone Surveys**

Telephone surveys require much lesser investment than face-to-face surveys. Depending on the required reach, telephone surveys cost as much or a little more than online surveys. Contacting respondents via the telephonic medium requires less effort and manpower than the face-to-face survey medium.

If interviewers are located at the same place, they can cross-check their questions to ensure error-free questions are asked to the target audience. The main drawback of conducting telephone surveys is that establishing a friendly equation with the respondent becomes challenging due to the bridge of the medium. Respondents are also highly likely to choose to remain anonymous in their feedback over the phone as the reliability associated with the researcher can be questioned.

For example, if a retail giant would like to understand purchasing decisions, they can conduct a telephonic, motivation, and buying experience survey to collect data about the entire purchasing experience.

- **Paper Surveys**

The other commonly used survey method is paper surveys. These surveys can be used where laptops, computers, and tablets cannot go, and hence they use the age-old method of data collection; pen and paper. This method helps collect survey data in field research and helps strengthen the number of responses collected and the validity of these responses.

A popular example or use case of a paper survey is a fast food restaurant survey where the fast-food chain would like to collect feedback on the dining experience of its patrons.

Types of survey data based on the frequency at which they are administered:

Surveys can be divided into 3 distinctive types on the basis of the frequency of their distribution. They are:

- **Cross-Sectional Surveys**

Cross-sectional surveys are an observational research method that analyzes data of variables collected at one given point of time across a sample population or a pre-defined subset. The survey data from this method helps the researcher understand what the respondent is feeling at a certain point in time. It helps measure opinions in a particular situation.

For example, if the researcher would like to understand movie rental habits, a survey can be conducted across demographics and geographical locations. The cross-sectional survey, for example, can help understand that males between 21-28 rent action movies and females between 35-45 rent romantic comedies. This survey data helps for the basis of a longitudinal study.

- **Longitudinal Surveys**

Longitudinal surveys are those surveys that help researchers to make an observation and collect data over an extended period of time. This survey data can be qualitative or quantitative in nature, and the survey creator does not interfere with the survey respondents.

For example, a longitudinal study can be carried out for years to help understand if mine workers are more prone to lung diseases. This study takes a year and discounts any pre-existing conditions.

- **Retrospective Surveys**

In retrospective surveys, researchers ask respondents to report events from the past. This survey method offers in-depth survey data but doesn't take as long to complete. By deploying this kind of survey, researchers can gather data based on past experiences and beliefs of people.

For example, if hikers are asked about a certain hike – the conditions of the hiking trail, ease of hike, weather conditions, trekking conditions, etc. after they have completed the trek, it is a retrospective study.

Survey Data Analysis

After the survey data has been collected, this data has to be analyzed to ensure it aids towards the end research objective. There are different ways of conducting this research and some steps to follow. They are as below:

Survey Data Analysis: Steps and Tips

There are four main steps of survey data analysis:

1. **Understand the most popular survey research questions:** The survey questions should align with the overall purpose of the survey. That is when the collected data will be effective in helping researchers. For example, if a seminar has been conducted, the researchers will send out a post-seminar feedback survey. The primary goal of this survey will be to understand whether the attendees are interested in attending future seminars. The question will be: “How likely are you to attend future seminars?” – Data collected for this question will decide the likelihood of success of future seminars.
2. **Filter obtained results using the cross-tabulation technique:** Understand the various categories in the target audience and their thoughts using cross-tabulation format. For example, if there are business owners, administrators, students, etc. who attend the seminar, the data about whether they would prefer attending future seminars or not can be represented using cross-tabulation.
3. **Evaluate the derived numbers:** Analyzing the gathered information is critical. How many of the attendees are of the opinion that they will be attending future seminars and how many will not – these facts need to be evaluated according to the results obtained from the sample.
4. **Draw conclusions:** Weave a story with the collected and analyzed data. What was the intention of the survey research, and how does the survey data suffice that objective? – Understand that and develop accurate, conclusive results.

Survey Data Analysis Methods

Conducting a survey without having access to the resultant data and the inability to drawing conclusions from the survey data is pointless. When you conduct a survey, it is imperative to have access to its analytics. It is tough to analyze using traditional survey methods like pen and paper and also requires additional manpower. Survey data analysis becomes much easier when using advanced online data collection methods with an online survey platform such as market research survey software or customer survey software.

Statistical analysis can be conducted on the survey data to make sense of the data that has been collected. There are multiple data analysis methods of quantitative data. Some of the commonly used types are:

- **Cross-tabulation:** Cross-tabulation is the most widely used data analysis methods. It uses a basic tabulation framework to make sense of data. This statistical analysis method helps tabulate data into easily understandable rows and columns, and this helps draw parallels between different research parameters. It contains data that is mutually exclusive or have some connection with each other.
- **Trend analysis:** Trend analysis is a statistical analysis method that provides the ability to look at survey-data over a long period of time. This method helps plot aggregated response data over time allows drawing a trend line of the change, if any, of perceptions over time about a common variable.
- **MaxDiff analysis:** The MaxDiff analysis method is used to gauge what a customer prefers in a product or a service across multiple parameters. For example, a product's feature list, the difference with the competition, ease of use and likert scale, pricing, etc. form the basis for maxdiff analysis. In a simplistic form, this method is also called the "best-worst" method. This method is very similar to conjoint analysis, but it is much easier to implement and can be interchangeably used.
- **Conjoint analysis:** As mentioned above, conjoint analysis is similar to maxdiff analysis, only differing in its complexity and the ability to collect and analyze advance survey data. This method analyzes each parameter behind a person's purchasing behavior. By using conjoint analysis, it is possible to understand what exactly is important to a customer and the aspects that are evaluated before purchase.
- **TURF analysis:** TURF analysis or Total Unduplicated Reach and Frequency analysis, is a statistical research methodology that assesses the total market reach of a product or service or a mix of both. This method is used by organizations to understand the frequency and the avenues at which their messaging reaches customers and prospective customers. This helps them tweak their go-to-market strategies.
- **Gap analysis:** Gap analysis uses a side-by-side matrix question type that helps measure the difference between expected performance and actual performance. This statistical method for survey data helps understand the things that have to be done to move performance from actual to planned performance.
- **SWOT analysis:** SWOT analysis, another widely used statistical method, organizes survey data into data that represents the strength, weaknesses, opportunities, and threats of an organization or product or service that provides a holistic picture of competition. This method helps to create effective business strategies.
- **Text analysis:** Text analysis is an advanced statistical method where intelligent tools make sense of and quantify or fashion qualitative and open-ended data into easily understandable data. This method is used when the survey data is unstructured.

Data Collection Method –

Interview techniques

There are more types of interviews than most people think. An interview is generally a qualitative research technique that involves asking open-ended questions to converse with respondents and collect elicited data about a subject.

The interviewer, in most cases, is the subject matter expert who intends to understand respondent opinions in a well-planned and executed series of star questions and answers.

Interviews are similar to focus groups and surveys for garnering information from the target market but are entirely different in their operation – focus groups are restricted to a small group of 6-10 individuals, whereas surveys are quantitative.

Interviews are conducted with a sample from a population, and the key characteristic they exhibit is their conversational tone.

What is an Interview?

An interview is a way to get information from a person by asking questions and hearing their answers.

An interview is a question-and-answer session where one person asks questions, and the other person answers those questions. It can be a one-on-one, two-way conversation, or there can be more than one interviewer and more than one participant.

The interview is the most important part of the whole selection process. It is used to decide if a person should be interviewed further, hired, or taken out of consideration. It is the main way to learn more about applicants and the basis for judging their job-related knowledge, research skills, and abilities.

Fundamental Types of Interviews in Research

A researcher has to conduct interviews with a group of participants at a juncture in the research where information can only be obtained by meeting and personally connecting with a section of their target audience. Interviews offer the researchers a platform to prompt their participants and obtain inputs in the desired detail. There are three fundamental types of interviews in research:



1. Structured Interviews:

Structured interviews are defined as research tools that could be more flexible in their operations are allows more or no scope of prompting the participants to obtain and analyze results. It is thus also known as a standardized interview and is significantly quantitative in its approach.

Questions in this interview are pre-decided according to the required detail of information. This can be used in a focus group interview and an in-person interview.

These interviews are excessively used in survey research with the intention of maintaining uniformity throughout all the interview sessions.

They can be closed-ended and open-ended – according to the type of target population. Closed-ended questions can be included to understand user preferences from a collection of answer options. In contrast, open-ended ones can be included to gain details about a particular section in the interview.

Example of a structured interview question:

Here's an example of a structured question for a job interview for a customer service job:

- Can you talk about what it was like to work in customer service?

- How do you deal with an angry or upset customer?
- How do you ensure that the information you give customers is correct?
- Tell us about when you went out of your way to help a customer.
- How do you handle a lot of customers or tasks at once?
- Can you talk about how you've used software or tools for customer service?
- How do you set priorities and use your time well while giving good customer service?
- Can you tell us about when you had to get a customer to calm down?
- How do you deal with a customer who wants something that goes against your company's rules?
- Tell me about a time when you had to deal with a hard customer or coworker.

Advantages of structured interviews:

- It focuses on the accuracy of different responses, due to which extremely organized data can be collected. Different respondents have different types of answers to the same structure of questions – answers obtained can be collectively analyzed.
- They can be used to get in touch with a large sample of the target population.
- The interview procedure is made easy due to the standardization offered by it.
- Replication across multiple samples becomes easy due to the same structure of the interview.
- As the scope of detail is already considered while designing the interview questions, better information can be obtained. The researcher can analyze the research problem comprehensively by asking accurate research questions.
- Since the structure of the interview is fixed, it often generates reliable results and is quick to execute.
- The relationship between the researcher and the respondent is not formal, due to which the researcher can clearly understand the margin of error in case the respondent either degree to be a part of the survey or is just not interested in providing the right information.

Disadvantages of structured interviews:

- The limited scope of assessment of obtained results.
- The accuracy of information overpowers the detail of information.
- Respondents are forced to select from the provided answer options.
- The researcher is expected to always adhere to the list of decided questions, irrespective of how interesting the conversation is turning out to be with the participants.
- A significant amount of time is required for a structured interview.

2. Semi-Structured Types of Interviews:

Semi-structured interviews offer a considerable amount of leeway to the researcher to probe the respondents, along with maintaining a basic interview structure. Even if it is a guided conversation between researchers and interviewees – appreciable flexibility is offered to the researchers. A researcher can be assured that multiple interview rounds will not be required in the presence of structure in this type of research interview.

Keeping the structure in mind, the researcher can follow any idea or take creative advantage of the entire interview. Additional respondent probing is always necessary to garner information for a research study. The best application of semi-structured interviews is when the researcher doesn't have time to conduct research and requires detailed information about the topic.

Example of a semi-structured interview question:

Here's an example of a semi-structured marketing job interviews question:

- Can you tell us about the marketing work you've done?
- What do you think are the most important parts of a marketing campaign that works?
- Tell me about a campaign you worked on that you're very proud of.
- How do you do research on the market and look at data to help you make marketing decisions?
- Can you tell us about a time when you had to change your marketing plan because of something that didn't go as planned?
- How do you figure out if a marketing campaign worked?
- Can you talk about how you've used social media to market?
- How do you ensure your marketing message gets through to the people you want to hear it?
- Can you tell us about a time when you had to run a marketing campaign on a small budget?
- How do you keep up with changes and trends in marketing?

Advantages of semi-structured interviews:

- Questions from semi-structured interview questions are prepared before the scheduled interview, giving the researcher time to prepare and analyze the questions.
- It is flexible to an extent while maintaining the research guidelines.
- Unlike a structured interview, researchers can express the interview questions in the preferred format.
- Reliable qualitative data can be collected via these interviews.
- The flexible structure of the interview.

Disadvantages of semi-structured interviews:

- Participants may question the reliability factor of these interviews due to the flexibility offered.
- Comparing two different answers becomes difficult as the guideline for conducting interviews is not entirely followed. No two questions will have the exact same structure, and the result will be an inability to compare and infer results.

3. Unstructured Interviews:

Also called in-depth interviews, unstructured interviews are usually described as conversations held with a purpose in mind – to gather data about the research study. These interviews have the least number of questions as they lean more towards a normal conversation but with an underlying subject.

The main objective of most researchers using unstructured interviews is to build a bond with the respondents, due to which there is a high chance that the respondents will be 100% truthful with their answers. There are no guidelines for the researchers to follow. So they can approach the participants ethically to gain as much information as possible about their research topic.

Since there are no guidelines for these interviews, a researcher is expected to keep their approach in check so that the respondents do not sway away from the main research motive.

For a researcher to obtain the desired outcome, he/she must keep the following factors in mind:

- The intent of the interview.
- The interview should primarily take into consideration the participant's interests and skills.
- All the conversations should be conducted within the permissible limits of research, and the researcher should try and stick by these limits.
- The researcher's skills and knowledge should match the interview's purpose.
- Researchers should understand the dos and don'ts of it.

Example of an unstructured interview question:

Here's an example of a question asked in an unstructured interview:

- Can you tell me about when you had to deal with something hard and how you did it?
- What are some of the things you're most proud of, and what did you learn from them?
- How do you deal with ambiguity or not knowing what to do at work?
- Can you describe how you lead and how you get your team going?

- Tell me about a time when you had to take a chance and how it turned out.
- What do you think are the most important qualities for success in this role?
- How do you deal with setbacks or failures, and what do you learn from them?
- Can you tell me about a time when you had to solve a problem by thinking outside the box?
- What do you think makes you different from the other people who want this job?
- Can you tell me about a time when you had to make a hard choice and how you made that choice?

Advantages of Unstructured Interviews:

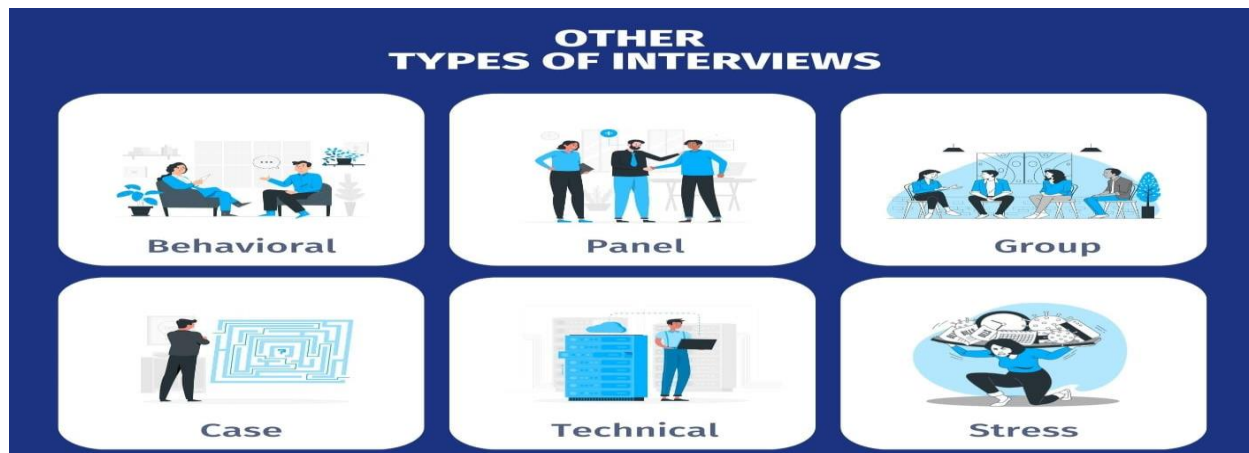
- Due to this type of interview's informal nature, it becomes extremely easy for researchers to try and develop a friendly rapport with the participants. This leads to gaining insights in extreme detail without much conscious effort.
- The participants can clarify all their doubts about the questions, and the researcher can take each opportunity to explain his/her intention for better answers.
- There are no questions that the researcher has to abide by, and this usually increases the flexibility of the entire research process.

Disadvantages of Unstructured Interviews:

- Researchers take time to execute these interviews because there is no structure to the interview process.
- The absence of a standardized set of questions and guidelines indicates that its reliability of it is questionable.
- The ethics involved in these interviews are often considered borderline upsetting.

Other Types of Interviews

Besides the 3 basic interview types, we have already mentioned there are more. Here are some other interview types that are commonly used in a job interview:



Behavioral Interview

During this type of interview, candidates are asked to give specific examples of how they have acted in the past. The idea behind this kind of interview is that what someone did in the past can be a sign of how they will act in the future. And by this interview, the company can also understand the interviewee's behavior through body language.

Panel Interview

During a panel interview, three or more interviewers usually ask questions and evaluate the candidate's answers as a group. This is a good way to get a full picture of a candidate's skills and suitability for the job.

Group Types of Interviews

Multiple people are interviewed at the same time in group interviews. This form of interview often focus groups that are utilized on entry-level positions or employment in customer service to examine how well candidates get along with others and function as a team.

Case Interview

During a case interview, candidates are given a business problem or scenario and asked to think about how to solve it. In the consulting and finance fields, this kind of interview is common.

Technical Interview

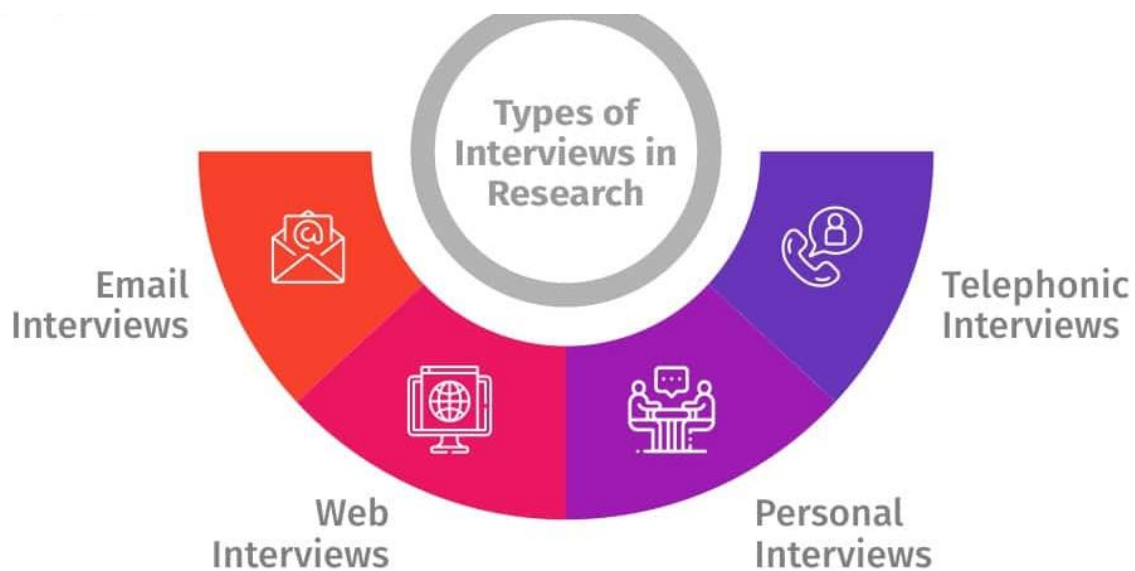
A candidate's technical skills and knowledge are tested during a technical interview, usually in fields like engineering or software development. Most of the time, candidates are asked to solve problems or complete technical tasks.

Stress Interview

During a stress interview, candidates are put under pressure or asked difficult or confrontational questions on purpose to see how they react in stressful situations. This kind of interview is used to see how well a candidate can deal with stress and hard situations.

Methods of Research Interviews:

There are four methods to conduct research interviews, each of which is peculiar in its application and can be used according to the research study requirement.



Personal Interviews:

Personal interviews are one of the most used types of interviews, where the questions are asked personally directly to the respondent as a form of an individual interview. One of the many in-person interviews is a lunch interview, which is frequently better suited for casual inquiries and discussions.

For this, a researcher can have a guide to online surveys to take note of the answers. A researcher can design his/her survey in such a way that they take notes of the comments or points of view that stands out from the interviewee. It can be a one-on-one interview as well.

Advantage:

- Higher response rate.
- When the interviewees and respondents are face-to-face, there is a way to adapt the questions if this is not understood.
- More complete answers can be obtained if there is doubt on both sides or a remarkable piece of information is detected.
- The researcher has an opportunity to detect and analyze the interviewee's body language at the time of asking the questions and taking notes about it.

Disadvantages:

- They are time-consuming and extremely expensive.
- They can generate distrust on the part of the interviewee since they may be self-conscious and not answer truthfully.
- Contacting the interviewees can be a real headache, either scheduling an appointment in workplaces or going from house to house and not finding anyone.
- Therefore, many interviews are conducted in public places like shopping centers or parks. Even consumer studies take advantage of these sites to conduct interviews or surveys and give incentives, gifts, and coupons. In short, There are great opportunities for online research in shopping centers.
- Among the advantages of conducting such types of interviews is that the respondents will have more fresh information if the interview is conducted in the context and with the appropriate stimuli so that researchers can have data from their experience at the scene of the events immediately and first hand. The interviewer can use an online survey through a mobile device that will undoubtedly facilitate the entire process.

Telephonic Type of Interviews:

Phonic interviews are widely used and easily combined with online surveys to conduct research effectively.

Advantages:

- To find the interviewees, it is enough to have their phone numbers on hand.
- They are usually lower cost.
- The information is collected quickly.
- Having a personal contact can also clarify doubts or give more details of the questions.

Disadvantages:

- Many times researchers observe that people do not answer phone calls because it is an unknown number for the respondent or simply already changed their place of residence and they cannot locate it, which causes a bias in the interview.
- Researchers also face that they simply do not want to answer and resort to pretexts such as they are busy to answer, they are sick, they do not have the authority to answer the questions asked, they have no interest in answering, or they are afraid of putting their security at risk.
- One of the aspects that should be taken care of in these types of interviews is the kindness with which the interviewers address the respondents in order to get them to cooperate more easily with their answers. Good communication is vital for the generation of better answers.

Email or Web Page Types of Interviews:

Online research is growing more and more because consumers are migrating to a more virtual world, and it is best for each researcher to adapt to this change.

The increase in people with Internet access has made it popular that interviews via email or web page stand out among the types of interviews most used today. For this nothing better than an online survey.

More and more consumers are turning to online shopping, which is why they are a great niche to be able to carry out an interview that will generate information for the correct decision-making.

Advantages of email surveys:

- Speed in obtaining data
- The respondents respond according to their time, when they want, and where they decide.
- Online surveys can be mixed with other research methods or using some of the previous interview models. They are tools that can perfectly complement and pay for the project.
- A researcher can use a variety of questions and logic to create graphs and reports immediately.

Disadvantages of email survey:

- Low response rates
- Limited access to certain populations
- Potential for spam filters
- Lack of personal touch

Questionnaire design:

What is a Questionnaire?

A questionnaire is a research instrument that consists of a set of questions or other types of prompts that aims to collect information from a respondent. A research questionnaire is typically a mix of close-ended questions and open-ended questions.

Open-ended, long-form questions offer the respondent the ability to elaborate on their thoughts. Research questionnaires were developed in 1838 by the Statistical Society of London.

The data collected from a data collection questionnaire can be both qualitative as well as quantitative in nature. A questionnaire may or may not be delivered in the form of a survey, but a survey always consists of a questionnaire.

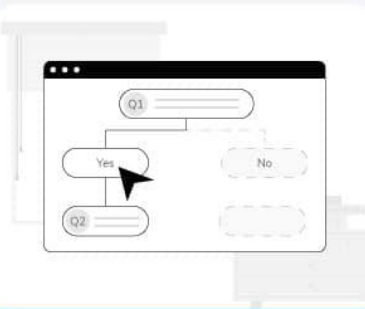
Advantages of a good questionnaire design

- With a survey questionnaire, you can gather a lot of data in less time.
- There is less chance of any bias creeping if you have a standard set of questions to be used for your target audience. You can apply logic to questions based on the respondents' answers, but the questionnaire will remain standard for a group of respondents that fall in the same segment.
- Surveying online survey software is quick and cost-effective. It offers you a rich set of features to design, distribute, and analyze the response data.
- It can be customized to reflect your brand voice. Thus, it can be used to reinforce your brand image.
- The responses can be compared with the historical data and understand the shift in respondents' choices and experiences.
- Respondents can answer the questionnaire without revealing their identity. Also, many survey software complies with significant data security and privacy regulations.

Advantages of a well-designed online questionnaire



Collect a large amount of data in less time.



There is less chance of bias with the help of logic questions.



Online survey software is fast and cost effective to design, distribute and analyze.



It can be customized to reflect your brand voice.



Responses can be compared with historical data.



Respondents can answer the questionnaire without revealing their identity.

Characteristics of a good questionnaire

Your survey design depends on the type of information you need to collect from respondents. Qualitative questionnaires are used when there is a need to collect exploratory information to help prove or disprove a hypothesis. Quantitative questionnaires are used to validate or test a previously generated hypothesis. However, most questionnaires follow some essential characteristics:

- **Uniformity:** Questionnaires are very useful to collect demographic information, personal opinions, facts, or attitudes from respondents. One of the most significant attributes of a research form is uniform design and standardization. Every respondent sees the same questions. This helps in data collection and statistical analysis of this data. For example, the retail store evaluation questionnaire template contains questions for evaluating retail store experiences. Questions relate to purchase value, range of options for product selections, and quality of merchandise. These questions are uniform for all customers.

- **Exploratory:** It should be exploratory to collect qualitative data. There is no restriction on questions that can be in your questionnaire. For example, you use a data collection questionnaire and send it to the female of the household to understand her spending and saving habits relative to the household income. Open-ended questions give you more insight and allow the respondents to explain their practices. A very structured question list could limit the data collection.
- **Question Sequence:** It typically follows a structured flow of questions to increase the number of responses. This sequence of questions is screening questions, warm-up questions, transition questions, skip questions, challenging questions, and classification questions. For example, our motivation and buying experience questionnaire template covers initial demographic questions and then asks for time spent in sections of the store and the rationale behind purchases.

Types & Definitions

As we explored before, questionnaires can be either structured or free-flowing. Let's take a closer look at what that entails for your surveys.

- **Structured Questionnaires:** Structured questionnaires collect quantitative data. The questionnaire is planned and designed to gather precise information. It also initiates a formal inquiry, supplements data, checks previously accumulated data, and helps validate any prior hypothesis.
- **Unstructured Questionnaires:** Unstructured questionnaires collect qualitative data. They use a basic structure and some branching questions but nothing that limits the responses of a respondent. The questions are more open-ended to collect specific data from participants.

Types of questions in a questionnaire

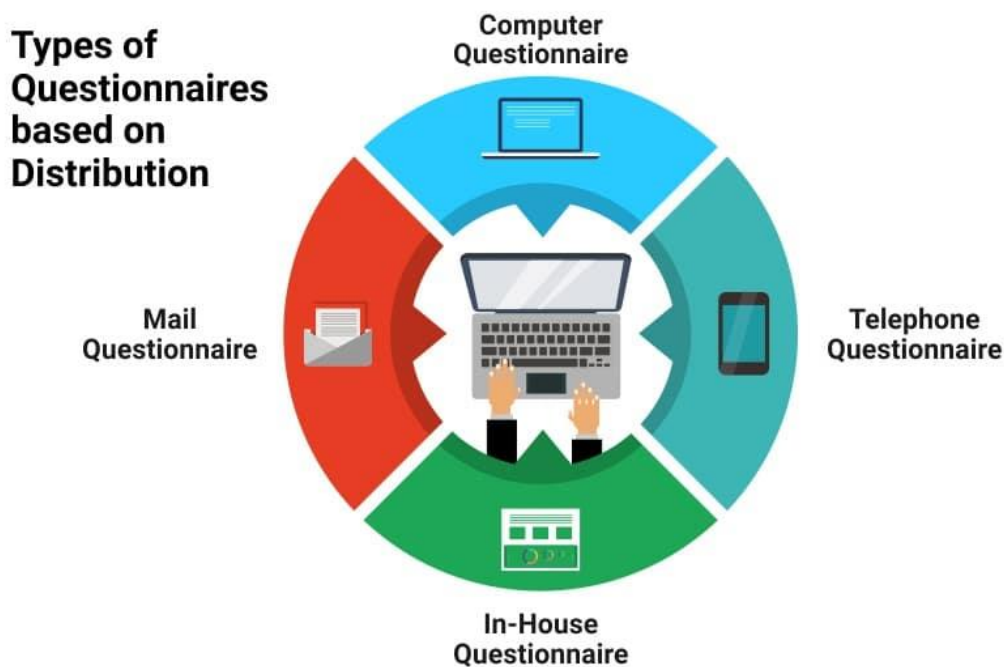
You can use multiple question types in a questionnaire. Using various question types can help increase responses to your research questionnaire as they tend to keep participants more engaged. The best customer satisfaction survey templates are the most commonly used for better insights and decision-making.

Some of the widely used types of questions are:

- **Open-Ended Questions:** Open-ended questions help collect qualitative data in a questionnaire where the respondent can answer in a free form with little to no restrictions.
- **Dichotomous Questions:** The dichotomous question is generally a “yes/no” close-ended question. This question is usually used in case of the need for necessary validation. It is the most natural form of a questionnaire.

- **Multiple-Choice Questions:** Multiple-choice questions are a close-ended question type in which a respondent has to select one (single-select multiple-choice question) or many (multi-select multiple choice question) responses from a given list of options. The multiple-choice question consists of an incomplete stem (question), right answer or answers, incorrect answers, close alternatives, and distractors. Of course, not all multiple-choice questions have all of the answer types. For example, you probably won't have the wrong or right answers if you're looking for customer opinion.
- **Scaling Questions:** These questions are based on the principles of the four measurement scales – nominal, ordinal, interval, and ratio. A few of the question types that utilize these scales' fundamental properties are rank order questions, Likert scale questions, semantic differential scale questions, and Stapel scale questions.
- **Pictorial Questions:** This question type is easy to use and encourages respondents to answer. It works similarly to a multiple-choice question. Respondents are asked a question, and the answer choices are images. This helps respondents choose an answer quickly without overthinking their answers, giving you more accurate data.

Types of Questionnaires



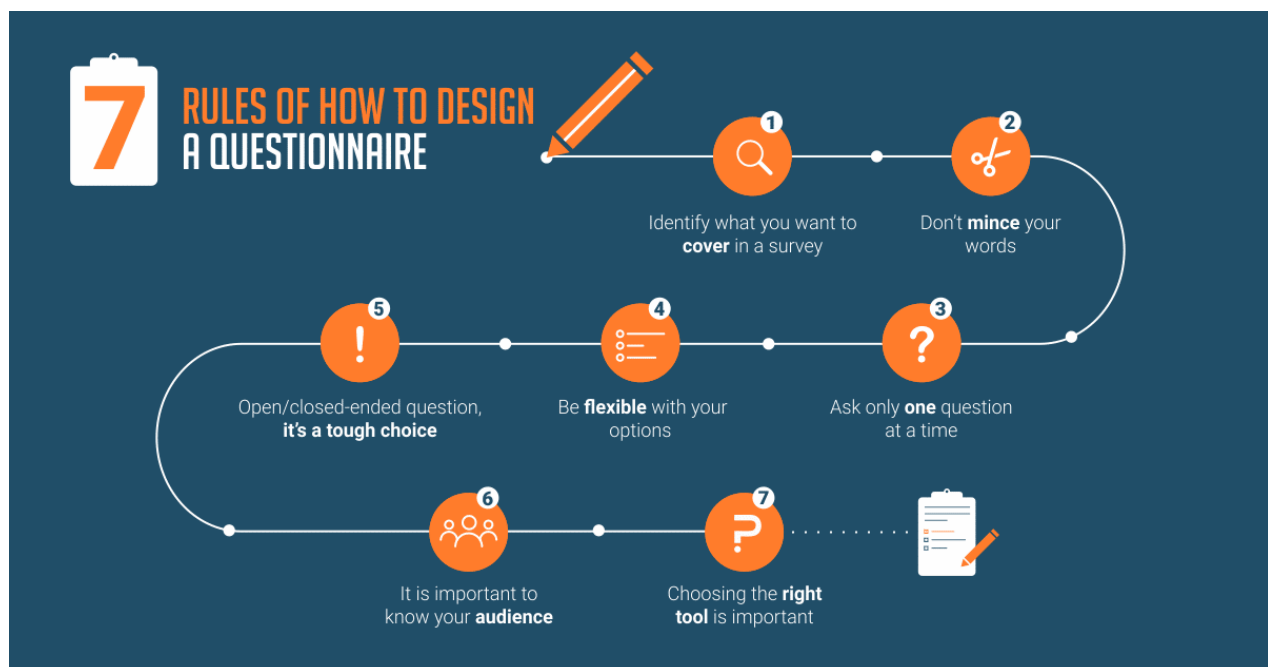
Questionnaires can be administered or distributed in the following forms:

- **Online Questionnaire:** In this type, respondents are sent the questionnaire via email or other online mediums. This method is generally cost-effective and time-efficient. Respondents can

also answer at leisure. Without the pressure to respond immediately, responses may be more accurate. The disadvantage, however, is that respondents can easily ignore these questionnaires. Read more about online surveys.

- **Telephone Questionnaire:** A researcher makes a phone call to a respondent to collect responses directly. Responses are quick once you have a respondent on the phone. However, a lot of times, the respondents hesitate to give out much information over the phone. It is also an expensive way of conducting research. You're usually not able to collect as many responses as other types of questionnaires, so your sample may not represent the broader population.
- **In-House Questionnaire:** This type is used by a researcher who visits the respondent's home or workplace. The advantage of this method is that the respondent is in a comfortable and natural environment, and in-depth data can be collected. The disadvantage, though, is that it is expensive and slow to conduct.
- **Mail Questionnaire:** These are starting to be obsolete but are still being used in some market research studies. This method involves a researcher sending a physical data collection questionnaire request to a respondent that can be filled in and sent back. The advantage of this method is that respondents can complete this on their own time to answer truthfully and entirely. The disadvantage is that this method is expensive and time-consuming. There is also a high risk of not collecting enough responses to make actionable insights from the data.

How to design a Questionnaire



Questionnaire design is a multistep process that requires attention to detail at every step.

Researchers are always hoping that the responses received for a survey questionnaire yield useable data. If the questionnaire is too complicated, there is a fair chance that the respondent might get confused and will drop out or answer inaccurately.

As a survey creator, you may want to pre-test the survey by administering it to a focus group during development. You can try out a few different questionnaire designs to determine which resonates best with your target audience. Pre-testing is a good practice as the survey creator can comprehend the initial stages if there are any changes required in the survey.

Steps Involved in Questionnaire Design

1. Identify the scope of your research:

Think about what your questionnaire is going to include before you start designing the look of it. The clarity of the topic is of utmost importance as this is the primary step in creating the questionnaire. Once you are clear on the purpose of the questionnaire, you can begin the design process.

2. Keep it simple:

The words or phrases you use while writing the questionnaire must be easy to understand. If the questions are unclear, the respondents may simply choose any answer and skew the data you collect.

3. Ask only one question at a time:

At times, a researcher may be tempted to add two similar questions. This might seem like an excellent way to consolidate answers to related issues, but it can confuse your respondents or lead to inaccurate data. If any of your questions contain the word “and,” take another look. This question likely has two parts, which can affect the quality of your data.

4. Be flexible with your options:

While designing, the survey creator needs to be flexible in terms of “option choice” for the respondents. Sometimes the respondents may not necessarily want to choose from the answer options provided by the survey creator. An “other” option often helps keep respondents engaged in the survey.

5. The open-ended or closed-ended question is a tough choice:

The survey creator might end up in a situation where they need to make distinct choices between open or close-ended questions. The question type should be carefully chosen as it defines the tone and importance of asking the question in the first place.

If the questionnaire requires the respondents to elaborate on their thoughts, an open-ended question is the best choice. If the surveyor wants a specific response, then close-ended questions should be their primary choice. The key to asking closed-ended questions is to generate data that is easy to analyze and spot trends.

6. It is essential to know your audience:

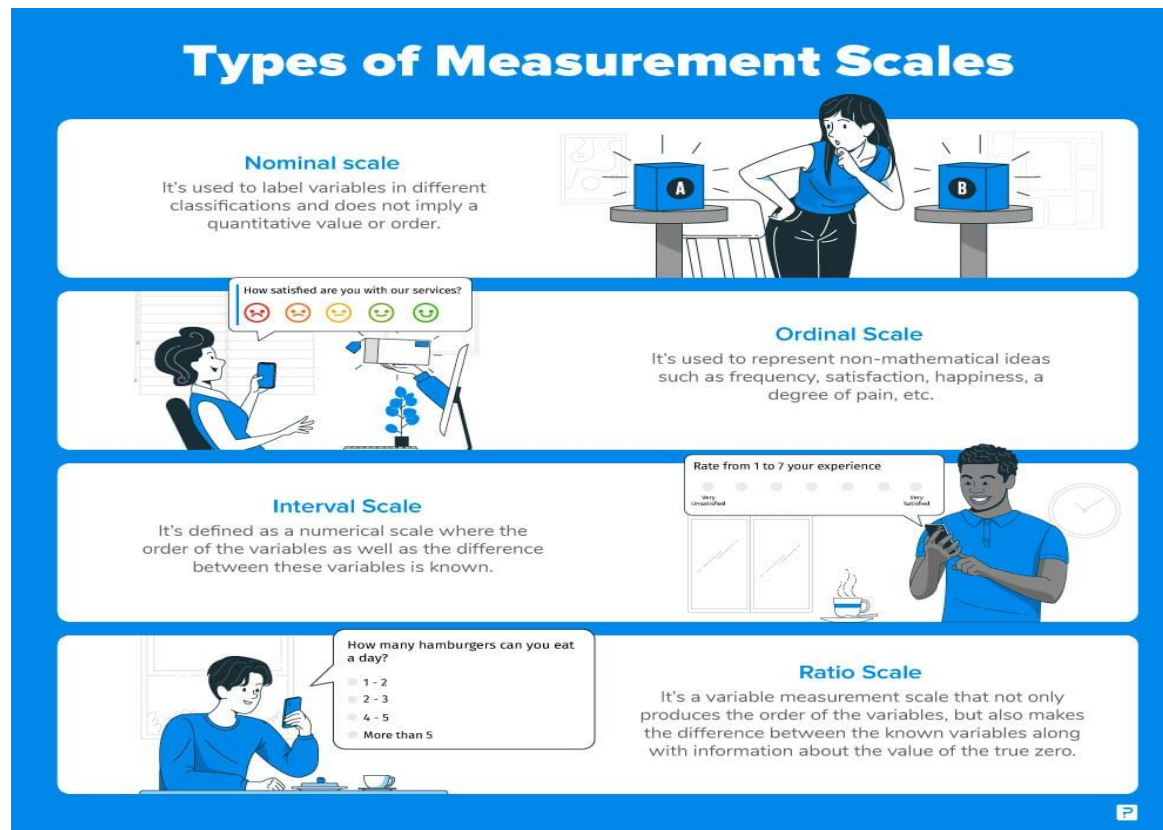
A researcher should know their target audience. For example, if the target audience speaks mostly Spanish, sending the questionnaire in any other language would lower the response rate and accuracy of data. Something that may seem clear to you may be confusing to your respondents. Use simple language and terminology that your respondents will understand, and avoid technical jargon and industry-specific language that might confuse your respondents.

For efficient market research, researchers need a representative sample collected using one of the many sampling techniques, such as a sample questionnaire. It is imperative to plan and define these target respondents based on the demographics required.

7. Choosing the right tool is essential:

Always save personal questions for last. Sensitive questions may cause respondents to drop off before completing. If these questions are at the end, the respondent has had time to become more comfortable with the interview and are more likely to answer personal or demographic questions.

Measurement and Scaling Techniques



Nominal Scale: 1st Level of Measurement

Nominal Scale, also called the categorical variable scale, is defined as a scale that labels variables into distinct classifications and doesn't involve a quantitative value or order. This scale is the simplest of the four variable measurement scales. Calculations done on these variables will be futile as the options have no numerical value.

There are cases where this scale is used for the purpose of classification – the numbers associated with variables of this scale are only tags for categorization or division. Calculations done on these numbers will be futile as they have no quantitative significance.

For a question such as:

Where do you live?

- 1- Suburbs

- 2- City
- 3- Town

Nominal scale is often used in research surveys and questionnaires where only variable labels hold significance.

For instance, a customer survey asking “Which brand of smartphones do you prefer?”

Options : “Apple”- 1 , “Samsung”-2, “OnePlus”-3.

- In this survey question, only the names of the brands are significant for the researcher conducting consumer research or netnography. There is no need for any specific order for these brands. However, while capturing nominal data, researchers conduct analysis based on the associated labels.
- In the above example, when a survey respondent selects Apple as their preferred brand, the data entered and associated will be “1”. This helped in quantifying and answering the final question – How many respondents selected Apple, how many selected Samsung, and how many went for OnePlus – and which one is the highest.
- This is the fundamental of quantitative research, and nominal scale is the most fundamental research scale.

Nominal Scale Data and Analysis

There are two primary ways in which nominal scale data can be collected:

1. By asking an open-ended question, the answers of which can be coded to a respective number of label decided by the researcher.
2. The other alternative to collect nominal data is to include a multiple choice question in which the answers will be labeled.

In both cases, the analysis of gathered data will happen using percentages or mode, i.e., the most common answer received for the question. It is possible for a single question to have more than one mode as it is possible for two common favorites can exist in a target population.

Nominal Scale Examples

- Gender
- Political preferences
- Place of residence

What is your Gender?	What is your Political preference?	Where do you live?
• M- Male • F- Female	• 1- Independent • 2- Democrat • 3- Republican	• 1- Suburbs • 2- City • 3- Town

Ordinal Scale: 2nd Level of Measurement

Ordinal Scale is defined as a variable measurement scale used to simply depict the order of variables and not the difference between each of the variables. These scales are generally used to depict non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc. It is quite straightforward to remember the implementation of this scale as ‘Ordinal’ sounds similar to ‘Order’, which is exactly the purpose of this scale.

Ordinal Scale maintains descriptive qualities along with an intrinsic order but is void of an origin of scale and thus, the distance between variables can’t be calculated. Descriptive qualities indicate tagging properties similar to the nominal scale, in addition to which, the ordinal scale also has a relative position of variables. Origin of this scale is absent due to which there is no fixed start or “true zero”.

Ordinal Data and Analysis

Ordinal scale data can be presented in tabular or graphical formats for a researcher to conduct a convenient analysis of collected data. Also, methods such as Mann-Whitney U test and Kruskal–Wallis H test can also be used to analyze ordinal data. These methods are generally implemented to compare two or more ordinal groups.

In the Mann-Whitney U test, researchers can conclude which variable of one group is bigger or smaller than another variable of a randomly selected group. While in the Kruskal–Wallis H test, researchers can analyze whether two or more ordinal groups have the same median or not.

Ordinal Scale Examples

Status at workplace, tournament team rankings, order of product quality, and order of agreement or satisfaction are some of the most common examples of the ordinal Scale. These scales are generally used in market research to gather and evaluate relative feedback about product satisfaction, changing perceptions with product upgrades, etc.

For example, a semantic differential scale question such as:

How satisfied are you with our services?

- Very Unsatisfied – 1
- Unsatisfied – 2
- Neutral – 3
- Satisfied – 4
- Very Satisfied – 5

1. Here, the order of variables is of prime importance and so is the labeling. Very unsatisfied will always be worse than unsatisfied and satisfied will be worse than very satisfied.
2. This is where ordinal scale is a step above nominal scale – the order is relevant to the results and so is their naming.
3. Analyzing results based on the order along with the name becomes a convenient process for the researcher.
4. If they intend to obtain more information than what they would collect using a nominal scale, they can use the ordinal scale.

This scale not only assigns values to the variables but also measures the rank or order of the variables, such as:

- Grades
- Satisfaction
- Happiness

How satisfied are you with our services?

- 1- Very Unsatisfied
- 2- Unsatisfied
- 3- Neural
- 4- Satisfied
- 5- Very Satisfied

Interval Scale: 3rd Level of Measurement

Interval Scale is defined as a numerical scale where the order of the variables is known as well as the difference between these variables. Variables that have familiar, constant, and computable differences are classified using the Interval scale. It is easy to remember the primary role of this scale

too, 'Interval' indicates 'distance between two entities', which is what Interval scale helps in achieving.

These scales are effective as they open doors for the statistical analysis of provided data. Mean, median, or mode can be used to calculate the central tendency in this scale. The only drawback of this scale is that there no pre-decided starting point or a true zero value.

Interval scale contains all the properties of the ordinal scale, in addition to which, it offers a calculation of the difference between variables. The main characteristic of this scale is the equidistant difference between objects.

For instance, consider a Celsius/Fahrenheit temperature scale –

- 80 degrees is always higher than 50 degrees and the difference between these two temperatures is the same as the difference between 70 degrees and 40 degrees.
- Also, the value of 0 is arbitrary because negative values of temperature do exist – which makes the Celsius/Fahrenheit temperature scale a classic example of an interval scale.
- Interval scale is often chosen in research cases where the difference between variables is a mandate – which can't be achieved using a nominal or ordinal scale. The Interval scale quantifies the difference between two variables whereas the other two scales are solely capable of associating qualitative observation values with variables.
- The mean and median values in an ordinal scale can be evaluated, unlike the previous two scales.
- In statistics, interval scale is frequently used as a numerical value can not only be assigned to variables but calculation on the basis of those values can also be carried out.

Even if interval scales are amazing, they do not calculate the “true zero” value which is why the next scale comes into the picture.

Interval Data and Analysis

All the techniques applicable to nominal and ordinal data analysis are applicable to Interval Data as well. Apart from those techniques, there are a few analysis methods such as descriptive statistics, correlation regression analysis which is extensively for analyzing interval data.

Descriptive analysis statistics is the term given to the analysis of numerical data which helps to describe, depict, or summarize data in a meaningful manner and it helps in calculation of mean, median, and mode.

Interval Scale Examples

- There are situations where attitude scales are considered to be interval scales.
- Apart from the temperature scale, time is also a very common example of an interval scale as the values are already established, constant, and measurable.
- Calendar years and time also fall under this category of measurement scales.
- Likert scale, Net Promoter Score, Semantic Differential Scale, Bipolar Matrix Table, etc. are the most-used interval scale examples.

The following questions fall under the Interval Scale category:

- What is your family income?
- What is the temperature in your city?

Ratio Scale: 4th Level of Measurement

Ratio Scale is defined as a variable measurement scale that not only produces the order of variables but also makes the difference between variables known along with information on the value of true zero. It is calculated by assuming that the variables have an option for zero, the difference between the two variables is the same and there is a specific order between the options.

With the option of true zero, varied inferential, and descriptive analysis techniques can be applied to the variables. In addition to the fact that the ratio scale does everything that a nominal, ordinal, and interval scale can do, it can also establish the value of absolute zero. The best examples of ratio scales are weight and height. In market research, a ratio scale is used to calculate market share, annual sales, the price of an upcoming product, the number of consumers, etc.

- Ratio scale provides the most detailed information as researchers and statisticians can calculate the central tendency using statistical techniques such as mean, median, mode, and methods such as geometric mean, the coefficient of variation, or harmonic mean can also be used on this scale.
- Ratio scale accommodates the characteristic of three other variable measurement scales, i.e. labeling the variables, the significance of the order of variables, and a calculable difference between variables (which are usually equidistant).
- Because of the existence of true zero value, the ratio scale doesn't have negative values.
- To decide when to use a ratio scale, the researcher must observe whether the variables have all the characteristics of an interval scale along with the presence of the absolute zero value.
- Mean, mode and median can be calculated using the ratio scale.

Ratio Data and Analysis

At a fundamental level, Ratio scale data is quantitative in nature due to which all quantitative analysis techniques such as SWOT, TURF, Cross-tabulation, Conjoint, etc. can be used to calculate ratio data. While some techniques such as SWOT and TURF will analyze ratio data in such a manner that researchers can create roadmaps of how to improve products or services and Cross-tabulation will be useful in understanding whether new features will be helpful to the target market or not.

Ratio Scale Examples

The following questions fall under the Ratio Scale category:

- What is your daughter's current height?
 - Less than 5 feet.
 - 5 feet 1 inch – 5 feet 5 inches
 - 5 feet 6 inches- 6 feet
 - More than 6 feet
- What is your weight in kilograms?
 - Less than 50 kilograms
 - 51- 70 kilograms
 - 71- 90 kilograms
 - 91-110 kilograms
 - More than 110 kilograms

Attitude measurement scale - Likert Scale

Likert Scale Survey Question

Even those new to survey research have been exposed to the ubiquitous Likert scale. What is a Likert Scale Survey Question?

Likert scale survey question and the strongly agree to strongly disagree framework is a staple in the pantry of questions we regularly use on surveys. In the years I have been involved in constructing surveys and teaching market research I have seen many techniques and question types rise and fall in favor, yet the Likert scale holds its own.

Organizational psychologist Rensis Likert developed the measure in order to assess the level of agreement or disagreement of a symmetric agree-disagree scale. In general, a series of statements each designed to view a construct from a slightly different perspective are leveraged. The power of

this technique is that it works across disciplines – it is just as applicable to a social science construct as it is a marketing one.

For example:

Ted's Pizza has excellent customer service. You get a lot of food for the price at Ted's Pizza. People come to me for information on new products. I enjoy sharing information with others. I consider myself knowledgeable on a variety of issues.

The first two statements measure the customer's perceptions about the business. Qualitative research such as focus groups or in-depth interviews can be useful in helping to generate a list of statements. The last three statements were centered on the individual and might be part of an opinion leader or early adopter scale. Just as easily you can create a scale with items touching upon political or social topics, religion, or other important issues of the day.

To what extent do you agree or disagree with the following statements?							
	SD	2	3	4	5	6	SA
I aspire to a better life for myself.	☐	☐	☐	☐	☐	☐	☐
I do not like going into new situations	☐	☐	☐	☐	☐	☐	☐
My future looks better than my past	☐	☐	☐	☐	☐	☐	☐

Likert scales usually have five, seven, or nine points, with five and seven points used most frequently. For example – Strongly Agree, Agree, No opinion, Disagree, and Strongly Disagree. Adding “Somewhat” to both sides creates the sixth and seventh points. The scales are anchored by strongly agree and strongly disagree. There is some research that indicates having the agree side shown first could inflate the scores. This can be tested by alternating the anchor points within a survey wave and comparing scores in the data analysis stage.

Likert scale is designed to measure attitudes and by nature are multi-item. Basic research tells us that multiple-item measures of a construct are inherently more stable and subject to less random variability than single-item measures. How many items is enough? If you are creating a new scale then you should create as many items as possible and let subsequent analysis narrow the field of contenders. This can be done through brainstorming sessions, focus groups or a review of existing literature.

Semantic Differential Scale

semantic differential scale is a survey or questionnaire rating scale that asks people to rate a product, company, brand, or any 'entity' within the frames of a multi-point rating option. These survey answering options are grammatically on opposite adjectives at each end. For example, love-hate, satisfied-unsatisfied, and likely to return-unlikely to return with intermediate options in between.

Surveys or questionnaires using the semantic differential question is the most reliable way to get information on people's emotional attitude towards a topic of interest.

Charles Egerton Osgood, a famous American psychologist, invented the semantic differential scale so that this 'connotative meaning' of emotional attitude towards entities can be recorded and put to good use.

Osgood conducted this research on an extensive database and found that three scales were commonly useful, irrespective of race or culture or difference in language:

- Estimate: Combination similar to good-bad.
- Authority: Pairs on the lines of powerful-weak.
- Activeness: Combos like active-passive.

Researchers can measure a wide variety of subjects using these combinations, like customers' outlooks about an upcoming product launch or employee satisfaction.

What is the semantic differential scale?

The ease-of-understanding and the popularity it comes with it makes it extremely reliable. The data collection is accurate due to the versatility that these survey questions come with.

Researchers use the semantic differential scale questions to ask respondents to rate products, organization, or services with multi-point questions with polar adjectives at the extremes of this scale like likely/ unlikely, happy/sad, loved the service/ hated the service.

Semantic differential scale examples & question types

1. Slider rating scale: Questions that feature a graphical slider give the respondent a more interactive way to answer the semantic differential scale question.

Which of the following categories best describes your last experience purchasing a product or service on our website?

Very Pleasant Somewhat Pleasant Neither Pleasant nor Unpleasant Somewhat Unpleasant Very Unpleasant

My experience was Somewhat Pleasant



2. Non-slider rating scale: The non-slider question uses typical radio buttons for a more traditional survey look and feel. Respondents are more used to answering.

Given your most recent experience, please rate the following question on a scale of 1 to 7.

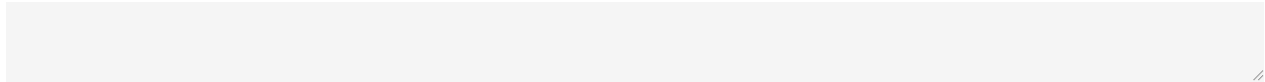
How likely are you to use our service again?

(1) Very Unlikely (2) (3) (4) (5) (6) (7) Very Likely

☐ ☐ ☐ ☐ ☐ ☐ ☐

3. Open-ended questions: These questions give the users ample freedom to express their emotions about your organization, products, or services.

Comments/Suggestions:



4. Ordering: The ordering questions offer the scope to rate the parameters that the respondents feel are best or worst according to their personal experiences.

Based on what you have seen, heard, and experienced, please rank the following brands according to their reliability. Place a "1" next to the brand that is most reliable, a "2" next to the brand that is next most reliable, and so on. Remember, no two cars can have the same ranking.

Ford	-- Select --	▼
BMW	-- Select --	▼
Chrysler	-- Select --	▼
Chevrolet	-- Select --	▼

5. Satisfaction rating: The easiest and eye-catching semantic differential scale questions are the satisfaction rating questions.

How satisfied are you with our services

Very Unsatisfied Unsatisfied Neutral Satisfied Very Satisfied

Thurstone Scale: Definition

Thurstone scale is defined as a unidimensional scale that is used to track respondent's behavior, attitude or feeling towards a subject. This scale consists of statements about a particular issue or topic where each statement has a numerical value that indicates the respondents attitude towards the topic as favorable or unfavorable. Respondents indicate the statements that they agree with, and an average is computed. A mean score of the agreements or disagreements is calculated as the attitude of the respondent towards the topic.

This scale was developed by Robert Thurstone to approximate measurement in equal-appearing interval levels. The Thurstone scale is built upon the fundamentals of the Likert scale, but this method of constructing an attitude scale not only takes value of each item of into consideration while evaluating the final attitude score but also caters to neutral items. The Guttman scale and the Bogardus social distance scale are also variations of unidimensional scales where elements can be ordered in a hierarchical manner.

There are three scales while talking about the Thurstone scale question but the most commonly used method is the equal appearing intervals and hence the scale is also called the equal-appearing interval scale. The two other are slightly more complex to develop but still result in the same agree/disagree quiz question type; they are the **method of successive intervals** and the **method of paired comparisons**.

How to conduct a Thurstone Scale Survey with an Example

An example of a Thurstone scale survey is to understand the attitude of employees in an organization towards diversity hiring in that organization. There are 2 distinct milestones in the Thurstone scale question; to derive the final questions and administer the Thurstone scale question and conduct its analysis.

Derive the Final Question

There are 5 distinctive steps to derive the final question. They are:

Step 1 – Develop statements: Develop a large number of agree/disagree statements on a certain topic. For example, if you wanted to find out people's attitudes towards the policy of diversity hiring in an organization, your statements may include:

- Policy on diversity hiring is wrong.

- Diversity hiring takes away jobs from deserving candidates.
- Diversity hiring brings different viewpoints to the team.
- Diversity hiring brings out the best in the local community.
- Diversity hiring aids in increasing the reputation of a brand

Statement	Median or Mean	IQR
43	1	1.25
21	1	1
16	1.5	1
3	3	2
6	5	2
28	6	1
37	7	3
9	7.5	1.5
18	9	2
26	11	2

Step 2 – Rank each statement with a panel of judges: The next step would be to have a panel of judges rate each item on a scale of 1 to 11 where 1 is the least favorable attitude towards the common vector – diversity hiring and 11 is an extremely favorable attitude. It is important to note that the judges are required to rate each option and not agree or disagree with them.

Step 3 – Calculate median and/or mean and Interquartile range (IQR): The data collected from all judges is to then be analyzed to draw up a table with the mean or median values in the ascending order. Using median or mean is a personal choice and the options

throw up accurate results in the use of either. If you have 50 statements, you need to have 50 means/medians and 50 IQR's.

Step 4 – Sorting the table: The data has to be sorted on the basis of smallest to largest median/mean and the IQR's for each median/mean have to be in a descending order. This can be represented as:

Step 5 – Selecting final variables or options: Select options on the basis of the above table. For example, you could select one item from each mean/median value. You want the statements with the most agreement between judges. For each median value, this is the item with the lowest interquartile range. This is a “Rule of Thumb”: you don't have to choose this item. If you decide it's poorly worded or ambiguous, choose the item above it (with the next lowest IQR).

Administering and Analysis of the Thurstone Scale Question

Once the final questions have been decided upon, they are shared with the respondents to select from agreement or disagreement. The ratings are shown in parenthesis but the rating is not shared with the actual respondents.

- **Administering and analysis by median or mean:** The question and the subsequent options can be administered to the respondents by using either the mean or the median score in the below format. The weights of the statements are summed and divided by the number of statements checked. If a respondent were to agree with statements 2, 5, 7 and 10; the attitude score is $10.5 + 2.5 + 4.5 + 6.0 = 23.5/4 = 5.8$. Dividing the number of statements puts this score at a little above the midway point of the 1-11 scale. This score indicates that the attitude is slightly favorable to diversity hiring in the organization.

Please check all those statements with which you agree.

	Agree	Disagree
1. Policy on diversity hiring is wrong (3.0)	☐	☐
2. Diversity hiring takes away jobs from deserving candidates. (10.5)	☐	☐
3. Diversity hiring brings different viewpoints to the team. (8.5)	☐	☐
4. Diversity hiring brings out the best in the local community. (1.0)	☐	☐
5. Diversity hiring aids in a increasing the reputation of a brand. (2.5)	☐	☐
6. Diversity hiring is detrimental to collaboration (5.5)	☐	☐
7. Such people are generally from the lower rungs of the society. (4.5)	☐	☐
8. Acquired diversity is more important than inherent diversity. (10.5)	☐	☐
9. Diversity hiring increases productivity in teams. (2.0)	☐	☐
10. Diversity hiring leads to interpersonal issues. (6.0)	☐	☐
11. Diversity hiring diminishes discriminatory behavior. (9.5)	☐	☐

- **Administering and analysis by simple count or percentage:** In the same example above, if the question is administered without the mean or median score, the calculation can be depicted by a simple count of agreement on the 1-11 scale or a percentage. If the respondent were to agree with statements 1, 4, 5, 6, 8, 9 and 11; the agreement count is 7 out of 11 which puts the percentage at 63.63% which means that the attitude is favorable towards diversity hiring.

Characteristics of a Thurstone Scale Question

Some distinct characteristics of a Thurstone scale question are:

- **They are two stepped:** A Thurstone scale question is never administered at the first iteration stage without the rating by judges. This is an important characteristic of this question as the options that a respondent sees are weighted and there is consensus of them being in the survey.

- **Mean or median is always calculated:** Since each option is weighted, the mean or median is calculated for each option. This also forms the basis of the selection method for use in the final survey.
- **Only agree or disagree options:** The respondent makes a selection only on the basis of the agreement or disagreement with the statement.

Use of a Thurstone Scale Survey

The Thurstone scale survey is used to measure the respondents attitudes on a particular subject. The scale can be applied to a wide range of market research surveys, in market research including:

- **Surveys that measure opinions:** The Thurstone scale question produces quantifiable data about the measures of strength of the respondents opinions.
- **Those that gauge attitudes or feelings:** This scale is used effectively in customer satisfaction to predict future purchasing trends and in employee engagement to calculate turnover.

Multi-Dimensional Scaling: Non comparative scaling techniques

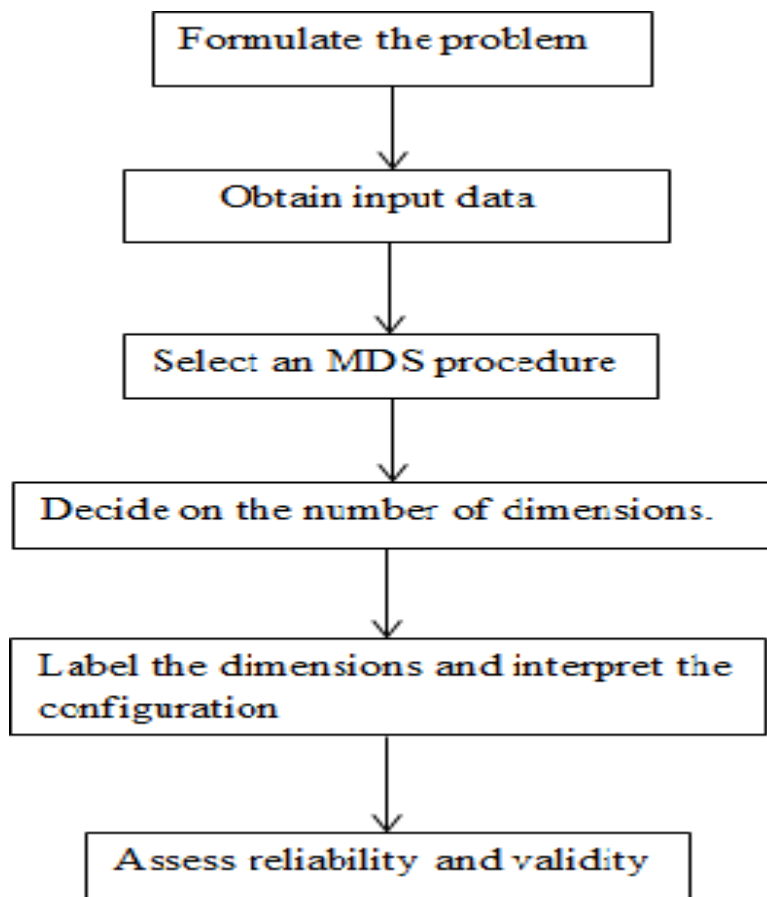
Multidimensional Scaling (MDS) is a class of procedures for representing perception and preferences of respondents spatially by means of a visual display. Perceived or psychological relationships among points in a multidimensional space. These geometric representations are often called spatial maps. The axes of the spatial map are assumed to denote the psychological bases or underlying dimensions respondents use to form perceptions and preferences for stimuli. MDS has been used in marketing to identify:

1. The number and nature of dimensions consumers use to perceive different brands in the marketplace
2. The positioning of current brands on these dimensions
3. The positioning of consumers ideal brand on these dimensions

Information by MDS has been used for a variety of marketing applications, including:

- i. Image measurement. Compare the customers' and noncustomers' perceptions of the firm with the firm's perceptions of itself and thus identify perceptual gaps.
- ii. Market segmentation. Position brands and consumers in the same space and thus identify groups of consumers with relatively homogeneous perceptions.
- iii. New product development. To look for gaps in the spatial map, which indicate potential opportunities for positioning new products. Also, to evaluate new product concepts and existing brands on attest basis to determine how consumers perceive the new concepts. The proportion of preferences for each new product is one indicator of its success.

- iv. Assessing advertising effectiveness. Spatial maps can be used to determine whether advertising has been successful in achieving the desired brand positioning.
- v. Pricing analysis. Spatial maps developed with and without pricing information can be compared to determine the impact of pricing.
- vi. Channel decisions. Judgments on compatibility of brands with different retail outlets could lead to spatial maps useful for making channel decisions.
- vii. Attitude scale construction. MDS techniques can be used to develop the appropriate dimensionality and configuration of the attitude space.



Formulate the problem

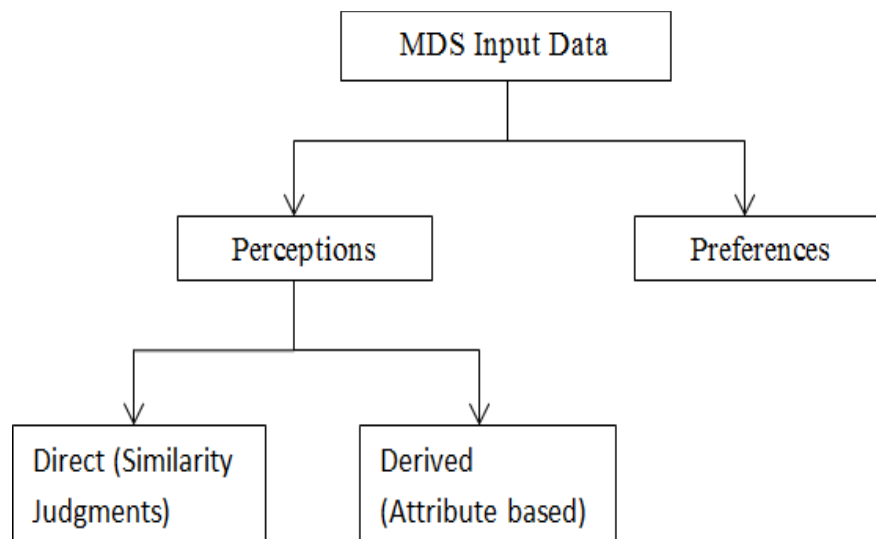
Formulating the problem requires that the researcher specify the purpose for which the MDS results would be used and select the brand or other stimuli to be included in analysis. The number of brand or stimuli selected and the specific brands included determine the nature of the resulting dimensions and configurations. At a minimum, eight brands or stimuli should be included so as to

obtain a well-defined spatial map. Including more than 25 brands is likely to be cumbersome and may result in respondent fatigue.

The decision regarding which specific brands or stimuli to include should be made carefully. Suppose a researcher is interested in obtaining consumer perceptions of automobiles. If luxury automobiles are not included in the stimuli set, this dimension may not emerge in the results. The choice of number and specific brand and stimuli to be included should be based on the statement of marketing research problem, theory and the judgments of researcher.

Obtain input data

As shown Figure 2, input data obtained from the respondents may be related to perception or preferences. Perception data, which may be direct or derived, is discussed first.



Input data for Multi-dimensional Scaling

Perception data: Direct Approaches.

In direct approaches to gathering perception data, the respondents are asked to judge how similar or dissimilar the various brands or stimuli are using their own criteria. Respondents are often required to rate all possible pair of brands or stimuli in terms of similarity on a Likert scale. These data are referred to as similarity judgments. For example, similarity judgment on all possible pairs of toothpaste brands may be obtained in the following manner.

Perception Data : Derived Approaches, Derived approaches to collect perception data are attribute-based approaches requiring the respondent to rate the brand or stimuli on the identified attributes

using semantic differential or Likert scale example, the different brands of toothpastes may be rated on attributes such as these

Sometimes an ideal brand is also included in the stimulus set. The respondent asked to evaluate their hypothetical ideal brand on the same set of attribute. If attribute rating are obtained, a similarity measure (such as Euclidean distance) is derived for pair of brand.

Direct vs. Derived Approaches:

Direct approaches have advantages thus researcher does not have to identify a set of salient attributes. Respondent makes stimuli judgment using their own criteria, as they would under normal circumstances. Disadvantages is that the criteria are influenced by the brands and stimuli being evaluated. If the various brands of automobiles being evaluated are in the same price range, then price will not emerge not as an important factor. It may be difficult to determine before analyze and how the individual respondent judgment should be combined. Furthermore, it is difficult to label the dimension of the spatial map.

The advantage of the attribute-based approach is that it is easy to identify respondent with homogeneous perceptions. The respondent can be clustered the advantage of the dimension based on rating. It is also easier to label the dimension. A disadvantage is that

the researcher identifies all salient attributes, a different task. The spatial map obtained depends on the attributes identified.

Preference data: Preference data order the brand or stimuli in terms of respondent preference for some property. A common way in which such data are obtained is preference rating. Respondents are required to rank the brand from the most preferred to the least preferred. Alternatively, respondent may be required to make paired comparison and indicate which brand in a pair they prefer. Another method is to obtain preference rating for the various brands. When spatial maps are based on preference data, distance implies difference in preference. The configuration derived from preference data may differ greatly from that obtained from similarity data. Two brands may be perceived as different in similarity map yet similar in a preference map and vice versa. For example, Crest and Pepsodent may be perceived by a group of respondents as very different brands and thus appear far apart on a perception map. However these two brands may be about equally preferred and appear close together on a preference map.

Select an MDS Procedure

Selection of a specific MDS procedure depends upon whether perception or preference data are being scaled, or whether the analysis requires both kinds of data. The nature of the input data is also a determining factor. Nonmetric MDS procedures assume that the input data are ordinal, but they result in the metric output. The distances in the resulting spatial map may be assumed to be interval scaled. These procedures find, in a given dimensionality, a spatial map whose rank orders of

estimated distances brands and stimuli best preserve or reproduce the input rank orders. In contrast, Metric MDS methods assume that input data are metric. Because the output is also metric, a stronger relationship between the output and input data is maintained, and the metric (interval or ratio) qualities of the input data are preserved. The metric and non-metric methods produce similar results.

Another factor influencing the selection of a procedure is whether the MDS analysis will be conducted at the individual respondent level or at an aggregate level. In individual level analysis, the data is analyzed separately for each respondent, resulting in a spatial map for each respondent. Although individual level analysis is useful from a research perspective, it is not appealing from managerial standpoint. Marketing strategies are typically formulated at the segment or aggregate level, rather than at the individual level. If the aggregate-level analysis is conducted, some assumption must be made in aggregating individual data. Typically it is assumed that all respondents use the same dimensions to evaluate the brand or stimuli, but that different respondents weight these common dimensions differentially.

Label the Dimensions and Interpret the Configuration

Once a spatial map is developed, the dimensions must be labeled and the configuration interpreted. Labeling the dimensions requires subjective judgment on the part of the researcher. The following guidelines can assist in this task:

1. Even if direct similarity judgments are obtained, ratings of the brands or researcher supplied attributes may still be collected. The axes may then be labeled for the attributes with which they are most closely aligned.
2. After providing direct similarities or preference data, the respondents may be asked to indicate the criteria used in making their evaluations. These criteria may then be subjectively related to the spatial map to label the dimensions.
3. If possible, the respondents can be shown their spatial maps and asked to label the dimensions by inspecting the configurations.
4. If objective characteristics of the brands are available (e.g., horsepower or miles per gallon for automobiles), these could be used as an aid in interpreting the subjective dimensions of spatial maps.

Assess Reliabilities and Validity

The input data, and consequently the MDS solutions, are invariably subject to sub random variability. Hence, it is necessary that some assessment be made of the relation and validity of MDS solutions. The following guidelines are suggested.

1. The index of fit, or R -square, should be examined. This is squared correlation index that indicates the proportional of variance of the optimally scaled data that can be accounted for the MDS procedure. Thus, it indicates how well the MDS model fits the input data. Although higher values of R -square are desirable, values of 0.60 or better are considered acceptable.

2. Stress values are also indicative of the quality of MDS solutions. Whereas *R*-square is a measure of goodness of fit, stress measures badness of fit, or the proportion of variance of the optimally scaled data that is not accounted for the MDS model. Stress values vary with the type of MDS procedure and the data being analyzed. For Kruskal's stress formula 1, the recommendations for evaluating stress values are as follows.

Stress (%)	Goodness of fit
20	poor
10	fair
5	good
2.5	excellent
0	perfect

3. If an aggregate-level analysis has been done, the original data should be split into two or more parts. MDS analysis should be conducted separately on each part and the results compared.
4. Stimuli can be selectively from the input data and the solutions determined for the remaining stimuli.
5. A random error term could be added to the input data. The resulting data are subjected to MDS analysis and the solutions compared.
6. The input data could be collected at two different points in time and the test-retest reliability different.

Assumption and limitations of MDS

It is worthwhile to point out some assumptions and limitations of MDS. It is assumed that the similarity of stimulus A to B is the same as the similarity of stimulus B to A there are some instances where the assumption may be violated. For example, Mexico is perceived as more similar to United States than the United States is to Mexico. MDS assumes that the distance (similarity) between two stimuli is some function of their partial similarities on each of several perceptual dimensions. Not much research has been done to test this assumption. When a spatial map is obtained, it is assured that inter point distances are ratio scaled and that the axes of the map are multi dimensionally interval scaled. A limitation of MDS is that dimension interpretation relating physical changes in brands or stimuli to changes in the perceptual map is difficult at best.

Module-5

Data Analysis and Report Writing: Editing, Coding, Classification, Tabulation, Validation. Analysis and Interpretation, Report writing and presentation of results, Importance of report writing, types of research reports, Report structure, Guidelines for effective documentation.

Data Analysis and Report Writing: Editing

The data which has been collected from various primary/secondary sources is RAW in nature, this means that there are likely chances of errors and inconsistencies in it.

Since data collected is of pivotal importance to policy and decision makers everywhere be it governmental departments, business organizations, health or educational institutions etc it would be better to have a team of experts at hand who know how to scrutinize, review and **edit** this data before it is finally fed into the data bases and the required statistics are generated.

How to define data editing?

The process through which the data is reviewed to check for consistency, adequacy, detect errors and outliers (values that are either too big or too small from the rest of the data) and the correction of errors within the data in order to maximize its usefulness for the purpose for which it was collected is called **data editing**.

Purposes and objectives of data editing

The basic purpose served by data editing is that it improves the quality, accuracy and adequacy of the collected data thereby making it more suitable for the purpose for which the data was collected. The following can therefore be identified as the main objectives of the data editing process:

- Detection of errors in the data that otherwise affect the validity of outputs.
- Validation of data for the purposes it was collected.
- Provision of information that that would help assess the overall level of accuracy of the data.
- Detection and identification of any inconsistencies in the data and outliers and to make adjustments for them.

Types of data editing

There are different types of data editing

- **Validity and completeness of data:** refers to correctness and completeness of obtained responses. This helps ensure that there are no missing values or empty fields in the data bases.
- **Range:** verifies that data within a field fall between the boundaries specified for the particular field.

- **Duplicate data entry:** this helps ensure that there is no repetition or duplication of data and each unit on the data base or register was filled only once.
- **Logical consistency:** through this type of editing connections between data fields or variables are taken into account.
- **Outliers:** this type of editing helps detect values that are too extreme or unusual so that they can be verified and checked.

Stages of data editing

The data editing process can be divided into three parts or stages

- The first stage is the stage where you set the rules for editing. This stage is further subdivided into two steps.

In step one, you provide instructions to desk editors who then check the data for coherence and consistency.

In step two, you set the rules by establishing logical relations between the variables according to various criteria. This set of rules is called automated validation rules and this type of editing seeks to detect errors during data entry and to screen them.

- The **manual desk editing** stage is a traditional method that is put into effect by a specialized editing team. The data, (if) on paper is checked after the data has been collected and before it is fed into the data bases. If however, electronic means have been used to collect the data, the forms entered into the database are revised individually.
- The **automated data editing** method makes use of computer programs and systems for checking the data all at once after it has been entered electronically. These programs and systems contain **Audit rules** which validate the data, detect errors and determine unacceptable responses.

Limitations to data editing

There are certain factors that can influence or pose limitations to the data editing process, these can be summed up as follows;

- Data editing can be influenced by the amount of time available, the budget, the presence or absence of other resources and also by the group of people involved in the editing process.
- The available computer software programs.
- Follow up with the respondents is of critical importance in the data editing process because they are often the best source of information in many cases. However, the respondents might feel this to be stressful and burdensome thereby causing limitations to the data editing process.

- Some types of data do not require extensive editing, therefore it would be better to keep in mind the intended uses of data and make sure that the more important part of data is kept free from all errors. In this way, the intended use of data does play an important role in influencing the data editing process.
- What you need to do is to establish the methods and procedures that must be followed while correcting or handling the data errors, in the survey plan, right at the start of the project otherwise the process would be of no or little use.
- Also remember that if you plan to edit your data manually, you must develop and document the methods that are to be followed. Your team must be trained, a method must be established to check their work progress and the impact of the edits on the original data must also be assessed.
- In case of automated editing, you would need to develop and document the rules for editing. You might also need to develop a software or customize an existing computer program as per your data editing demands or requirements.

General guidelines for data editing

There are certain things that must be kept in mind when editing the data, they are as follows;

- Who should make or set the editing rules? The answer to this question would be that such rules should be made by professionals who are **experts** in data collection, questionnaire design and analysis.
- The editing rules need to be consistent and free from any contradictions.
- When setting the editing rules, it must be established whether the variable is qualitative or quantitative because the rules for editing either one are different from the other.
- Give enough time to each of the various stages of the process, that is, data collection entry and analysis and at the end of each make a quick check to see that all the necessary edits have been made and that there are no empty places within the questionnaire form.
- The questionnaire must be edited in full during the early stages of editing. If however it is found that some errors remain, a sample of forms should be subjected to re editing. The size of the sample is determined according to the expected number of the remaining errors.
- You also need to re run the desk editing stage to ensure that the data is almost free from all errors.
- The questionnaire must be subjected to desk editing and also to automated rules within the built in data editing/computer software programs.

Coding

Introduction

When a researcher has completed collecting information or data, this information is ready to be processed and analyzed. Quantitative data is information that is measurable and focuses on numerical values, unlike qualitative data which is more descriptive. During the data processing step, the collected data is transformed into a form that is appropriate to manipulate and analyze.

The process in which raw data is transformed into a standardized form suitable for machine processing and analysis is called coding. In other words, coding is the act of assigning numerical values to a set of data in order to make the analysis simpler. Coding can be used to quantify both manifest content i.e. the tangible or concrete surface content (data), and latent content i.e. the underlying meaning behind this information. The difference between manifest content and latent content is very important when it comes to survey research.

It is advisable to do a pilot or a pretest of the instrument of data collection as it would help uncover the potential problems with the study and accordingly help make changes in the tool. It will also give the researcher an idea of how the data will look. On the basis of this, the researcher can work out the layout of the codebook keeping in mind the responses collected for each variable, guiding him to provide enough variables to capture all the richness, complexity, and variety of data that has been collected. Depending on what shape the data comes in, the researcher will have to decide how to code this information, with the help of one, two, or multiple variables.

Data Coding

Though the preparation of a codebook commences prior to actual data collection, after designing the instrument and pre testing it, data coding as a step in the research process takes place after the completion of data collection, simultaneously during data entry. It is important to keep the following points in mind during coding of data:

- **Identification variables:** - Unique identification of the respondent/ questionnaire/ response sheet is extremely important as it helps the researcher verify and check data. The identification variable is a unique number corresponding to every respondent which has to be accommodated in a special field at the beginning of each record. For example, 001, 002, etc. may be used as identification variables
- **Code categories:-** Code categories should be mutually exclusive, exhaustive, and precisely defined. Each interview response should fit into one and only one category. Ambiguity will cause coding difficulties and problems with the interpretation of the data also. An example of this would be while recording literacy levels of youth in a slum community, the coding should include not just those who have gone through the formal system of education, but also those who have participated in non-formal education programs and are therefore literate.

- **Preserving original information:-** Data once coded is retained and becomes final; hence it is important to code as much detail as possible by recording the original data rather than collapsing or bracketing the information. With original or detailed data, the research analyst can determine other meaningful relationships between variables beyond those which are selected primarily or restricted by the entering or coding data. Hence, occupation of women in a slum community being surveyed could include the home based small enterprises (involved in small scale business like making snacks, Knick knacks, etc.) each as a separate code.
- **Closed-ended questions:-** Responses to survey questions that are pre coded in the questionnaire should retain this coding scheme in the machine-readable data to avoid errors and confusion. For example, in the above mentioned study in case the women respondents may be pre divided as per their age and marital status, this pre coding should be retained.
- **Open-ended questions:-** For open-ended items, investigators can either use a predetermined coding scheme or review the initial survey responses to construct a coding scheme based on major categories that emerge. Any coding scheme and its derivation should be reported in study documentation. Increasingly, investigators submit the full verbatim text of responses to open-ended questions to archives so that users can code these responses themselves. However, such responses may contain sensitive information and may involve the risk of identification; they must therefore be reviewed prior to disclosure.
- **Check-coding:-** Check-coding provides an important means of quality control in the coding process. In this process some cases are repeated with an independent coder in order to verify the coding assigned and rule out discrepancies and ambiguities if any.
- **Series of responses:-** If a series of responses requires more than one field, organizing the responses into meaningful major classifications becomes helpful. Responses within each major category are assigned the same first digit. Secondary digits can distinguish specific responses within the major categories. Such a coding scheme permits analysis of the data using broad groupings or more detailed categories.

Dealing with Missing Data

There are various situations wherein the data may be missing and each of these would need to be coded differently. Some of these situations are listed below:

- **Refusal to answer or No response** - In such a scenario the respondent explicitly refuses to answer a question or does not answer it when he or she should have. This may be with regard to questions that may be deemed personal and sensitive or too private by the respondent or even in a case where there was lack of clarity with regard to a particular question.
- **Don't know responses** - In this case the respondent was unable to answer a question, either because he or she had no opinion or because the required information was not available (e.g., a respondent could not provide family income for the previous year).

- **Processing error** - In this case, there is no answer to the question, although the subject provided one. This would indicate an error on the part of the interviewer, incorrect coding or other such problems, despite the respondent providing a response to the particular question.
- **Not applicable** - The subject was never asked a question for a particular reason. While this may be a result of skip patterns following filter questions, for example, subjects who are not married are not asked questions pertaining to their children. Other examples of inapplicability are sets of items asked only of random subsamples and those asked of one member of a household but not another.
- **No match** - This situation arises when data are drawn from different sources (for example, a survey questionnaire and an administrative database), and information from one source cannot be located
- **No data available** - The question should have been asked to the respondent, but for a reason other than those listed above, no answer was given or recorded. This may be due to an error on the part of the interviewer.

Classification,

Meaning of Classification of Data

- It is the process of arranging data into homogeneous (similar) groups according to their common characteristics.
- Raw data cannot be easily understood, and it is not fit for further analysis and interpretation. Arrangement of data helps users in comparison and analysis.
- For example, the population of a town can be grouped according to sex, age, marital status, etc.

Definition of classification given by Professor. Secrist - “Classification is the process of arranging data into sequences according to their common characteristics or separating them into different related parts.”

Classification of data

The method of arranging data into homogeneous classes according to the common features present in the data is known as classification.

A planned data analysis system makes the fundamental data easy to find and recover. This can be of particular interest for legal discovery, risk management, and compliance. Written methods and sets of guidelines for data classification should determine what levels and measures the company will use to organise data and define the roles of employees within the business regarding input stewardship.

Once a data -classification scheme has been designed, the security standards that stipulate proper approaching practices for each division and the storage criteria that determines the data's lifecycle demands should be discussed.

Objectives of Data Classification

The primary objectives of data classification are:

- To consolidate the volume of data in such a way that similarities and differences can be quickly understood. Figures can consequently be ordered in sections with common traits.
- To aid comparison.
- To point out the important characteristics of the data at a flash.
- To give importance to the prominent data collected while separating the optional elements.
- To allow a statistical method of the materials gathered.

Tabulation

What is the meaning of Tabulation?

Tabulation is the systematic and logical representation of figures in rows and columns to ease comparison and statistical analysis. It eases comparison by bringing related information closer to each other and helps further in statistical research and interpretation. In other words, tabulation is a method of arranging or organizing data in a tabular form. The tabulation process may be simple or complex depending upon the type of categorization.

Tabulation Definition

Tabulation is defined as the process of placing classified data in tabular form. A table is a systematic arrangement of statistical information in rows and columns. The rows of a table are the horizontal arrangement of data whereas the columns of a table are the vertical arrangement of data.

What are the Essential Parts of a Table?

To tabulate data correctly, one must learn about the eight essential parts of a table. These are as follows –

1. Table Number –

This is the first part of a table and is given on top of any table to facilitate easy identification and for further reference.

2. Title of the Table –

One of the most important parts of any table is its title. The title is either placed just below the table number or at its right. It is imperative for the title to be brief, crisp and carefully-worded to describe the tables' contents effectively.

3. Headnote –

The headnote of a table is presented in the portion just below the title. It provides information about the unit of data in the table, like “amount in Rupees” or “quantity in kilograms”, etc.

4. Column Headings or Captions –

The headings of the columns are referred to as the caption. It consists of one or more column heads. A caption should be brief, short, and self-explanatory. Column heading is written in the middle of a column in small letters.

5. Row Headings or Stubs –

The title of each horizontal row is called a stub.

6. Body of a Table –

This is the portion that contains the numeric information collected from investigated facts. The data in the body is presented in rows which are read horizontally from left to right and in columns, read vertically from top to bottom.

7. Footnote –

Given at the bottom of a table above the source note, a footnote is used to state any fact that is not clear from the table's title, headings, caption or stub. For instance, if a table represents the profit earned by a company, a footnote can be used to state if said profit is earned before, or after tax calculations.

8. Source Note –

As its name suggests, a source note refers to the source from where the table's information has been collected.

Tabulation of Data: Illustration

Following is a tabulation of data to represent the total number of boys and girls in classes V, VI and VII in XYZ School.

Table Number – 1**Gender-Wise Distribution of Students in Classes- V, VI and VII of XYZ School
(Head Note)**

Gender	V	VI	VII	Total
Boys	50	60	65	175
Girls	45	50	60	155
Total	95	110	125	330

Types of Tabulation**1. Simple Tabulation or One-way Tabulation**

When the data in the table are tabulated to one characteristic, it is termed as a simple tabulation or one-way tabulation.

For example, Data tabulation of all the people of the World is classified according to one single characteristic like religion.

2. Double Tabulation or Two-way Tabulation

When the data in the table are tabulated considering two different characteristics at a time, then it is defined as a double tabulation or two-way tabulation.

For example, Data tabulation of all the people of the World is classified by two different characteristics like religion and sex.

3. Complex Tabulation

When the data in the table are tabulated according to many characteristics, it is referred to as a complex tabulation.

For example, Data tabulation of all the people of the World is classified by three or more characteristics like religion, sex, and literacy, etc.

What are the Objectives of Tabulation?

Tabulation essentially bridges the gap between the collection of data and analysing them. The primary objectives of tabulation are given below –

- **For Simplification of Complex Data –**

When any information is tabulated, the volume of raw data is compressed and presented in a much more simplified manner. This facilitates easy comprehension and analysis of previously complex data.

- **To Highlight Important Information –**

Representing any data in tabular form increases the scope to highlight important information. Since data is presented in a concise manner without any textual explanation, any crucial information is automatically highlighted without difficulty.

- **To Enable Easy Comparison –**

When data is presented in an orderly fashion in rows and columns, it becomes easier to compare between them on the basis of several parameters. For example, it becomes easier to determine the month when a country has received the maximum amount of rainfall if the data is presented in a table. Otherwise, there always remains room for making a mistake in processing the data correctly.

- **To Help in the Statistical Analysis of Data –**

Statistical analysis involves computing correlation, average, dispersion, etc. of data. When information is presented in an organised manner in a table, statistical analysis becomes a lot simpler.

- **Saves Space**

Even though it might not seem as important as the other objective of tabulation, saving space without sacrificing the quality of data can be extremely helpful in the long run. Additionally, a table helps to present facts in a much more concise manner than page after page of text.

How is Data in Tabulation Executed?

With tabulation, data can be carried out manually or with the help of a computer. Mostly, the execution of data tabulation depends on the cost, type, and size of the study, availability of computers, time at disposal, and such other factors.

If tabulation is carried out on a computer, the answers are converted in numeric form. Whereas in the case of tabulation by hand, one can utilize lists, tally, card sort, and count methods.

Following are the categories of tabulating data by hand:

- **Direct Tally Method –**

Here, codes are first written down in tally sheets. Then a stroke is marked against codes to denote response. After every fourth stroke, the fifth response is given by putting a horizontal or diagonal line through the stroke.

- **Card Sort and Count Method –**

This is perhaps the most efficient hand tabulation method. Here the data is recorded in cards of various sizes and shapes with the help of a series of holes. Next, cards belonging to each of the categories are segregated and counted, and their frequency is recorded. This way, a total of 40 items can be included in a single page.

- **List and Tally Method –**

With this method, a large number of questionnaires are listed in one sheet. The responses to each question are then entered into rows, and the code corresponding to each question is represented in columns.

What are the Rules of Tabulation?

There are a few general rules that have to be followed while constructing tables. These are –

- The tables illustrated should be self-explanatory, simple and attractive. There should be no need for further explanation (details). If the volume of information is substantial, it is best to put them down in multiple tables instead of a single one. This reduces the chances of mistakes and defeats the purpose of forming a table. However, each table formed should also be complete in itself and serve the purpose of analysis.
- The number of rows and columns should be kept minimal to present information in a crisp and concise manner.
- Before tabulating, data should be approximated, wherever necessary.
- Stubs and captions should be self-explanatory and should not require the help of footnotes to be comprehended.
- If certain positions of data collected cannot be tabulated under any stub or captions, they should be put down in a separate table under the heading `` miscellaneous.
- Quantity and quality of data should not be compromised under any scenario while forming a table.

Validation

What is Data Validation?

Data validation refers to the process of ensuring the accuracy and quality of data. It is implemented by building several checks into a system or report to ensure the logical consistency of input and stored data.

In automated systems, data is entered with minimal or no human supervision. Therefore, it is necessary to ensure that the data that enters the system is correct and meets the desired quality standards. The data will be of little use if it is not entered properly and can create bigger downstream reporting issues. Unstructured data, even if entered correctly, will incur related costs for cleaning, transforming, and storage.

Types of Data Validation

There are many types of data validation. Most data validation procedures will perform one or more of these checks to ensure that the data is correct before storing it in the database. Common types of data validation checks include:

1. Data Type Check

A data type check confirms that the data entered has the correct data type. For example, a field might only accept numeric data. If this is the case, then any data containing other characters such as letters or special symbols should be rejected by the system.

2. Code Check

A code check ensures that a field is selected from a valid list of values or follows certain formatting rules. For example, it is easier to verify that a postal code is valid by checking it against a list of valid codes. The same concept can be applied to other items such as country codes and NAICS industry codes.

3. Range Check

A range check will verify whether input data falls within a predefined range. For example, latitude and longitude are commonly used in geographic data. A latitude value should be between -90 and 90, while a longitude value must be between -180 and 180. Any values out of this range are invalid.

4. Format Check

Many data types follow a certain predefined format. A common use case is date columns that are stored in a fixed format like “YYYY-MM-DD” or “DD-MM-YYYY.” A data validation procedure that ensures dates are in the proper format helps maintain consistency across data and through time.

5. Consistency Check

A consistency check is a type of logical check that confirms the data’s been entered in a logically consistent way. An example is checking if the delivery date is after the shipping date for a parcel.

6. Uniqueness Check

Some data like IDs or e-mail addresses are unique by nature. A database should likely have unique entries on these fields. A uniqueness check ensures that an item is not entered multiple times into a database.

Practical Example

Consider the example of a retailer that collects data on its stores but fails to create a proper check on the postal code. The oversight could make it difficult to leverage the data for information and business intelligence. Several problems can occur if the postal code is not entered or entered improperly.

It can be difficult to define the location of the store in some mapping software. A store postal code will also help generate insights about the neighborhood where the store is located. Without a data check on the postal code, it is more likely to lose the value of data. It will result in further costs if the data needs to be recollected or the postal code needs to be manually entered.

A simple solution to the problem would be to put a check in place that ensures a valid postal code is entered. The solution could be a dropdown menu or an auto-complete form that allows the user to choose the postal code from a list of valid codes. Such a type of data validation is called a code validation or code check.

Data Validation in Excel

The following example is an introduction to data validation in Excel. The data validation button under the data tab provides the user with different types of data validation checks based on the data type in the cell. It also allows the user to define custom validation checks using Excel

formulas. The data validation can be found in the Data Tools section of the Data tab in the ribbon of Excel:

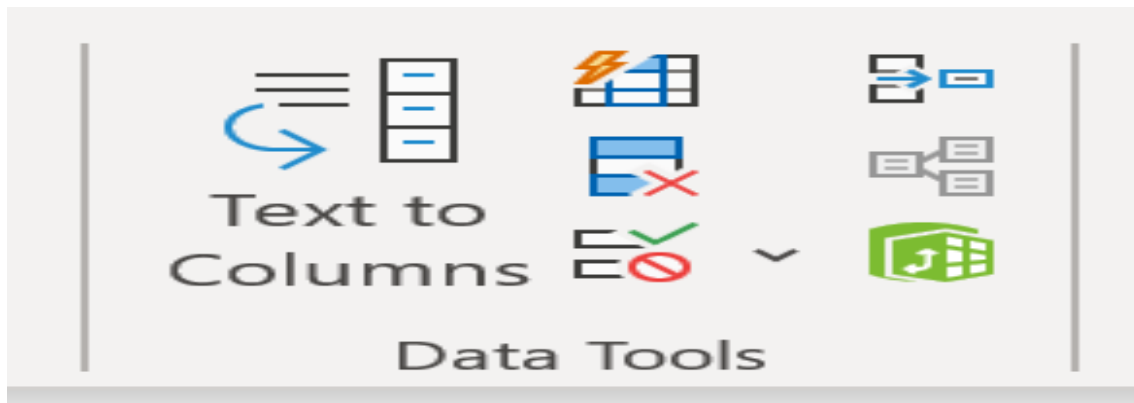


Fig. 1: Data validation tool in Excel

Data Entry Task

The example below illustrates a case of data entry, where the province must be entered for every store location. Since stores are only located in certain provinces, any incorrect entry should be caught.

It is accomplished in Excel using a two-fold data validation. First, the relevant provinces are incorporated into a drop-down menu that allows the user to select from a list of valid provinces.

	A	B
1	Store ID	Province
2	5001	AB
3	5024	ON
4	5047	BC
5	5070	AB
6	5093	SK
7	5116	NS
8	5139	NY
9	5162	BC
10	5185	AB
11		ON

Fig. 2: First level of data validation

Second, if the user inputs a wrong province by mistake, such as “NY” instead of “NS,” the system warns the user of the incorrect input.

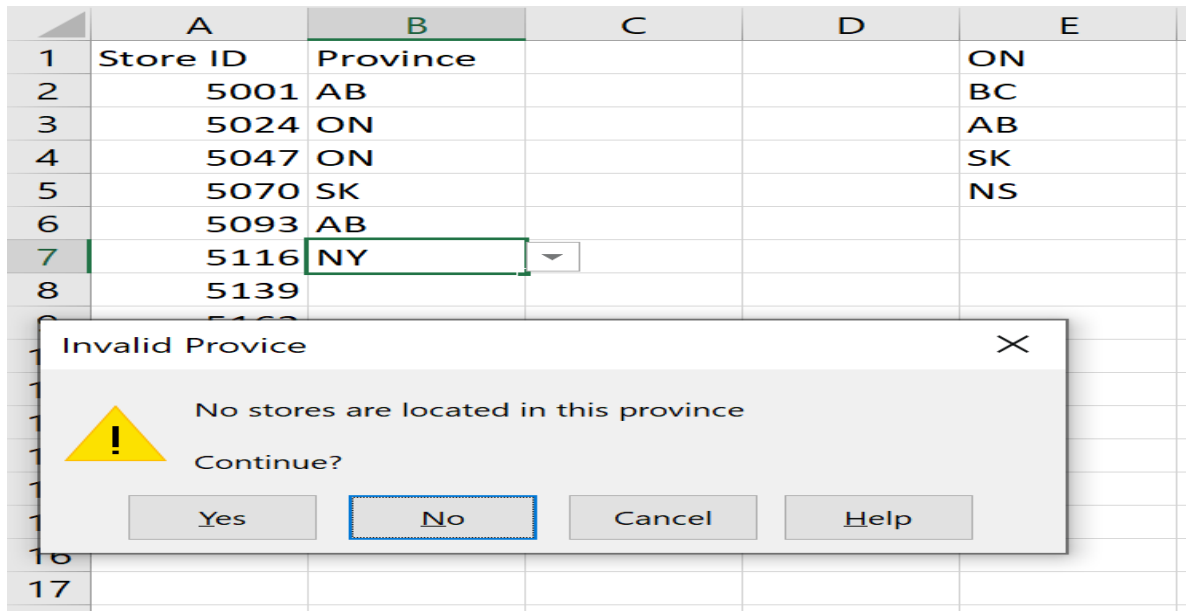


Fig. 3: Second level of data validation

Further, if the user ignores the warning, an analysis can be conducted using the data validation feature in Excel that identifies incorrect inputs.

	A	B	
1	Store ID	Province	
2	5001	AB	
3	5024	ON	
4	5047	ON	
5	5070	SK	
6	5093	AB	
7	5116	NY	
8	5139	BC	
9	5162	AB	
10	5185	ON	
11			

Fig. 4: Final level of data validation

Analysis and Interpretation

What is Data Analysis?

Data analysis is defined as a process of cleaning, transforming, and modeling data to discover useful information for business decision-making. The purpose of Data Analysis is to extract useful information from data and taking the decision based upon the data analysis.

A simple example of Data analysis is whenever we take any decision in our day-to-day life is by thinking about what happened last time or what will happen by choosing that particular decision. This is nothing but analyzing our past or future and making decisions based on it. For that, we gather memories of our past or dreams of our future. So that is nothing but data analysis. Now same thing analyst does for business purposes, is called Data Analysis.

Why Data Analysis?

To grow your business even to grow in your life, sometimes all you need to do is Analysis!

If your business is not growing, then you have to look back and acknowledge your mistakes and make a plan again without repeating those mistakes. And even if your business is growing, then you have to look forward to making the business to grow more. All you need to do is analyze your business data and business processes.

Data Analysis Tools



Data analysis tools make it easier for users to process and manipulate data, analyze the relationships and correlations between data sets, and it also helps to identify patterns and trends for interpretation. Here is a complete list of tools used for data analysis in research.

Types of Data Analysis: Techniques and Methods

There are several **types of Data Analysis** techniques that exist based on business and technology. However, the major Data Analysis methods are:

- Text Analysis
- Statistical Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

Text Analysis

Text Analysis is also referred to as Data Mining. It is one of the methods of data analysis to discover a pattern in large data sets using databases or data mining tools. It used to transform raw data into business information. Business Intelligence tools are present in the market which is used to take strategic business decisions. Overall it offers a way to extract and examine data and deriving patterns and finally interpretation of the data.

Statistical Analysis

Statistical Analysis shows “What happen?” by using past data in the form of dashboards. Statistical Analysis includes collection, Analysis, interpretation, presentation, and modeling of data. It analyses a set of data or a sample of data. There are two categories of this type of Analysis – Descriptive Analysis and Inferential Analysis.

Descriptive Analysis

analyses complete data or a sample of summarized numerical data. It shows mean and deviation for continuous data whereas percentage and frequency for categorical data.

Inferential Analysis

analyses sample from complete data. In this type of Analysis, you can find different conclusions from the same data by selecting different samples.

A good data interpretation process is key to making your data usable. It will help you make sure you’re drawing the correct conclusions and acting on your information.

No matter what, data is everywhere in the modern world. There are two groups and organizations: those drowning in data or not using it appropriately and those benefiting.

Diagnostic Analysis

Diagnostic Analysis shows “Why did it happen?” by finding the cause from the insight found in Statistical Analysis. This Analysis is useful to identify behavior patterns of data. If a new problem arrives in your business process, then you can look into this Analysis to find similar patterns of that problem. And it may have chances to use similar prescriptions for the new problems.

Predictive Analysis

Predictive Analysis shows “what is likely to happen” by using previous data. The simplest data analysis example is like if last year I bought two dresses based on my savings and if this year my salary is increasing double then I can buy four dresses. But of course it’s not easy like this because

you have to think about other circumstances like chances of prices of clothes is increased this year or maybe instead of dresses you want to buy a new bike, or you need to buy a house!

So here, this Analysis makes predictions about future outcomes based on current or past data. Forecasting is just an estimate. Its accuracy is based on how much detailed information you have and how much you dig in it.

Prescriptive Analysis

Prescriptive Analysis combines the insight from all previous Analysis to determine which action to take in a current problem or decision. Most data-driven companies are utilizing Prescriptive Analysis because predictive and descriptive Analysis are not enough to improve data performance. Based on current situations and problems, they analyze the data and make decisions.

Data Analysis Process

The **Data Analysis Process** is nothing but gathering information by using a proper application or tool which allows you to explore the data and find a pattern in it. Based on that information and data, you can make decisions, or you can get ultimate conclusions.

Data Analysis consists of the following phases:

- Data Requirement Gathering
- Data Collection
- Data Cleaning
- Data Analysis
- Data Interpretation
- Data Visualization

Data Requirement Gathering

First of all, you have to think about why do you want to do this data analysis? All you need to find out the purpose or aim of doing the Analysis of data. You have to decide which type of data analysis you wanted to do! In this phase, you have to decide what to analyze and how to measure it, you have to understand why you are investigating and what measures you have to use to do this Analysis.

Data Collection

After requirement gathering, you will get a clear idea about what things you have to measure and what should be your findings. Now it's time to collect your data based on requirements. Once you collect your data, remember that the collected data must be processed or organized for Analysis. As you collected data from various sources, you must have to keep a log with a collection date and source of the data.

Data Cleaning

Now whatever data is collected may not be useful or irrelevant to your aim of Analysis, hence it should be cleaned. The data which is collected may contain duplicate records, white spaces or errors. The data should be cleaned and error free. This phase must be done before Analysis because based on data cleaning, your output of Analysis will be closer to your expected outcome.

Data Analysis

Once the data is collected, cleaned, and processed, it is ready for Analysis. As you manipulate data, you may find you have the exact information you need, or you might need to collect more data. During this phase, you can use data analysis tools and software which will help you to understand, interpret, and derive conclusions based on the requirements.

Data Interpretation

After analyzing your data, it's finally time to interpret your results. You can choose the way to express or communicate your data analysis either you can use simply in words or maybe a table or chart. Then use the results of your data analysis process to decide your best course of action.

Data Visualization

Data visualization is very common in your day to day life; they often appear in the form of charts and graphs. In other words, data shown graphically so that it will be easier for the human brain to understand and process it. Data visualization often used to discover unknown facts and trends. By observing relationships and comparing datasets, you can find a way to find out meaningful information.

What is Data Interpretation

Data interpretation is the process of reviewing data and arriving at relevant conclusions using various analytical research methods. Data analysis assists researchers in categorizing, manipulating, and summarizing data to answer critical questions.

In business terms, the interpretation of data is the execution of various processes. This process analyzes and revises data to gain insights and recognize emerging patterns and behaviors. These conclusions will assist you as a manager in making an informed decision based on numbers while having all of the facts at your disposal.

Importance of Data Interpretation

Raw data is useless unless it's interpreted. Data interpretation is important to businesses and people. The collected data helps make informed decisions.

Make better decisions

Any decision is based on the information that is available at the time. People used to think that many diseases were caused by bad blood, which was one of the four humors. So, the solution was to get rid of the bad blood. We now know that things like viruses, bacteria, and immune responses can cause illness and can act accordingly.

In the same way, when you know how to collect and understand data well, you can make better decisions. You can confidently choose a path for your organization or even your life instead of working with assumptions.

The most important thing is to follow a transparent process to reduce mistakes and tiredness when making decisions.

Find trends and take action

Another practical use of data interpretation is to get ahead of trends before they reach their peak. Some people have made a living by researching industries, spotting trends, and then making big bets on them.

With the proper data interpretations and a little bit of work, you can catch the start of trends and use them to help your business or yourself grow.

Better resource allocation

The last importance of data interpretation we will discuss is the ability to use people, tools, money, etc., more efficiently. For example, If you know via strong data interpretation that a market is underserved, you'll go after it with more energy and win.

In the same way, you may find out that a market you thought was a good fit is actually bad. This could be because the market is too big for your products to serve, there is too much competition, or something else.

No matter what, you can move the resources you need faster and better to get better results.

What are the steps in interpreting data?

Here are some steps to interpreting data correctly.

1. Gather the data

The very first step in data interpretation is gathering all relevant data. You can do this by first visualizing it in a bar, graph, or pie chart. This step aims to analyze the data accurately and without bias. Now is the time to recall how you conducted your research.

Here are two question patterns that will help you to understand better.

- Were there any flaws or changes that occurred during the data collection process?
- Have you saved any observatory notes or indicators?

You can proceed to the next stage when you have all of your data.

2. Develop your discoveries

This is a summary of your findings. Here, you thoroughly examine the data to identify trends, patterns, or behavior. If you are researching a group of people using a sample population, this is the section where you examine behavioral patterns. You can compare these deductions to previous data sets, similar data sets, or general hypotheses in your industry. This step's goal is to compare these deductions before drawing any conclusions.

3. Draw Conclusions

After you've developed your findings from your data sets, you can draw conclusions based on your discovered trends. Your findings should address the questions that prompted your research. If they do not respond, inquire about why; it may produce additional research or questions.

4. Give recommendations

The interpretation procedure of data comes to a close with this stage. Every research conclusion must include a recommendation. As recommendations are a summary of your findings and conclusions, they should be brief. There are only two options for recommendations; you can either recommend a course of action or suggest additional research.

Data interpretation examples

Here are two examples of data interpretations to help you understand it better:

Example 1

Let's say your users fall into four age groups. So a company can see which age group likes their content or product. Based on bar charts or pie charts, they can develop a marketing strategy to reach uninvolved groups or an outreach strategy to grow their core user base.

Example 2

Another example of data analysis is the use of recruitment CRM by businesses. They utilize it to find candidates, track their progress, and manage their entire hiring process to determine how they can better automate their workflow.

Report writing and presentation of results,

Report writing

Reports are usually spread across a vast horizon of topics but are focused on communicating information about a particular topic and a niche target market. The primary motive of research reports is to convey integral details about a study for marketers to consider while designing new strategies.

Certain events, facts, and other information based on incidents need to be relayed to the people in charge, and creating research reports is the most effective communication tool. Ideal research reports are extremely accurate in the offered information with a clear objective and conclusion. These reports should have a clean and structured format to relay information effectively.

What are Research Reports?

Research reports are recorded data prepared by researchers or statisticians after analyzing the information gathered by conducting organized research, typically in the form of surveys or qualitative methods.

A research report is a reliable source to recount details about a conducted research. It is most often considered to be a true testimony of all the work done to garner specificities of research.

The various sections of a research report are:

1. Summary
2. Background/Introduction
3. Implemented Methods
4. Results based on Analysis
5. Deliberation
6. Conclusion

Components of Research Reports

Research is imperative for launching a new product/service or a new feature. The markets today are extremely volatile and competitive due to new entrants every day who may or may not provide effective products. An organization needs to make the right decisions at the right time to be relevant in such a market with updated products that suffice customer demands.

The details of a research report may change with the purpose of research but the main components of a report will remain constant. The research approach of the market researcher also influences the style of writing reports. Here are seven main components of a productive research report:

- **Research Report Summary:** The entire objective along with the overview of research are to be included in a summary which is a couple of paragraphs in length. All the multiple components of the research are explained in brief under the report summary. It should be interesting enough to capture all the key elements of the report.
- **Research Introduction:** There always is a primary goal that the researcher is trying to achieve through a report. In the introduction section, he/she can cover answers related to this goal and establish a thesis which will be included to strive and answer it in detail. This section should answer an integral question: “What is the current situation of the goal?”. After the research design was conducted, did the organization conclude the goal successfully or they are still a work in progress – provide such details in the introduction part of the research report.
- **Research Methodology:** This is the most important section of the report where all the important information lies. The readers can gain data for the topic along with analyzing the quality of provided content and the research can also be approved by other market researchers. Thus, this section needs to be highly informative with each aspect of research discussed in detail. Information needs to be expressed in chronological order according to its priority and importance. Researchers should include references in case they gained information from existing techniques.

- **Research Results:** A short description of the results along with calculations conducted to achieve the goal will form this section of results. Usually, the exposition after data analysis is carried out in the discussion part of the report.
- **Research Discussion:** The results are discussed in extreme detail in this section along with a comparative analysis of reports that could probably exist in the same domain. Any abnormality uncovered during research will be deliberated in the discussion section. While writing research reports, the researcher will have to connect the dots on how the results will be applicable in the real world.
- **Research References and Conclusion:** Conclude all the research findings along with mentioning each and every author, article or any content piece from where references were taken.

Tips for Writing Research Reports

Writing research reports in the manner can lead to all the efforts going down the drain. Here are 15 tips for writing impactful research reports:

- **Prepare the context before starting to write and start from the basics:** This was always taught to us in school – be well-prepared before taking a plunge into new topics. The order of survey questions might not be the ideal or most effective order for writing research reports. The idea is to start with a broader topic and work towards a more specific one and focus on a conclusion or support, which a research should support with the facts. The most difficult thing to do in reporting, without a doubt is to start. Start with the title, the introduction, then document the first discoveries and continue from that. Once the marketers have the information well documented, they can write a general conclusion.
- **Keep the target audience in mind while selecting a format that is clear, logical and obvious to them:** Will the research reports be presented to decision makers or other researchers? What are the general perceptions around that topic? This requires more care and diligence. A researcher will need a significant amount of information to start writing the research report. Be consistent with the wording, the numbering of the annexes and so on. Follow the approved format of the company for the delivery of research reports and demonstrate the integrity of the project with the objectives of the company.
- **Have a clear research objective:** A researcher should read the entire proposal again, and make sure that the data they provide contributes to the objectives that were raised from the beginning. Remember that speculations are for conversations, not for research reports, if a researcher speculates, they directly question their own research.
- **Establish a working model:** Each study must have an internal logic, which will have to be established in the report and in the evidence. The researchers' worst nightmare is to be required to write research reports and realize that key questions were not included.

- **Gather all the information about the research topic.** Who are the competitors of our customers? Talk to other researchers who have studied the subject of research, know the language of the industry. Misuse of the terms can discourage the readers of research reports from reading further.
- **Read aloud while writing.** While reading the report, if the researcher hears something inappropriate, for example, if they stumble over the words when reading them, surely the reader will too. If the researcher can't put an idea in a single sentence, then it is very long and they must change it so that the idea is clear to everyone.
- **Check grammar and spelling.** Without a doubt, good practices help to understand the report. Use verbs in the present tense. Consider using the present tense, which makes the results sound more immediate. Find new words and other ways of saying things. Have fun with the language whenever possible.
- **Discuss only the discoveries that are significant.** If some data are not really significant, do not mention them. Remember that not everything is truly important or essential within research reports.
- **Try and stick to the survey questions.** For example, do not say that the people surveyed "were worried" about an research issue, when there are different degrees of concern.
- **The graphs must be clear enough so that they understand themselves.** Do not let graphs lead the reader to make mistakes: give them a title, include the indications, the size of the sample, and the correct wording of the question.
- **Be clear with messages.** A researcher should always write every section of the report with an accuracy of details and language.
- **Be creative with titles**– Particularly in segmentation studies choose names "that give life to research". Such names can survive for a long time after the initial investigation.
- **Create an effective conclusion:** The conclusion in the research reports is the most difficult to write, but it is an incredible opportunity to excel. Make a precise summary. Sometimes it helps to start the conclusion with something specific, then it describes the most important part of the study, and finally, it provides the implications of the conclusions.
- **Get a couple more pair of eyes to read the report.** Writers have trouble detecting their own mistakes. But they are responsible for what is presented. Ensure it has been approved by colleagues or friends before sending the final draft out.

Presentation of results

Written Presentation of Results

Once you've gone through the process of doing communication research – using a quantitative, qualitative, or critical/rhetorical methodological approach – the final step is to communicate it.

The major style manuals (the APA Manual, the MLA Handbook, and Turabian) are very helpful in documenting the structure of writing a study, and are highly recommended for consultation. But, no matter what style manual you may use, there are some common elements to the structure of an academic communication research paper.

Title Page:

This is simple: Your Paper's Title, Your Name, Your Institutional Affiliation (e.g., University), and the Date, each on separate lines, centered on the page. Try to make your title both descriptive (i.e., it gives the reader an idea what the study is about) and interesting (i.e., it is catchy enough to get one's attention).

For example, the title, "The uncritical idealization of a compensated psychopath character in a popular book series," would not be an inaccurate title for a published study, but it is rather vague and exceedingly boring. That study's author fortunately chose the title, "A boyfriend to die for: Edward Cullen as compensated psychopath in Stephanie Meyer's *Twilight*," which is more precisely descriptive, and much more interesting (Merskin, 2011). The use of the colon in academic titles can help authors accomplish both objectives: a catchy but relevant phrase, followed by a more clear explanation of the article's topic.

Abstract:

In some instances, you might be asked to write an abstract, which is a summary of your paper that can range in length from 75 to 250 words. If it is a published paper, it is useful to include key search terms in this brief description of the paper (the title may already have a few of these terms as well). Although this may be the last thing you write, make it one of the best things you write, because this may be the first thing your audience reads about the paper (and may be the only thing read if it is written badly). Summarize the problem/research question, your methodological approach, your results and conclusions, and the significance of the paper in the abstract.

Quantitative and qualitative studies will most typically use the rest of the section titles noted below. Critical/rhetorical studies will include many of the same steps, but will often have different headings. For example, a critical/rhetorical paper will have an introduction, definition of terms, and literature review, followed by an analysis (often divided into sections by areas of investigation) and ending with a conclusion/implications section. Because critical/rhetorical research is much more descriptive, the subheadings in such a paper are often times not generic subheads like "literature review," but instead descriptive subheadings that apply to the topic at hand, as seen in the schematic below. Because many journals expect the article to follow typical research paper headings of introduction, literature review, methods, results, and discussion, we discuss these sections briefly next.

Social Scientific Project

- Introduction
- Literature Review
- Method
- Results
- Discussion
- References

Critical Projects

- Introduction
- Definition of Terms
- Literature Review
- Analysis (divide sections by areas of investigation)
- Conclusions*

*All sections may not be named this way

Introduction:

As you read social scientific journals (see chapter 1 for examples), you will find that they tend to get into the research question quickly and succinctly. Journal articles from the humanities tradition tend to be more descriptive in the introduction. But, in either case, it is good to begin with some kind of brief anecdote that gets the reader engaged in your work and lets the reader understand why this is an interesting topic. From that point, state your research question, define the problem (see Chapter One) with an overview of what we do and don't know, and finally state what you will do, or what you want to find out. The introduction thus builds the case for your topic, and is the beginning of building your argument, as we noted in chapter 1.

By the end of the Introduction, the reader should know what your topic is, why it is a significant communication topic, and why it is necessary that you investigate it (e.g., it could be there is gap in literature, you will conduct valuable exploratory research, or you will provide a new model for solving some professional or social problem).

Literature Review:

The literature review summarizes and organizes the relevant books, articles, and other research in this area. It sets up both quantitative and qualitative studies, showing the need for the study. For critical/rhetorical research, the literature review often incorporates the description of the historical

context and heuristic vocabulary, with key terms defined in this section of the paper. For more detail on writing a literature review, see Appendix 1.

Methods:

The methods of your paper are the processes that govern your research, where the researcher explains what s/he did to solve the problem. As you have seen throughout this book, in communication studies, there are a number of different types of research methods. For example, in quantitative research, one might conduct surveys, experiments, or content analysis. In qualitative research, one might instead use interviews and observations. Critical/rhetorical studies methods are more about the interpretation of texts or the study of popular culture as communication. In creative communication research, the method may be an interpretive performance studies or filmmaking. Other methods used sometimes alone, or in combination with other methods, include legal research, historical research, and political economy research.

In quantitative and qualitative research papers, the methods will be most likely described according to the APA manual standards. At the very least, the methods will include a description of participants, data collection, and data analysis, with specific details on each of these elements. For example, in an experiment, the researcher will describe the number of participants, the materials used, the design of the experiment, the procedure of the experiment, and what statistics will be used to address the hypotheses/research questions.

Critical/rhetorical researchers rarely have a specific section called "methods," as opposed to quantitative and qualitative researchers, but rather demonstrate the method they use for analysis throughout the writing of their piece.

Helping your reader understand the methods you used for your study is important not only for your own study's credibility, but also for possible replication of your study by other researchers. A good guideline to keep in mind is **transparency**. You want to be as clear as possible in describing the decisions you made in designing your study, gathering and analyzing your data so that the reader can retrace your steps and understand how you came to the conclusions you formed. A research study can be very good, but if it is not clearly described so that others can see how the results were determined or obtained, then the quality of the study and its potential contributions are lost.

Results:

After you completed your study, your findings will be listed in the results section. Particularly in a quantitative study, the results section is for revisiting your hypotheses and reporting whether or not your results supported them, and the statistical significance of the results. Whether your study supported or contradicted your hypotheses, it's always helpful to fully report what your results

were. The researcher usually organizes the results of his/her results section by research question or hypothesis, stating the results for each one, using statistics to show how the research question or hypothesis was answered in the study.

The qualitative results section also may be organized by research question, but usually is organized by themes which emerged from the data collected. The researcher provides rich details from her/his observations and interviews, with detailed quotations provided to illustrate the themes identified. Sometimes the results section is combined with the discussion section.

Critical/rhetorical researchers would include their analysis often with different subheadings in what would be considered a "results" section, yet not labeled specifically this way.

Discussion:

In the discussion section, the researcher gives an appraisal of the results. Here is where the researcher considers the results, particularly in light of the literature review, and explains what the findings mean. If the results confirmed or corresponded with the findings of other literature, then that should be stated. If the results didn't support the findings of previous studies, then the researcher should develop an explanation of why the study turned out this way. Sometimes, this section is called a "conclusion" by researchers.

References:

In this section, all of the literature cited in the text should have full references in alphabetical order. Appendices: Appendix material includes items like questionnaires used in the study, photographs, documents, etc. An alphabetical letter is assigned for each piece (e.g. Appendix A, Appendix B), with a second line of title describing what the appendix contains (e.g. Participant Informed Consent, or New York Times Speech Coverage). They should be organized consistently with the order in which they are referenced in the text of the paper. The page numbers for appendices are consecutive with the paper and reference list.

Tables/Figures:

Tables and figures are referenced in the text, but included at the end of the study and numbered consecutively. (Check with your professor; some like to have tables and figures inserted within the paper's main text.) Tables generally are data in a table format, whereas figures are diagrams (such as a pie chart) and drawings (such as a flow chart).

Importance of report writing

- 1. Decision Making Tool:** Today's complex business organizations require thousands of information. Reports provide the required information. A large number of important decisions in business or any other areas are taken on the basis of information presented in the report.
- 2. Investigation:** Whenever there is any problem, a committee or commission or study group investigate the problem to find out the reason behind the problem and present the found output with or without the recommendation in the form of a report.
- 3. Evaluation:** Large scale organization are engaged in multidimensional activities. It is not possible for a single top executive to keep personal watch on what others are doing. So, the executive depends on reports to evaluate the performance of various departments or units.
- 4. Quick Location:** There is no denying the fact that business executives need information for quick decision making. As top executive are found to be busy for various purposes, they need vital sources of information. Such sources can be business report.
- 5. Development of Skill:** Report writing skill develops the power of designing, Organization co-ordination, judgement and communication. Therefore such skill acts as a catalyst.
- 6. Neutral Presentation of Facts:** Facts are required to be presented in a neutral way. Such presentation is ensured through a report as it investigates, explains and evaluates and much more about any fact independently.
- 7. Professional Advancement:** Report also plays a major role in professional progress and advancement.
For promotion to the rank and file position, satisfactory job performance is enough to help a person. But for promotion to high level position, intellectual ability is highly required. Such ability can be expressed through the report submitted to higher authority.
- 8. Proper Control:** Whether activities are happening according to plan or not is expressed through a report. So, controlling such activities are implemented based on the information of a report.
- 9. A Managerial Tool:** Various reports make activities easy for the managers. For planning, organizing, coordinating, motivating and controlling, A manager may find help from a report which acts as a source of information.
- 10. Encountering Advance and Complex Situation:** In a large business organization, there is always some sort of labor problems which may bring complex situations. To tackle those situation, managers take the help of a report.

Types of research reports

1. Thesis

Thesis is a type of research report. A thesis is a long-form research document that presents the findings and conclusions of an original research study conducted by a student as part of a graduate or postgraduate program. It is typically written by a student pursuing a

higher degree, such as a Master's or Doctoral degree, although it can also be written by researchers or scholars in other fields.

2. Research Paper

Research paper is a type of research report. A research paper is a document that presents the results of a research study or investigation. Research papers can be written in a variety of fields, including science, social science, humanities, and business. They typically follow a standard format that includes an introduction, literature review, methodology, results, discussion, and conclusion sections.

3. Technical Report

A technical report is a detailed report that provides information about a specific technical or scientific problem or project. Technical reports are often used in engineering, science, and other technical fields to document research and development work.

4. Progress Report

A progress report provides an update on the progress of a research project or program over a specific period of time. Progress reports are typically used to communicate the status of a project to stakeholders, funders, or project managers.

5. Feasibility Report

A feasibility report assesses the feasibility of a proposed project or plan, providing an analysis of the potential risks, benefits, and costs associated with the project. Feasibility reports are often used in business, engineering, and other fields to determine the viability of a project before it is undertaken.

6. Field Report

A field report documents observations and findings from fieldwork, which is research conducted in the natural environment or setting. Field reports are often used in anthropology, ecology, and other social and natural sciences.

7. Experimental Report

An experimental report documents the results of a scientific experiment, including the hypothesis, methods, results, and conclusions. Experimental

reports are often used in biology, chemistry, and other sciences to communicate the results of laboratory experiments.

8. Case Study Report

A case study report provides an in-depth analysis of a specific case or situation, often used in psychology, social work, and other fields to document and understand complex cases or phenomena.

9. Literature Review Report

A literature review report synthesizes and summarizes existing research on a specific topic, providing an overview of the current state of knowledge on the subject. Literature review reports are often used in social sciences, education, and other fields to identify gaps in the literature and guide future research.

Report structure

(A)-Front Matter

- 1) Letter of transmittal (Cover letter)
- 2) Title page
- 3) Executive summary
- 4) Table of content
- 5) List of figures/ Illustration

(B)-Body of the report

- 6) Introduction
- 7) Findings and Documentation
- 8) Conclusion and Recommendation

(C)-Back Matter

- 9) Appendices, References, and Glossary

Note: The letter of transmittal is an optional element of reporting which may or may not be included depending on the type of report prepared.

(A)Front Matter

The front matter consists of all the opening components of a business report. This includes a title page, executive summary, table of contents, and a list of figures/illustrations. A letter of transmittal, which serves as an official cover letter, may also be included.

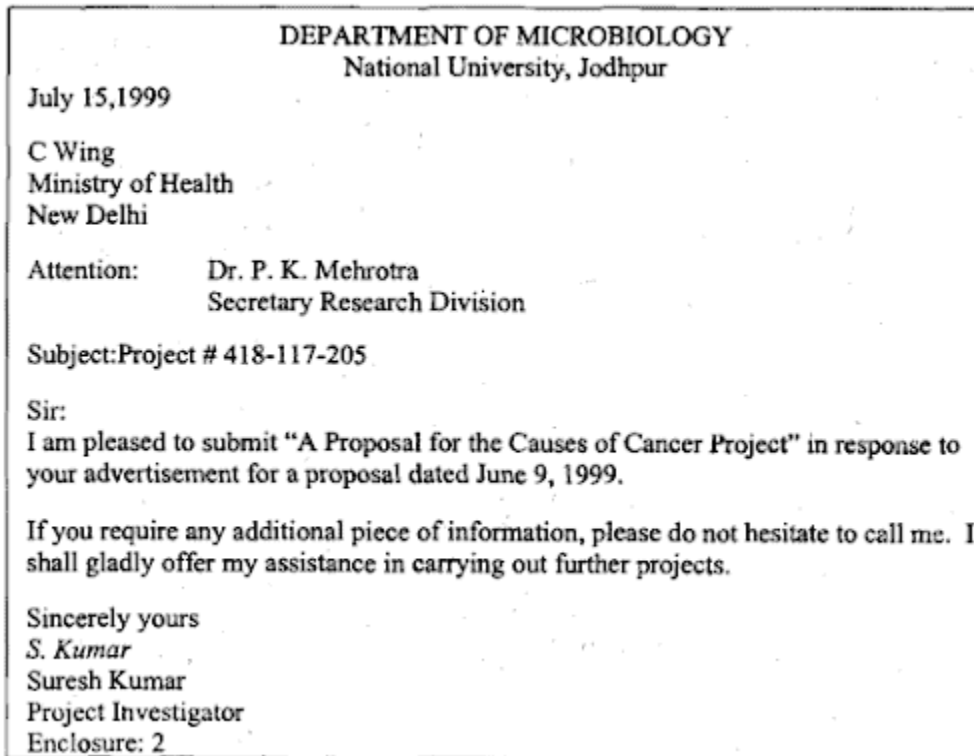
The following elements are included in the front matter of a report:

1) Letter of Transmittal (Cover):

The letter of transmittal serves as a customary cover page. Its purpose is to state the title of the report and if necessary, briefly preview conclusions, state the purpose, and mention key highlights.

It mentions who commissioned the report and appreciates the efforts of those involved in the project. This letter is attached to the top of the report and is written in a courteous tone.

An Example of the Letter of Transmittal



An example of the letter of transmittal.

2) Title Page:

As the name suggests, the title page is the opening page of a business report. It comprises the following:

1. The title of the subject
2. Name of the writer
3. Name of the receiver, title, and organization
4. Purpose of the report
5. Date of submission

The title of a business report should clearly convey the type of report and its purpose. It should be written in capital letters and centered in alignment.

An Example of the Title Page

REMOTE SENSING IN INDIA; A STUDY A Paper Presented to Prof. U. C. Nag Computer Centre, Meerut Open University by Vinay Kumar In Partial Fulfilment Of the Requirements for the Postgraduate Diploma in Remote Sensing On 23 March 1999

3) Executive Summary:

The executive summary or the abstract is a major element of a business report. It is written for busy executives who may only read this part of the report. It offers a summary of the entire report and its findings.

It includes:

1. Purpose of the report
2. Scope of the investigation
3. The methods used to conduct the investigation
4. Key findings along with essential statistics
5. Conclusions, analysis, and recommendations

[R&D] Q2 explorations into new watch series — Executive Summary

Problem statement

In recent customer feedback sessions, 52% of customers have expressed a need for a simpler and cheaper version of our product. In surveys of customers who have chosen competitor watches, price is mentioned 87% of the time. To best serve our existing customers, and to branch into new markets, we need to develop a series of watches that we can sell at an appropriate price point for this market.

Proposed solution

Our new watch series will begin at 20% cheaper than our current cheapest option, with the potential for 40%+ cheaper options depending on material and movement. In order to offer these prices, we will do the following:

- Offer watches in new materials, including potentially silicone or wood
- Use high-quality quartz movement instead of in-house automatic movement
- Introduce customizable band options, with a focus on choice and flexibility over traditional luxury

Note that every watch will still be rigorously quality controlled in order to maintain the same world-class speed and precision of our current offerings.

Value

With new offerings that are between 20% and 40% cheaper than our current cheapest option, we expect to be able to break into the casual watch market, while still supporting our luxury brand. That will help us hit FY22's Objective 3: Expanding the brand. These new offerings have the potential to bring in upwards of three million dollars in profits annually, which will help us hit FY22's Objective 1: 7 million dollars in annual profit.

Early customer feedback sessions indicate that cheaper options will not impact the value or prestige of the luxury brand, though this is a risk that should be factored in during design. In order to mitigate that risk, the product marketing team will begin working on their go-to-market strategy six months before the launch.

Final thoughts & next steps

Cheaper and varied offerings not only allow us to break into a new market—it will also expand our brand in a positive way. With the attention from these new offerings, plus the anticipated demand for cheaper watches, we expect to increase market share by 2% annually. For more information, read our [go-to-market strategy](#) and [customer feedback documentation](#).

Source: asana.com

Note: You can further check out the following blog by Hubspots on powerfull executive summary for more details.

4) Table of Contents:

For longer and more detailed information, a table of contents is a necessary element of report writing. It provides the readers with an overview of the content present. This helps them easily locate particular sections of a report. All headings and subheadings are listed along with page numbers in the exact order of appearance.



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5) List of Figures/Illustrations:

The graphic aids (charts, graphs, photographs, diagrams, etc) used in a business report are organized in their own separate table of contents. A list of figures or list of illustrations helps the readers locate the specific page where this information is presented.

If a report has only tables, it is called a list of tables. If it has only figures with no tables, it is termed a list of figures. And if it contains both tables and figures, it is collectively called a list of illustrations.

An Example of a List of Illustrations		
LIST OF ILLUSTRATIONS		
Figures		
Figure 1	Compilation-execution of FORTRAN program	3
Figure 2	Example of EXEC statement to invoke compiler	6
Figure 3	Example of compilation feature	8
Figure 4	Text of FORTXCL catalogued procedure	9
Figure 5	Structure of segment	14
Figure 6	Composition of error-control table	21
Tables		
Table 1	Job control statements	2
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Table 3	Contents of object list	7
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Table 5	ANS control characters for line feed control	11
Table 6	Standard values of error-control table	16
(Source, Markel 1984: 247)		

(B)Body of a Report

The body contains the meaty content of the report. After the introductory sections, is where the substance of the report lies. The body of a report may vary according to the type of business report. For example, The essential elements of a technical report will differ from the content of a progress report.

The following sections make up the body of a report:

6) Introduction:

The introduction is the ‘beginning’ of the report. This is the section where the structure of the report is detailed. Along with this, key questions are answered such as “what is the report about ” and “how can it be used”.

This is where the page numbering of a report starts. The introduction of a report is used to:

- Define the main subject or problem
- Provide brief background information
- State the overall purpose of the report
- State theoretical foundations, scope, methodology, and any limitations

7) Findings and Documentation:

This is one of the main elements of a report. It is also usually the longest section of a report. The factual findings are presented along with an interpretation of the data collected.

The key findings of the investigation are explained in detail. The writer mentions all the sources used to formulate the report. This helps in the authentication of the report and allows the readers to visit the source material for further information.

All significant information gathered is documented in this section. Both positive and negative points are mentioned to form conclusions and recommend solutions.

The relevant questions pertaining to the report, its purpose, addressing problems and analysis of information are present in the findings and documentation.

8) Conclusion:

The findings and purpose of the report are summarized in the conclusion. The information presented in the report is analyzed and the advantages, limitations, and actions that need to be taken are stated in this section.

The writer must avoid including new information in the conclusion. It must only be relevant to the data presented in the rest of the report.

9) Recommendations:

The recommendation is one of the prime elements of the structure of a business report, this part includes the recommended actions that need to be taken based on the findings of the report. They are relevant to:

- What must be done
- Who will do it
- How and when they will do it
- Where it will be done

Recommendations can be made for improvement, implementation of new ideas, or change which will help the company's performance.

4.1 Suggestions

- In order to have interactive sessions, teacher should have an interaction with student even through informal channel when needed in order to keep sense of belonging. The informal channel may include (instant messages online chat groups, audio calls, private video calls ...) with equal importance given to formal channels such as (online platforms, email ...).
- At the end students and teacher should collaborate and work on developing different kind of activates and incentives in order to reduce the boredom and increase participation in class.

4.2 Conclusions

- Pandemic had turned out to be a disaster for the planet, providing chance and opportunity to restructure current ongoing educational system dependent on classrooms. Though there have been some major challenges such as creating a connection within, teaching, curriculum assessment, and learning in school and non-school environments to meet the 'new normal'.
- Learn ways to empower students and teachers and parents to manage new challenges in 'new normal' schooling. On the other hand, the sudden changes due to external factor in online education has given advantage to students for picking up their skill areas and joining the online education platform to enhance their skill.

(C)Back Matter

Following the front matter and the body of the report, we have the back matter. This section of the report highlights auxiliary and supporting segments such as glossary, references, and appendices.

10) Glossary:

The glossary serves as a dictionary providing definitions of technical terms used in the report. The words are listed in alphabetical order.

The glossary is listed before the appendix and if very brief, it can be placed just after the table of contents.

11) References:

References are also one of the essential components of a business report because a report is only considered authentic if all accurate sources are listed. All references from where the data was collected are mentioned in this section. Data collected from third-party sources such as stats, facts, and journals must be credited to their original sources.

References-

https://www.researchgate.net/publication/323335958_Literature_Review_The_effectiveness_of_e-learning_for_impacting_quality_education_to_students (R. Safiyeh).

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12) Appendices:

The appendix is the last element of the business report structure that contains supporting information that would be too bulky to include in the body of the report. This is considered an optional section that may or may not be included in a report.

It contains specialized information such as maps, questionnaires, interviews, letters, etc.

Guidelines for effective documentation

Rule 1. Never consider the time you invest in writing or planning documentation as wasted time

Like any well-designed project, the production of good documentation requires clear planning and a considerable investment of time and resources. This may seem like a pain to you, especially when you are overloaded with other work. Rather than viewing documentation and the lengthy meetings with the "non-specialised" technical writer who needs to be explained step by step the simplest things, as a necessary evil, you should see it as a tool for helping you clarify your product to yourself as well as to others.

One of the problems of design is that when you become bogged down in the gritty details and mechanics you may lose sight of the forest, for all the trees. Documentation not only provides you with a roadmap of the forest, it also presents an overall perspective, an opportunity to step out of the logger's mindset and see the larger plan or goal. Where are you going? Are you missing anything? Have you included everything that needs to be included?

Your discussions with your writers may help identify problems that you were not aware. The documentation process helps you to place what you are doing in perspective and keep in focus the ultimate purpose or use of the product being designed.

Preferably an outline should be produced that covers the major sections of the documentation. The outline is designed to lead the reader in a logical manner through the task, procedure or description being described.

Make sure your time allotment for a project takes into consideration time devoted to documentation.

Rule 2. Don't take what you say or write for granted

What may seem clear and obvious to you may not appear so to someone from outside the company, who is not familiar with the concepts and jargon that you use daily.

When a writer asks you what you mean by something, the answer may seem obvious, but it may not be. Language needs to be taken apart, examined for its basic concepts and then reassembled. Often, drafts need to be rewritten several times, to clarify the meaning and focus the text to fit the reader's needs.

Information that has been assembled from a technical or development perspective needs to be presented in a form suitable for the purpose of the document and the intended reader. Readers of manuals or brochures will probably not need to know any technical details, other than on a broad, conceptual level. This requires removing information not relevant to the reader.

Rule 3. Provide plenty of feedback and input

The basic formula for documentation is:

Quality of input = quality and clarity of output

Remember this formula the next time you give feedback. Writers depend on receiving good quality information. You are responsible for ensuring that the information you provide is comprehensive and accurate.

Feedback should be concise and to the point. It should:

- Refer to specific items
- Provide clear guidelines for what needs to be changed
- Preferably be presented as a list of to do items or as text to be edited/changed in the document

You may not be sure of the precise problem with the documentation. Something may feel wrong or be missing, or a description of a concept or procedure may feel incomplete. Trust your gut feeling and explore this further. You know the product. Ask yourself or the writer “What do you mean by this?” Try rephrasing unclear sections in a way that communicates the way that you see the product.

Use diagrams and examples to explain what you mean.

Finally, if a writer requests information or feedback, don’t procrastinate and delay. This holds up the entire documentation process, since the writer cannot proceed until your feedback or information has been received.

Rule 4. Be prepared for several drafts

Clear documentation, like any other activity, requires constant rethinking and reshaping. Concepts, terminology and processes need to be identified, written about, clarified, rewritten about, discussed and honed. Writing is a cyclical process. Each round of feedback produces a document that is clearer to read, more comprehensive and accurate. The more input received from different perspectives and different readers, the more reliable and accurate will be the nature of the information you produce.

Unfortunately, the myth of dashing off a draft in one sitting, which contains everything that needs to be said in a clear way, is impractical. It would be similar to asking a software engineer to produce a program in one draft.

Rule 5. Be on friendly terms with your writer

The writer assigned to your project is there to help you. He/she is not an added nuisance that is a drain on your resources and time. The writer’s purpose is to take the problems and considerations

that go with documentation off your hands.

The task of documenting a project is both lengthy and time- consuming – technical writers are employed to take over responsibility, allowing you to devote your energies where they are most needed.

Don't cut short or avoid your writer, for fear of wasting your time. Your patience will pay off.

Rule 6. Make sure all stages of design are well-documented

It pays to invest time and energy at the start of a project in clarifying the concepts of your design, through white papers, product proposals, presentations, diagrams and system specification documents. The more complete the documentation, the more information will be available at a later stage for user-orientated manuals and other external documents.

Rule 7. Involve your writer in all stages of the product design

Your writer is a valuable resource. Don't involve him/her at the last stage, in the final rush to complete the project.

The writer should be involved in all stages of a project. An experienced writer will be able to provide a bird's eye view of the entire project and its components. Such a writer can integrate and synthesise information from different sources, written over different time periods. A writer who has been involved in all stages of documentation will be able to fill in gaps and make critical decisions about the contents of a manual, with the minimum of investment from the product manager and engineers.

Rule 8. Writing is team work

The final documentation product should be the result of the combined input and feedback from the entire team: engineers, product managers, writers and quality assurance staff. Each professional member of the team can add something from his or her own perspective that can contribute to the value of the documentation and the project. Remember this when you are asked to offer feedback or if it is your task to coordinate and ensure that other members of the team contribute their feedback.

Rule 9. Be Flexible in your expectations and demands

Be prepared to be flexible in your expectations and demands. You may have to share a writer with several other product managers, each with their own deadlines. A busy writer, with several projects on hand, may not be in a position to drop everything else and devote all their time to your project. It is therefore important that you plan your documentation needs in advance, together with the writer. This will enable him/her to schedule time for you.

Module-6

Intellectual Property Rights: Meaning and Concepts of Intellectual Property, Nature and Characteristics of Intellectual Property, Origin and Development of Intellectual Property, Kinds of Intellectual Property, Intellectual Property System in India, IPRs- Invention and Creativity- Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs)- **A brief summary of:** Patents, Copyrights, Trademarks, TRIPS and TRIMS , Industrial Designs- Integrated Circuits-Geographical Indications-Establishment of WIPO- Application and Procedures.

Intellectual Property Rights: Meaning

Intellectual property rights are the rights given to each and every person for the creation of new things according to their minds. IPR usually give the creator a complete right over the use of his/her creation for a certain period of time.

Intellectual property rights are the legal rights that cover the benefits given to individuals who are the owners and inventors of work and have created something unique with their intellectual creativity or capability. Every person related to areas such as literature, music, invention, etc., can be granted such rights, which can then be used in their business practices by them.

The creator/inventor gets complete rights against any misuse or use of work without his/her prior information. However, the rights are issued for a limited period of time to maintain equilibrium.

What is the Definition of Intellectual Property Rights?

The definition of intellectual property rights is any and all rights associated with intangible assets owned by a person or company and protected against use without consent. Intangible assets refer to non-physical property, including right of ownership in intellectual property . Examples of intellectual property rights include:

- Patents
- Domain names
- Industrial design
- Confidential information
- Inventions
- Moral rights
- Database rights
- Works of authorship
- Service marks
- Logos
- Trademarks
- Design rights
- Business or trade names

- Commercial secrets
- Computer software

What Are the Types of Intellectual Property?

There are four main types of intellectual property rights, including patents, trademarks, copyrights, and trade secrets. Owners of intellectual property frequently use more than one of these types of intellectual property law to protect the same intangible assets. For instance, trademark law protects a product's name, whereas copyright law covers its tagline.

1. Patents

The U.S. Patent and Trademark Office grants property rights to original inventions, from processes to machines. Patent law protects inventions from use by others and gives exclusive rights to one or more inventors. Technology companies commonly use patents, as seen in the patent for the first computer to protect their investment in creating new and innovative products. The three types of patents consist of:

- **Design patents:** Protection for the aesthetics of a device or invention. Ornamental design patents include a product's shape (Coca-Cola bottle), emojis, fonts, or any other distinct visual traits.
- **Plant patents:** Safeguards for new varieties of plants. An example of a plant patent is pest-free versions of fruit trees. But inventors may also want a design patent if the tree has unique visual properties.
- **Utility patents:** Protection for a product that serves a practical purpose and is useful. IP examples include vehicle safety systems, software, and pharmaceuticals. This was the first, and is still the largest, area of patent law.

2. Trademarks

Trademarks protect logos, sounds, words, colors, or symbols used by a company to distinguish its service or product. Trademark examples include the Twitter logo, McDonald's golden arches, and the font used by Dunkin'.

Although patents protect one product, trademarks may cover a group of products. The Lanham Act, also called the Trademark Act of 1946, governs trademarks, infringement, and service marks.

3. Copyrights

Copyright law protects the rights of the original creator of original works of intellectual property. Unlike patents, copyrights must be tangible. For instance, you can't copyright an idea. But you can write down an original speech, poem, or song and get a copyright.

Once someone creates an original work of authorship (OWA), the author automatically owns the copyright. But, registering with the U.S. Copyright Office gives owners a head-start in the legal system.

4. Trade Secrets

Trade secrets are a company's intellectual property that isn't public, has economic value, and carries information. They may be a formula, recipe, or process used to gain a competitive advantage.

To qualify as a trade secret, companies must work to protect proprietary information actively. Once the information is public knowledge, then it's no longer protected under trade secrets laws. According to 18 USC § 1839(3), assets may be tangible or intangible, and a trade secret can involve information that's:

- Business
- Financial
- Technical
- Economic
- Scientific
- Engineering

Two well-known examples include the recipe for Coca-Cola and Google's search algorithm. Although a patent is public, trade secrets remain unavailable to anyone but the owner.

What Are Some Examples of Violations of Intellectual Property?

The significant violations of intellectual property consist of infringement, counterfeiting, and misappropriation of trade secrets. Violations of intellectual property include:

- Creating a logo or name meant to confuse buyers into thinking they're buying the original brand
- Recording video or music without authorization or copying copyrighted materials (yes, even on a photocopier, for private use)
- Copying another person's patent and marketing it as a new patent
- Manufacturing patented goods without a license to do so

Since intellectual property can be bought, sold, or leased out, it offers many protections equal to real property ownership. Likewise, similar remedies exist. A dispute may end with property confiscation, an order of monetary damages, or cease and desist orders.

Concept of Intellectual Property Rights

Intellectual property rights are the rights given to persons over the creations of their minds, such as inventions, literary and artistic works, designs, and symbols, names, and images used in commerce. Intellectual property is the product of human intellect including creativity, concepts, inventions, industrial models, trademarks, songs, literature, symbols, names, brands, etc.

Intellectual property rights do not differ from other property rights. They allow their owner to completely benefit from his/ her product which was initially an idea that developed and crystallized. They also entitled him/her to prevent others from using dealing with or tampering with his / her product without prior permission from him/ her. He/ she can in fact legally sue them and force them to stop and compensate for any damages. It is believed that IPR initially started in North Italy during the Renaissance era. In 1474, Venice issued a law regulating patent protection that granted an exclusive right for the owner. The copyright dates back to 1440 A.D. when Johannes Gutenberg invented the printing press with replaceable/ moveable wooden or metal letters. Late in the 19th century, a number of countries felt the necessity of laying down

laws regulating IPR. Globally two conventions constituting the basis for the IPR system worldwide had been signed; the Paris convention for the protection of Industrial property (1883) Berne Convention for the Protection of Literary and Artistic works (1886).

Protection of IPR allows the innovator, brand owner, patent holder, and copyright holder to benefit from his/her work, labor, and investment, which does not mean monopoly of the intellect. Such rights are set out in the International Declaration of Human Rights, which provides for the right to benefit from the protection of the moral and physical interests resulting from the right holder's work; literal or artistic product.

Nature & characteristics of Intellectual Property Right:

(1) Intangible property:

IP[1] does not cover the created physical object but retains the conceptual development behind the physical object. Intellectual property law does not deal with the material object in which works of the mind have been represented. The main feature which distinguishes IP from other types of property is its intangibility.

IP is about a person's ability to produce a new idea and put it before the public. Intellectual property: The product does not have any property as such but the strength, judgement, and initiative of the mind that is included in the product. This generates more theoretical value. Intangible product. IPR[2] is awarded in a specific authorship work, mark, design, etc. For the integration[3] of ideas.

(2) Rights & duties

IP not only gives rise to rights of ownership but also to duties. In relation to his work/product, the IP owner has the right to conduct such actions. He is entitled exclusively to produce, copy the work, market the work, etc. He has a negative right also to rule out the exercise of his statutory rights for third parties. In this way, IPR law grants the proprietor's [4] exclusive right to exclusion [5].

The reciprocal requirement must not be infringed on the right by all others. The privilege resulting from the application of IP law gives the owner the right to use the job. Such a right does not extend to others except the IP holders.

(3) creation of statute:

Intellectual property is derived from common law, and it is covered under specific laws.

In accordance with relevant legislation, IPRs are statutory rights. Intellectual property, to put it differently, is statute formation. The right holder is protected by proposals, technical

solutions or any other knowledge conveyed in a legally acceptable manner and subject in some instances to registration procedures.

In addition in the case of certain types of IPR, as is the case for patents and product designs the registration of the work is compulsory According to the applicable legislative requirements, whereas it is voluntary to register with respect to certain other forms of IPR such as trademarks, copyrights or geographical indications.

Some other formalities must be met to obtain such IP privileges. For example, microorganism patent. Similarly, it is a must for the divulgation[6] of an invention to secure a patent. IPR award is exclusively conditional on all statutory requirements.

(4) Territoriality

Intellectual property laws are mainly territorial and apply only within the relevant competence. Although the TRIPS agreement sets the minimum standard in its respective municipal laws for all nations, the IP laws around the world are not harmoniously [7]United. Full IPRs legislation is not in effect.

To order to secure and implement their IPR, developers and inventors of different kind of IPs have to comply with their national law and jurisdictions. The extent of security depends on the national legislation in question.

(5) Assignable

They should obviously be granted[8] because they are privileges. A dichotomy between rights to intellectual property and the actual form in which the work is embedded is conceivable. IP may be bought, sold or licensed, employed, or affiliated.

(6) Dynamism

The IPR is constantly changing. The IP sector is also developing accordingly, as technology in all areas of human activities is changing exponentially. In accordance with the demand for scientific and technological advancement the scope of its defense is being extended and new items are being added to the IPRs sector.

Copyright and protection of plant variety are mentioned as a few of the current developments in the field of IPR. Technological progress and social development require a steady reassessment of the IPR system. In the technological era, developments particularly in the field of IT and biotechnology, require the regular review of IPRs.

(7) subject to public policy

They are exposed to the profound incarnation[9] of public policy. IP seeks to maintain and find a suitable reconciliation between two competing interests. On one hand, customers try to take up works without much trouble, and on the other hand, the owners of intellectual rights need to be adequately compensated.

(8) subject matter of IPR protection

Intellectual property rights eligibility depends on the protection issue. Also, products specifically identified and acknowledged in the Law as the subject of protection are entitled to intellectual property rights. While the minimum requirements laid down in law may be included, protection may still be defined if it is expressly omitted from the subject matter entitled to protection.

Origin of Intellectual Property (IP)

The Intellectual Property (IP) is a term referred to work or inventions that are a result of some individual's original creativity. It basically, means the legal rights arising out of an intellectual activity in any field like literary, industrial and artistic or industrial etc. It includes the results that are the creations of one's creative mind. Prior to the General Agreement on Tariffs and Trade (GATT), the Intellectual Property and its related rights were not a subject, to any international trade negotiations. There's a long history of the Intellectual Property, which is way complex but, also fascinating. It is traced all the way to 500 BCE, when Sybaris, a Greek State made it possible for the citizens of their state, to obtain a patent for one year, for "any new refinement in luxury". Since, then we can conclude that Patent, Copyright and Trademark laws have become more complicated over the centuries but, the intent remains the same.

The laws and legislation procedures relating to the IPR have their roots in Europe. The trend of Patents started in the 14th Century, they were technologically less advanced than England. However, the first ever known Copyrights appeared to be in Italy where, Venice was considered the cradle of Intellectual Property systems. While, the Patents are about 150 years old concept, as first introduction was based on, the British Patent System. To foster creativity and to ensure the possibility for the inventor to make benefits of their creativity.

Development of Intellectual Property

With the ongoing development around it was necessary to protect the rights of the owner of intellectual property since without protecting the innovative work that a person has spent his time on, there would be no promotion of invention and creativeness and without such development it would be impossible to expect the market grow beyond a close boundary.

Development can only be persisted if the rights of the producers and the one who are sweating over the invention are getting protection for their hard work. Without providing exclusivity to them, the market would copy each other and there would be no mark of quality and would be tough for the sellers as well as the buyers in the market to choose product of a quality.

With liberalization the challenge was bigger than before since, not only protection was needed to be given to the product in Indian market but also to protect the intellectual property rights in the international market to achieve exclusivity over the production and distribution of the same.

The same requirement was guaranteed by demonstrating India's commitment to the WTO under the Trade Related Intellectual Property Rights Agreement (TRIPS), government took a leap of steps to make sure India's stance on Intellectual Property laws being acceptable on international level.

On every level, Government of India has made its efforts to make it possible to made Indian laws pertaining to IPR are recognized at global level since accession to the WTO.

Some of the developments are;

It started with the government of India replacing Trade and Merchandise Marks Act, 1958 with the Trade Marks Act, 1999 to bring the Indian trademarks law in persistence of international practice and to make sure India is keeping up its commitment to TRIPS agreement.

The following were the changes;

- Service marks were introduced to the protection, for the first time, through registration.
- Graphic representations, shapes and combinations of colors were induced in the definition of trademark.

An unprecedented step was taken by the trademark registry by issuing an order allowing alterations that may be made to an application for trademark registration. It included "substantial alterations", which would not be allowed and other alteration that may include clerical error/alteration would be accepted by the registrar.

- Interestingly the period of registration was also increased from 7 year to 10 years.
- Under the new law, both registered and unregistered trademarks can be assigned with or without the goodwill of the business.

India also enacted Geographical Identification of goods (Registration and Protection) Act 1999 which provides for registration and better protection of G.I. relating to products that would facilitate identification of goods associated with a place of production, quality, quantity or other distinctive characteristics of such goods.

Examples of such products which are now uniquely protected under the GIG act are Basmati rice, Darjeeling tea, Alphonso mangoes, Malabar Pepper, Hyderabad Grapes etc. which are known in the international market for the quality and place of its production. These goods are of such high quality and place of production is so relevant to them, which made them stand different from any other product in the market which made such requirement of protection for them necessary.

There were constant amendments in copyright law to accommodate with the pace of development in the field and to comply with the WIPO copyright treaty and WIPO performance and phonograms treaty, which made sure that Indian laws are kept in loop for development in relation to newly introduced technological protection measures, which indeed ensured the fair use survives in the digital world by providing unique provisions for fair use, also amended author friendly changes, unique provisions for disabled, etc. to streamline copyright administration.

Recently, Government of India notified the Copyright (Amendment) Rules, 2021 on March 30, 2021, amending the Copyright rules 2013. Although, the draft of this amendment was made public for opinion in 2019. The bill is similar to draft in many dimensions. Main highlight of which is 'copyright board' being replaced by 'appellate board'.

Plenty of emphasis has been given to the amendments, which are identical to the draft, based on the functioning of copyright societies in India. This includes provisions relating to formation of traceable systems for collection and distribution of royalties, the system of dealing with royalties for works whose authors cannot be located or identified, and the facility to search from the database of a copyright society.

Recent amendments are also been made in the patent laws since The Patent (Amendment) Rules 2020 (Revised Rules 2020) came into force from 20 October 2020. Before the Revised Rules 2020, patentees and licensees were supposed to submit statement of commercial working for each annum (January to December). Since, income statements in India are generally generated for a financial year (from 1 April to 31 March), the Revised Rules 2020 will significantly ease filing of statement of commercial working of a patented invention for every financial year.

The amendments make several procedural changes such as replacing the publication of notifications in the Official Gazette with Copyright Journal, and recognition of electronic means of operation.

The ambit of IPR is ever growing and these amendments are necessary to keep up with the pace of development as it requires. Changes are imperative to this field, especially, when world is competing it has become duty of state to protect the rights of the inventors and to promote more of the welfare of the society, more of such changes are needed to be done in near future.

Kinds of Intellectual Property

Copyright

The term 'copyright' concerns the rights of the creators/authors of literary and artistic works. A copyright is also called a 'literary right' or 'author's right'. Copyright gives an author exclusive rights to his creation and prevents the copying and unauthorized publishing of his work. Copyright protection begins at the very moment a work is created and expressed in some tangible form. Copyright protection is granted to a work that is an original creation. Also, the protection extends only to expressions. Mere ideas without any tangible expression are not granted legal protection and do not form the subject matter of copyright. Copyright protects the following two rights of the author:

1. Economic rights i.e., the right of the owner to derive financial benefit from the use of their works by others. For instance, the right to prohibit or authorise reproduction of the work in various forms, the right to prohibit unauthorised translation of the work, etc.
2. Moral rights i.e., protection of non-economic interests of the author. For instance, the right to oppose changes to work and the right to claim authorship, etc.

What kind of works can be protected under copyright?

The following categories of works typically come under copyright protection:

- Literary works such as novels, plays, poems, and newspaper articles;
- Computer programs and databases;
- Films, musical compositions, and choreography;
- Artistic works such as photographs, paintings, drawings, and sculpture;
- Architecture and advertisements, maps, and technical drawings.

In India, the term of copyright protection extends throughout the lifetime of the author and then 60 years after his death.

Law relating to copyright in India : the Copyright Act, 1957

The Copyright Act, 1957 is a comprehensive legislation dealing with copyrights in India. The Act regulates the various aspects relating to copyright regime in India such as:

- Registration of copyright
- Publication, term of copyright
- Assignment, and licence of copyright
- Special rights of broadcasting organisation and performer's rights
- Infringement of copyright and remedies thereof
- Establishment of copyright authorities and copyright societies
- International Copyright

The term of copyright protection provided under the Act for the various categories of works is given below:

1. Literary, dramatic, musical and artistic works: Life of the author plus 60 years after death.
2. Anonymous and pseudonymous works: 60 years from the date of publication. However, if the identity of the author is disclosed before the expiry of that 60 years, then the term of protection shall be life of the author plus 60 years after death.
3. Posthumous works: 60 years from publication.
4. Cinematograph films: 60 years from publication.
5. Sound recordings: 60 years from publication.
6. Government work: 60 years from publication.
7. Works of public undertakings: 60 years from publication.
8. Works of international organisations: 60 years from the publication of the work.

Copyright infringement

Section 51 of the Copyright Act, 1957 provides for 'What constitutes copyright infringement'. Copyright is said to be infringed:

1. when a person does something that the owner of the copyright has the exclusive right to do, or permits for profit the use of any place for the purpose of the communication of the work to the public, where such communication constitutes an infringement of the copyright in the work, without a licence or in violation of the conditions of the licence.
2. When any person makes for sale or hire, sells or lets for hire, or displays or offers for sale or hire, or distributes either for the purpose of trade or to such an extent as

to prejudice the owner of the copyright, or exhibits in public, or imports into India any infringing copies of the work.

Section 52 enlists the acts which do not constitute an infringement of copyright such as fair dealing in any work for personal, private use or for research, reproducing any work for the purpose of a judicial proceeding or replication by a teacher or a pupil in the course of teaching etc.

It is pertinent to note that the Copyright Act provides for both civil and criminal remedies against infringement of copyright.

How to register a copyright in India

The Registrar of Copyrights maintains a Register of Copyrights wherein he enters the names or titles of works and the names and addresses of authors, publishers and owners of the copyright. This entering or recording of names and other particulars of the copyright owners in the register of copyrights is called Registration of copyright.

The procedure for registration of copyright in India is provided under Section 45 of the Copyright Act, 1957 read with Chapter XIII of the Copyright Rules, 2013.

Steps to register copyright

1. Filing of application: The author/publisher/owner or any other person interested in the copyright can make an application (Form-XIV of Copyright Rules) for registration of copyright to the Registrar of Copyrights. Such application must be accompanied by the prescribed fee for entering particulars of the work in the Register of Copyrights.

Also, an application for registration of copyright shall be in respect of one work only. It should be signed only by the applicant, who may be the owner or author of the right. In case, the application is made by the owner of the copyright, an original copy of a no-objection certificate issued by the author in the favour of the owner has to be submitted.

2. Application for registration of copyright in an unpublished work: An application for registration of an unpublished work should be accompanied by two copies of the work.

3. Application for registration regarding an artistic work that is being used or could be used in connection with any goods or services: In case the application for registration is regarding an artistic work that is or can be used in relation to any goods or services, the application must include a statement along with a Certificate from the Registrar of Trademarks that no trademark identical to or deceptively similar to such artistic work has been registered under the Trademarks Act, 1999 or no such application has been made.

4. Application for registration in respect of an artistic work which is capable of being registered as a design: In this case, the application must be supported by an affidavit declaring that:

1. The design has not been registered under the Designs Act, 2000, and
2. That it has not been applied to an article through an industrial process and reproduced more than 50 times.
3. Mode of filing the application: The application for registration of copyright can be filed in following modes:

1. By visiting the Copyright Office in person; or
2. By post; or
3. By online facility
i.e., <https://www.copyright.gov.in/UserRegistration/frmLoginPage.aspx>.

6. Notice of application: The person applying for registration of copyright has to give the notice of the application to every person who claims to have, or has any interest in the subject matter of the copyright or who is disputing the rights of the applicant to the copyright.

7. Entering of particulars in Register of Copyright: A thirty day period is given for filing of objections and if no objections to the registration are received by the Registrar, and on being satisfied that the particulars stated in the application are correct, the Registrar of Copyright shall enter such particulars in the Register of Copyrights.

8. Completion of registration process: The registration process is complete when a copy of the entries made in the register of copyrights is signed and issued by the Registrar of Copyrights or by the Deputy Registrar of Copyrights. Also, every entry made by the Registrar of Copyrights has to be published by him in the prescribed manner.

Need and benefits of registration of copyright

The registration of copyright is optional. However, the registration of copyright offers several advantages to the author or owner of copyright. This can be discerned from Section 48 of the Copyright Act. Section 48 provides that the register of copyright is prima facie evidence of the particulars entered therein and shall be admissible in evidence in all courts. Thus, a person who has got the copyright registered in his name is generally presumed to be the author/owner of the work. Registration of copyright is beneficial due to the following reasons:

- It allows the owner to protect his work from being used in an unauthorised manner.

- It becomes easier to claim ownership and royalties for your work when it is to be used or adapted in any manner.
- Copyright registration specifies the date of publication.
- Registration of copyright in your name might work in your favour in case of any claim of copyright infringement.

Patents

A patent is an exclusive right granted for an invention or innovation, which might be a product, a method or a process, that introduces a novel way of doing something or offers a new technical solution to a problem. In other words, it is a right of monopoly granted to a person who has invented:

1. a new and useful article, or
2. improvement of an existing article, or
3. a new process of making an article.

A patent is granted for inventions having industrial and commercial value. It is the exclusive right to manufacture the new article/manufacture the article with the invented process for a limited period of time (usually 20 years from the filing date of the application) in exchange for disclosure of the invention. A patent owner can sell his patent or grant licence to others to exploit the same.

Criteria for patentability of an invention

1. It should be novel.
2. It should have inventive steps or it must be non-obvious.
3. It should be capable of Industrial application.

What kind of protection is given by patents

- The patent owner possesses the exclusive right to prevent others from commercially exploiting the patented invention.
- Third parties are prevented from manufacturing, using, distributing, selling etc. the patented invention/product without the consent of the patent owner.

Patent law in India: the Patents Act, 1970

The invention of a person can be patented only if the procedure and other requirements prescribed in the Patents Act, 1970 are fulfilled. The Patent Act, 1970 provides for a detailed procedure for obtaining a patent, right from the filing of an application to the grant of a patent. The Act also contains provisions for rights and obligations of the patentee, term of the patent, transfer of patent, surrender, revocation, and restoration of patent, infringement of patent, and remedies thereof. The Act provides for patent protection for a period of 20 years after which the technology or invention goes to the public domain.

Section 3 of the Act provides a list of non-patentable inventions for which no patent could be granted. Under Section 4, the inventions relating to atomic energy are also declared as non-patentable.

It is worth mentioning that earlier no product patent could be granted for medicine, food items and chemicals and only the process of manufacturing medicines, food items and chemicals could be patented. However, after the Patent (Amendment) Act, 2005 product patents can be issued for manufacturing these products.

Patent infringement and remedies

Any violation of the rights of the patentee constitutes infringement of patent such as a colorable imitation of your invention or taking of the essential features of your invention. Under the Patents Act, Sections 47 and 107-A provides for the acts that shall not be considered as an infringement of patent. For example, the import of any machine or other articles by or on behalf of the government or the manufacturing or use of a patented process by or on behalf of the government does not constitute patent infringement. The various remedies available against patent infringement are as follows:

- Injunction
- Damages or account of profits
- Delivery up or destruction of infringing goods
- Certificate of validity

Procedure of obtaining a patent in India

You can file a patent application at the Patent Office in physical mode or in electronic mode.

Following are the steps involved in obtaining a patent:

1. **Filing of application**

- Place of filing patent application: A patent application has to be filed at the head office of the patent office or the branch office, within whose territorial limits:
 1. Applicant normally resides or has a domicile, or
 2. Applicant has a place of business, or
 3. At the place where the invention actually originated.
- Mode of filing application: You can submit the patent application through post or by hand. You can also opt for e-filing through <https://ipindiaonline.gov.in/epatentfiling/user/frmLogin.aspx>.
- Who can file the application: Following persons either alone or jointly can file the patent application:
 1. Any person claiming to be the true and first inventor of the invention;
 2. Assignee of the above in respect of the right to make such an application;
 3. The legal representative of any deceased person who was entitled to make such an application immediately before his death.
- Form of application: Every patent application shall be for one invention only.
- Every application must specify that the applicant possesses the invention and identify the individual claiming to be the true and first inventor. If the individual claiming to be the true and first inventor is not the applicant or one of the applicants, the application must state that the applicant believes the person so listed/named to be the true and first inventor.
- Application must be accompanied by a provisional or a complete specification.

2. **Filing of provisional and complete specification**

- What is patent specification: A patent specification is a technical document describing the invention. The provisional specification gives the initial description of the invention on the filing of the patent application. Whereas a complete specification gives full and sufficient detail of an invention in such a manner that a person skilled in the art can use the invention when he reads such a description.
- If the patent application is accompanied by a provisional specification, the complete specification has to be filed within 12 months from the date of the filing of such

application. In case it is not filed within the said period, the application is deemed to have been abandoned.

3. **Claim of priority date:** Priority date is the date on which the patentee claims his invention. There shall be a priority date for each claim of a complete specification. Generally, the priority date is the date of filing of the provisional specification provided the claims contained therein are fairly based on the description of the invention as given in the provisional specification. But when the patent application is accompanied by complete specification or if any application is post-dated to the date of filing of complete specification, in that case the priority date shall be the date of filing of the complete specification.
4. **Amendment of specification:** The applicant may amend the application, the complete specification and other documents before or after the grant of the patent. Such amendment shall be in accordance with the procedure prescribed as regards to the permission of the Controller and publication of the amendment.
5. **Publication and Examination of application**
 - The patent application shall not be open to the public until the expiry of 18 months from the date of filing of the application or the date of the priority of the application. However, applicants may request the Controller to publish the application at an earlier date.
 - The application is published within one month after the expiry of the said period of 18 months.
 - Thereafter, a request has to be made by the applicant or other interested persons for examination of the application. Such a request shall be made within 48 months from the date of priority of the application or from the date of filing of application, whichever is earlier. If the request is not made within the prescribed period, the application is treated as withdrawn.
6. **Time for putting application in order for grant:** The applicant must comply with all the requirements imposed on him by or under the Act in relation to the application within 12 months from the date on which the Controller forwarded to the applicant the first statement of objections to the application, complete specification, or other documents related thereto.
7. **Opposition to grant of patent**
 - Pre-grant opposition: Before the patent has been granted, any person may, in writing, represent by way of opposition to the Controller against the grant of the patent.

- Post-grant opposition: After the grant of the patent but before the expiry of 1 year from the date of publication of grant of patent, any interested person may give notice of opposition to the Controller. Thereafter, the Controller constitutes the Opposition Board and the patent may be revoked on the basis of the report of the Board.

8. Grant of patent

- If the application for patent is found to be in order for grant of patent, the patent shall be granted.
- On the grant of patent, the Controller publishes the fact of such grant and thereupon the application and other documents shall be open for public inspection.

What are the benefits of patent registration

1. Patent registration ensures the complete protection of your patent/invention against any unauthorised use for a period of 20 years.
2. Patent registration allows you to enjoy monopoly in the market as regards your invention during the period of patent protection.
3. Patent registration confers exclusive right to exploit the patent on patentee or his licensee or assignee.
4. You can licence the patent and gain royalties for the same.

Trademarks and service marks

A trademark is a symbol that is used to distinguish the goods of one enterprise from its competitors. A trademark may consist of a single letter, logo, symbol, design, or numerals and three-dimensional features such as shape and packaging, etc. Section 2(zb) of the Trademarks Act, 1999 defines “trademark” as a mark capable of graphical representation and which can be used to distinguish the goods or services of one person from those of others. A trademark may include the shape of goods, their packaging, and a combination of colours. Hence, distinctiveness is the hallmark of a trademark.

Trademarks used in connection with services such as tourism, banking, etc., are called Service Marks.

The owner has the exclusive right to the use of a registered trademark. There are 45 classes of trademarks, consisting of 34 classes of products and 11 classes for services.

What is the function/purpose of a trademark

- A trademark is a symbol that identifies a product and its source.
- It reflects the goodwill of a business.
- It assures the consumer about the established quality of the product.
- It serves as an advertisement for the product.
- A registered trademark provides legal protection to your brand.
- It helps to establish a dedicated consumer base by preventing others from imitating your brand.

Law regulating to trademarks in India: the Trademarks Act, 1999

The Trademarks Act, 1999 was enacted to provide for the registration and better protection of trademarks for goods and services, as well as to prevent the use of fraudulent marks. The Act contains provisions regarding:

1. Registration of trademarks
2. Effect of registration
3. Rights of the trademark holder
4. Special provisions relating to protection of trademarks through international registration under the Madrid Protocol
5. Use of trademark and registered users
6. Collective marks
7. Certification of trademarks
8. Assignment and transmission of trademark
9. Infringement and passing off action in trademark and legal remedies thereof, etc.

A trademark is registered for 10 years but it can be periodically renewed and can be used for an indefinite period.

Infringement of trademark

In order to constitute infringement of a registered trademark, following conditions are required to be fulfilled:

1. The person is not authorised to use the trademark.
2. The infringing trademark is similar/identical/deceptively similar to the already registered trademark.
3. The infringing trademark must be used in the course of regular trade in which the registered proprietor or user is already engaged.
4. The infringing trademark must be printed represented usually in advertisement, invoices or bills. Mere oral use of a trademark is not infringement.
5. Using either the whole of the registered trademark or an adopted one by making a few additions and alterations.

Section 29 of the Trademarks Act provides for the common forms of trademark infringement. For instance, the advertisement of a registered trademark of another for promotion of one's trade amounts to infringement. Following remedies are available to the trademark owner against infringement of his trademark:

1. Filing suit for infringement
2. Criminal remedies

Process of registration of trademark in India

The various steps involved in registration of trademark are as follows:

Application for registration

- A person claiming to be the proprietor of a trademark used or proposed to be used by him, who desires to register it has to file an application with the Registrar for registration of his trademark. Such an application shall be made in writing and must be accompanied by prescribed fees.
- A single application may be made for registration of a trademark for different classes of goods and services.
- The application has to be filed in the office of the Trade Marks Registry within whose territorial limits the principal place of business in India of the applicant is situated.

Refusal, acceptance and withdrawal of acceptance

- The Registrar may accept or reject an application after it has been received. The application may be accepted with or without amendments, modifications, conditions and limitations.

- If after acceptance, but before registration, the Registrar discovers that the application was erroneously accepted, he may withdraw the acceptance.

Advertisement of application

- The Registrar shall after acceptance of the application, cause the application to be advertised in the prescribed manner.
- The application is advertised in the Trademark Journal for the purpose of inviting objections from interested persons.
- The Registrar may cause the application to be advertised before acceptance in certain cases.

Opposition to registration

- Any person may within 4 months from the date of the advertisement give a written notice of his opposition to the registration. The notice of such registration is given to the applicant and thereafter evidence is submitted to the Registrar. After hearing the parties, the Registrar decides as to whether the registration is to be permitted or not.

Registration

- If the application for trademark registration is accepted and not opposed, or if opposed, the objection is ruled in the applicant's favour, the Registrar must register the trademark within 18 months of the filing of the application.
- The date of registration of a trademark is the date of making of the said application.
- On registration of a trademark, the Registrar shall issue to the applicant a certificate in the prescribed form of the registration thereof, sealed with the seal of the Trademarks Registry.

Benefits of trademark registration

1. A registered trademark is an intangible asset that adds value to the business.
2. Trademark registration aids in creating brand value and gaining a strong position in the market.
3. Registration of a trademark is prima facie evidence of its validity.
4. The registered trademark holder has the exclusive right to use that mark and to obtain relief in case of infringement of trademark.

5. Trademark registration is for a period of 10 years and can be renewed as well.
6. A registered proprietor of a trademark has the right to transfer his right through licence or assignment of his trademark.

Industrial designs

An industrial design means the ornamental or visual aspects of an article. It may consist of three-dimensional features, for instance, the shape of an article, or two-dimensional features, such as lines, patterns, or colour. An industrial design is purely aesthetic, non-functional, and has no utility. It is necessary to provide legal protection to the creative originality of an industrial design to prevent others from copying it.

Type of protection provided by industrial design

The owner of registered industrial design reserves the right to prevent others from manufacturing, selling, or importing articles bearing or embodying a design which is a copy of or is substantially similar to the protected design.

Kinds of products that can come under Industrial design protection

- Products of industry and handicraft items
- Household goods
- Lighting equipment
- Jewellery
- Electronic devices
- Textiles, etc.

Law relating to designs in India: the Designs Act, 2000

The Designs Act, 2000 seeks to promote the creation of novel, original designs along with balancing competing interests by granting the time-bound monopoly right to use registered industrial design by the owner. The Act contains provisions regarding registration of designs, copyright in registered designs, industrial and international exhibitions, restoration of lapsed designs, the penalty for infringement of registered designs, etc.

Geographical Indications (GI)

A geographical indication (GI) is used to identify goods having a specific geographical origin. These indications denote quality, reputation, or other characteristics of such goods essentially attributable to their geographical origin. Generally, geographical indications are used for foodstuffs, agricultural products, wine, industrial products and handicrafts. Examples of GI include Basmati Rice, Darjeeling Tea etc.

Benefits of registration of GI

- Confers legal protection to domestic/national GI which in turn boosts exports.
- Prevents others from making unauthorised use of a Registered Geographical Indication.
- Promotes the economic well-being of producers of items produced in a specific geographic area.

Law relating to GI in India: the Geographical Indications of Goods (Registration and Protection) Act, 1999

The Geographical Indications of Goods (Registration and Protection) Act, 1999 provides for the registration and better protection of geographical indications relating to goods. The Act contains provisions relating to the establishment of a Geographical Indications Registry, registration of geographical indications of goods, rights conferred by registration, registration of authorised users of registered geographical indications, provisions for renewal, rectification and restoration of geographical indications, and prohibition of registration of geographical indication as a trade mark, etc.

Trade Secrets

Trade Secrets are IP rights on confidential information which may be sold or licensed. A trade secret refers to any confidential business information and may include designs, drawings, plans, business strategies, R & D related information, etc. In order to qualify as a trade secret, the information should be commercially valuable i.e. useful in a trade or business, known to a small number of people, and subject to reasonable steps taken by the rightful holder of the information to keep it secret.

Types of trade secrets

- Technical information such as information regarding manufacturing processes, designs, drawings of computer programs, etc.

- Commercial information, such as distribution methods, advertising strategies, etc.
- Financial information, formulas, recipes, secret combination of elements, source codes, etc.

Layout designs of integrated circuits

Integrated circuits are used in products such as television, radio, mobile, washing machine, and data processing instruments. The layout designs of integrated circuits not only reduce the space but also enhance the capacity and performance of the system. In India, the Semiconductor Integrated Circuit Layout Design Act, 2000 regulates the registration, use, and protection of original and distinct layout designs.

The Semiconductor Integrated Circuits Layout Designs Act, 2000

The Act deals with the protection of Semiconductor Integrated Circuits layout designs. It has been enacted to give effect to Section 6 in Part II of the TRIPS Agreement relating to Layout-Design (Topographies) of Integrated Circuits. The Act contains provisions relating to registration of Semiconductor Integrated Circuits layout designs including the procedure and duration of registration, the effect of registration, assignment and transmission of registered layout-design, use of layout-design, and penalty for infringement of layout-design, etc.

The Intellectual Property System in India

The origins of India's IPR system date back to British colonial rule, when as a colony the state enacted various rules and enforcement mechanisms pertaining to IP rights. Post-independence, India retained elements of these structures while updating some guiding regulations and other bureaucratic structures. As India moved toward liberalization, privatization, and globalization in the 1990s and later, Indian policymakers made further adjustments to keep up with growing needs of domestic and international stakeholders. Indian IPR Laws fully conform to the Agreement on Trade Related Aspects of Intellectual Property Rights under WTO aegis.

IPR Policy 2016

Adopted in May 2016, the IPR Policy is a giant leap by the Government of India to spur creativity and stimulate innovation. It lays the roadmap for the future of IPRs in India. The Policy seeks to reinforce the IPR framework in the country that will create public awareness about economic, social and cultural benefits of IPRs among all sections of the society, stimulate IPR generation and commercialization, modernize and strengthen service-oriented IPR administration (see **Annexure I**) as also the enforcement and adjudicatory mechanisms for combating IPR infringements.

Vision Statement of the Policy

(To create) An India where creativity and innovation are stimulated by Intellectual Property for the benefit of all; an India where intellectual property promotes advancement in science and technology, arts and culture, traditional knowledge and biodiversity resources; an India where knowledge is the main driver of development, and knowledge owned is transformed into knowledge shared.

The Policy lays down seven objectives which are elaborated with steps to be undertaken by the identified nodal Ministry/ Department. The objectives of the Policy are:

- 1) **IPR Awareness: Outreach and Promotion** - To create public awareness about the economic, social and cultural benefits of IPRs among all sections.
- 2) **Generation of IPRs - To stimulate the generation of IPRs:** India has a large talent pool of scientific and technological talent spread over R&D institutions, enterprises, universities and technical institutes. There is a need to tap this fertile knowledge resource and stimulate the creation of IP assets. (see **Annexures II and III** for no. of applications for patents and top patentees.)
- 3) **Legal and Legislative Framework** - To have strong and effective IPR laws, which balance the interests of rights owners with larger public interest.
- 4) **Administration and Management** - To modernize and strengthen service- oriented IPR administration.
- 5) **Commercialization of IPR** - Get value for IPRs through commercialization.
- 6) **Enforcement and Adjudication** - To strengthen the enforcement and adjudicatory mechanisms for combating IPR infringements.
- 7) **Human Capital Development** - To strengthen and expand human resources, institutions and capacities for teaching, training, research and skill building in IPRs.

The present IPR Policy aims to integrate IPR as a policy and strategic tool in national development plans. It foresees a coordinated and integrated development of IP system in India and the need for a holistic approach to be taken on IP legal, administrative, institutional and enforcement related matters.

IPRs- Invention and Creativity

What is innovation?

So **what does innovation really mean?** And how does it differ from **invention** and **creativity** in a business? Let's start from a definition of innovation, invention and creativity:

- **Innovation** is the process of turning a new concept into commercial success or widespread use.

- **Invention** is the creation of a new idea or concept.
- **Creativity** is the act of turning new and imaginative ideas into reality.

Creativity, invention, and innovation are all interrelated and necessary for growth to occur. We can follow any successful company and see an investment of time and effort into these three concepts. Based on our experience and research conducted, we have discovered these insights:

Creativity

Creativity is the act of channeling imagination into something concrete. It is the very first stage of design, where ideas start to actually take form, and a plan can be developed.

Examples of creativity:

- in the case of Uber, creativity was necessary in producing the **concept of ride sharing** and of making driving jobs accessible to any individual.
- another example can be the **Google Glass**, which although were very creative at the time (new, different in the market), were not necessarily innovative (monetary return over investment made by the company).

Invention

Invention is the physical creation of a new concept or idea. Also, from Nesta's innovation policy toolkit, it is defined as The creation of an idea to do or make something without verification that it works, or is commercially valuable. Invention cannot occur without creativity, but just creativity is not enough to properly develop an idea.

Examples of invention:

- **product designs, business models, or working prototypes.**
- Uber channeled invention in creating a solid and working business model, based upon the ideas they had formed in the creation stage.

Innovation

Innovation ties everything together in business. According to British Standards, **Innovation is the successful exploitation of new ideas.** It involves turning an invention into a commercial success and encouraging its widespread use. This is the stage where bold ideas, once properly

developed, are brought to the public eye. This is what Uber has achieved, now that its product is in use all over the world and benefits people everywhere.

Examples of innovation:

- successful **products and services** like Care of, Amazon Scout, Impossible Food, Google's self-driving taxis, ...
- successful **process innovations** like Amazon predictive orders, or additive manufacturing processes.

Once again, Innovation cannot happen without creativity or invention, and neither creativity nor invention is useful in business if not properly executed.

As can be clearly seen in the example of Uber, **all the three steps are crucial in order to create a profitable business.**

Uber achieved innovation through creative ideas, a profitable business plan, and widespread marketing, all of which allowed the product to completely change the car sharing industry. These three stages are steps that will allow you to create and commercialize your product, and make your business grow and succeed.

Importance of IPR

Positive Impacts :

IPRs play very important role in the progress and development of the society. IPRs not only provide incentive to the creator of his creation but also lead to a healthy competition among creators which ultimately leads to the progress of the society. Some of the positive impacts of IPR are:

IPRs are important for free flow for energy for enhancing invention and research. IPRs provide incentive to the individuals for new creations. IPRs provide due recognition to the creators and inventors. These laws provide them both the means and incentive to create newer works, products and services

Intellectual Property Rights enhance innovation and creativity by protecting the rights of inventors and artists. Since the filing of patents requires the disclosure of information that would enable others to replicate the inventor's discovery, others can use and build upon this shared knowledge to create newer and/or better products. IPRs ensure material reward for intellectual property.

IPRs ensure the availability of the genuine and original products. Intellectual Property rights, such as patent and copyrights, are an important means used by firms to help protect their investments in innovation (Shankar Narayanan, 2010).

IPRs may be helpful in the solution to global challenges like in the field of alternate sources of energy, new products to the farmers and development of low cost drugs for poor people.

IPRs are necessary to stimulate economic growth. Protection of intellectual property rights is essential in maintaining economic growth. They encourage fair trading which would contribute to economic and social development. Effective enforcement of intellectual property rights is critical to sustaining economic growth across all industries and globally.

Negative Impacts:

Though the purpose of IPRs is to enhance innovation and creativity by protecting the rights of creators and leading to the growth and advancement of human beings but do IPRs actually play a constructive role in the progress and development of society and mankind or it is a form of intellectual protectionism or a form of a temporary monopoly enforced by the state, is an issue of great concern.

They are considered to benefitting concentrated i.e. IPRs confer authority over resources to a few. The few gain power over the goals of many. The objective of IPRs is to protect the public interest but, in fact, the public interest is harmed. Pharmaceutical product prices form substantial portion of health care costs and strong intellectual property protection is one of the major reasons for high health care costs (Agatha, 2013). For example, patent of life saving drugs had allowed the countries to charge higher than the marginal cost of the production in the name of cost of research and development. This has led to an increase in the cost of drugs and they are not affordable by the poor sections of the society. When there is control by some particular group, IPRs actually discourage invention. Patent of the ideas may prevent the owner of a property to utilize it according to his wishes. In order to serve the purpose IPRs must focus on the needs of the poor and developing countries but in reality intellectual properties tend to be governed by economic goals preventing the progress of poor. Thus, IPRs are considered as intellectual protectionism, intellectual monopoly or government-granted monopoly by which the public interest is harmed and progress is stopped.

Protection of Intellectual Property Rights (IPRs)

Intellectual property rights (IPR) promote innovation and creativity, which in turn generates jobs and improves competitiveness. Intellectual property rights enable authors, artists, designers, inventors

and other IPR users to benefit when others use their creations and inventions. The EU needs an attractive, affordable and efficient intellectual property rights (IPR) system to compete on the global scale. This is particularly important for SMEs who do not have the same level of resources to manage their IP portfolio as bigger companies.

Patents: An exclusive right granted for a new technical invention. A patent holder can grant a licence to somebody wishing to produce copies of the invention against payment of a fee (or royalty), thus obtaining a return on the investment. A **standard essential patent (SEPs)** is a patent essential to implement a specific industry standard or technical solution. For example, smartphones or tablets connect to the internet via standardised technologies such as WiFi or 4G, which are protected by SEPs.

Trademarks (or 'brands'): A word, logo or symbol that competitor companies may not use once it has been protected.

Copyright ensures that authors, composers, artists, film makers and other creators receive recognition, payment and protection for their works such as books, films, musical works, paintings, photographs

Designs: The outward appearance of a product;

e.g. the shape of a smartphone, vacuum cleaner, loud speaker, lamp, armchair or a car

Geographical indications: The name of a location associated with a product (e.g. Champagne or Prosciutto di Parma) which gives an assurance that the product has been produced in a particular place, often according to traditional practices or recipes.

Trade secret: A valuable piece of know-how and business information that is treated as confidential because it gives the company a competitive advantage (for example, a recipe, the results of a marketing study, the launch date of a new product, or the price offered in a bidding procedure).

A brief summary of: Patents

What Is a Patent?

A patent is the granting of a property right by a sovereign authority to an inventor. This grant provides the inventor exclusive rights to the patented process, design, or invention for a designated period in exchange for a comprehensive disclosure of the invention. They are a form of incorporeal right.

Government agencies typically handle and approve applications for patents. In the United States, the U.S. Patent and Trademark Office (USPTO), which is part of the Department of Commerce, handles applications and grants approvals.

Understanding Patents

Most patents are valid for 20 years in the U.S. from the date the application was filed with the USPTO, although there are circumstances where exceptions are made to extend a patent's term. U.S. patents are only valid in the United States and U.S. Territories. If seeking protection outside of the United States, it is important to research the intellectual property rights of other nations and apply for protection with their governing authorities.

According to the U.S. Patent and Trademark Office, a patent can be granted to any person who:

Invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent, subject to the conditions and requirements of the law.¹

Types of Patents

There are three types of patents available in the United States: utility patents, design patents, and plant patents. Each has its own specifications and durations.

Utility Patents

Utility patents, or patents for invention, issue legal protection to people who invent a new and useful process, an article of manufacture, a machine, or a composition of matter. Utility patents are the most common type of patent, with more than 90% of patents issued by the U.S. government belonging to this category.² A utility patent lasts for 20 years from the date of filing as long as maintenance fees are paid. Maintenance fees are surcharges applied to utility patent applications filed after December 12, 1980.³

United States Patents and Trademarks Office. "Maintain Your Patent."

Design Patents

Design patents are patents issued for original, new, and ornamental designs for manufactured products. Design patents protect the design or look of something. They require the invention to which the design belongs to be original and useful. Design patents last for 15 years for applications filed after May 13, 2015. For applications filed before May 13, 2015, patents last for 14 years from the date of the filing. Maintenance fees do not apply to design patents.²

Plant Patents

Plant patents go to anyone who produces, discovers, and invents a new kind of plant capable of reproduction. These patents are granted for 20 years from the date of filing and no maintenance fees apply.²

Patents provide an incentive for companies or individuals to continue developing innovative products or services without the fear of infringement. For example, large pharmaceutical companies can spend billions of dollars on research and development. Without patents, their drugs and medicines could be duplicated and sold by companies that didn't research or invest the needed capital for R&D.

In other words, patents protect the intellectual property of companies to help their profitability. However, patents also serve as bragging rights for companies demonstrating their innovativeness.

How to Apply for a Patent

Before making a formal application, an applicant should research the Patent and Trademark Office's database to see if another person or institution has claimed a patent for a similar invention. The invention must be different from or an improvement upon a previous design to be considered for a patent. Applicants need to take care to maintain accurate records of the design process and the steps taken to create the invention. Enforcing the patent is up to the person or entity that applied for the patent.

To apply for a patent in the United States, the applicant submits specific documents and pays associated fees. Written documentation includes drawings, descriptions, and claims of the item to be patented. A formal oath or declaration confirming the authenticity of the invention or improvement of an existing invention must be signed and submitted by the inventor. After fee payment, the application is reviewed and either approved or denied.

Examples of Patents

One of the most notable patents in the past 40 years was the personal computer filed in 1980 by Steve Jobs and three other employees of Apple Inc.⁵

King C. Gillette patented the razor in 1904 and was dubbed a "safety razor."⁶ Garrett Morgan was granted a patent for the traffic light in 1923.⁷ The patent for the television was issued in 1930 to Philo Taylor Farnsworth for the "first television system."⁸

United States Patent and Trademark Office. "Patent 1,773,980."

At age 20, Farnsworth had created the first electric television image and went on to invent an early model of the electronic microscope.

IPR Status of India

India currently ranks 60th out of 127 countries on the Global innovation Index (GII) 2017, co-published by World-Intellectual Property Organization (WIPO), Cornell University and INSEAD¹³ with Confederation of Indian Industry & others as Knowledge Partners

India ranks 14 overall in the presence of global R&D companies, considerably better than comparable groups of lower- and upper-middle-income economies. India also surpasses most other middle-income economies in science and engineering graduates, gross capital formation, Gross Expenditure on Research and Development (GERD) performed by business, research talent, on the input side; quality of scientific publications, growth rate of GDP per worker, high-tech and ICT services exports, creative goods exports, high-tech manufactures, and IP receipts on the output side.

A brief summary of: Copyrights

What Is Copyright?

Copyright refers to the legal right of the owner of intellectual property. In simpler terms, copyright is the right to copy. This means that the original creators of products and anyone they give authorization to are the only ones with the exclusive right to reproduce the work.

Copyright law gives creators of original material the exclusive right to further use and duplicate that material for a given amount of time. Once a copyright expires, the copyrighted item becomes public domain.

How Copyrighting Works

When someone creates a product that is viewed as original and that required significant mental activity to create, this product becomes an intellectual property that must be protected from unauthorized duplication. Examples of unique creations include:

- Novels
- Art
- Poetry
- Musical lyrics and compositions
- Computer software
- Graphic designs,
- Film
- Original architectural designs
- Website content

One safeguard that can be used to legally protect an original creation is copyright. Under copyright law, a work is considered original if the author created it from independent thinking void of duplication. This type of work is known as an Original Work of Authorship (OWA).

Anyone with an original work of authorship automatically has the copyright to that work, preventing anyone else from using or replicating it. The copyright can be registered voluntarily by the original owner if they would like to get an upper hand in the legal system in the event that the need arises.

Not all types of work can be copyrighted. A copyright does not protect ideas, discoveries, concepts, or theories. Brand names, logos, slogans, domain names, and titles also cannot be protected under copyright law. For an original work to be copyrighted, it has to be in tangible form. This means that any speech, discoveries, musical scores, or ideas have to be written down in physical form in order to be protected by copyright.

In the U.S., original owners are protected by copyright laws all of their lives until 70 years after their death. If the original author of the copyrighted material is a corporation, the copyright protection period is 95 years from the date of publication or 120 years, whichever expires first.

U.S. copyright law has experienced a number of amendments and changes that have altered the duration of copyright protection. The "life of the author plus 70 years" protection can be attributed to the 1998 Copyright Term Extension Act, (also known as the Mickey Mouse Protection Act or Sonny Bono Act), which generally increased copyright protections by 20 years.

A brief summary of: Trademarks

Trade mark is a mark applied to any product or service that sets one producer's products or services apart from similar products made by another company. Any product or service with a trade mark on it gives the buyer the idea that it has that quality.

Definition:

Trade Mark **Sec 2(1)(zb)** Mark capable of represented graphically and capable of distinguishing goods/services of one person from those of others. It shows a connection between goods and person.

Procedure for Registration of Trade Mark

Under Section 3 (the "Controller General of Patent, Design and Trade Marks"), the government established the Trade Marks Registry, where applications for trademarks may be filed. Several

classes of goods or services are eligible for trademark grants. The registration of a trademark serves as prima facie proof that all legal requirements have been duly met. At the Trade Mark Registry's head office, a register of trademarks is kept. Any trade mark may be submitted for registration under Section 18 by the individual claiming ownership. These applications may be turned down for one of two reasons: Absolute Justifications for Rejection According to Section 9, the Registrar may reject a trademark application on the following grounds in its exclusive discretion:

If mark is devoid of any distinctive character, e.g. where any character of any language or any shape has not been especially designed or visible.

- Any mark which shows just kind, quality, quantity or intended purpose of any goods, e.g. trademark of any goods can't be "500 gram"
- Mark consisting of marks or indications which are customary in current language, e.g. OMG (abbreviation of Oh My God) cannot be trade mark.
- However where trademark which bears distinctive character because of its use, or is well-known to public is allowed, e.g. "Sugar-free" mark shows just quality but still allowed because this product is well-known to public
- If a mark is capable to deceive or cause confusion to public, or hurt religious susceptibility, or is obscene, it will not be registered → Mark consisting of shape of natural goods (e.g. Neem leaf) or such shape which is necessary to obtain a given technical result shall not be registered.

Grounds for Rejection

Sec 11 (as amended by Trade Mark (Amendment) Act, 2010) provides that Registrar will reject the trademark application on relative grounds which are:

- Where mark is identical and affixed to similar goods or service, e.g. Lux washing powder
- Where mark is similar and affixed to identical or similar goods or service, e.g. Nirama washing powder (Original is Nirma)
- Where mark is identical or similar and affixed to goods which are not similar, if such mark is well-known to public e.g. Lux shoes

- Where mark's use is prevented by law of passing-off or copyright However, Registrar shall not reject any application for trade mark on the grounds specified in point c and d supra, unless an objection is raised by proprietor of earlier trade mark.

Sec 14 provides that if any mark falsely suggests a connection with a living person or a person who died within 20 years prior to application of trade mark, then Registrar can demand consent of such person or his legal representative.

Registered Trade Mark

According to Section 20, the Registrar must publish a trade mark application whenever it is submitted to him in the manner specified. Within three months of the date the application was published, anyone may object. Trademark is granted if application has not faced objection or if opposition has been rejected in the applicant's favour. Trademarks have a 10-year term with the option to periodically renew them for additional 10-year terms.

The Registrar may delete a trade mark if it is not renewed after 10 years, however it may be reinstated within a year of the removal date. The Registrar may permit a mark to be registered as an associated trade mark when the owner of a trade mark applies for another mark that is the same as or similar to an earlier registered trade mark.

Initial registration is prima facie evidence of its legality; nevertheless, the owner of a registered mark cannot intervene if someone is using a mark that is confusingly similar to one that was registered earlier. Owners of unregistered marks may only bring passing-off claims and not infringement claims. In general, the order of use determines who owns a mark. Priority can be established simply by a prior sale of the item.

Infringement of Registered Trade Mark

Sec 29 provides that if a person who is not registered proprietor of mark uses a mark which is identical or deceptively similar to registered trade mark, it is called infringement. Further, registered trade mark is infringed if:

1. Mark is identical and is used in respect of similar goods/services, or
2. Mark is similar and is used in identical or similar goods/services, or
3. Mark is identical and is used in respect of identical goods, and It is likely to cause confusion on the part of public. Further, following action is also called infringement;

4. Mark is identical or similar to registered trade mark and used in respect of goods which are not similar,
5. Using someone else's trade mark as his trade name as dealing with similar goods.
6. Affixing someone else's trade mark to goods or packing or using such mark in business papers or advertisements.

What is Not Infringement of Trade Mark:

Where mark is used with honest business practice without taking unfair advantage, it is not deemed to be infringement. Following acts are not infringement:

1. If trade mark is used to indicate kind, quality or quantity etc of any goods/service.
2. If trade mark was registered subject to certain conditions.
3. Where mark is used in such types of goods/services for which owner of that mark had impliedly consented to its use.
4. Registered trade mark can be used in relation to parts and accessories to other goods/services.

Offence and Penalties:

Following penalty provisions are prescribed under this Act:

1. Penalty of imprisonment of 6 months to 3 years and fine upto Rs 2 lakh for applying false trade mark or false trade description.
2. Any removal or sale of cotton yarn or cotton thread which are not marked in accordance with provisions of this Act is liable to forfeited.
3. Any person who falsely represents a trade mark to be registered trade mark is punished under this Act.
4. In case of offence is committed by Company, every person in charge of the Company, as well as the Company shall be deemed to be liable. Where such contravention is committed with consent or with negligence on the part or director, manager or secretary, they shall also be deemed to be guilty.

A brief summary of: TRIPS and TRIMS

What is TRIPS?

The Trade-Related Aspects of Intellectual Property Rights (TRIPS) adheres to member states' core trade principles concerning intellectual property; all WTO members must abide by it. The TRIPS Agreement stipulates reasonable exceptions and constraints to reconcile intellectual property rights with community health and economic progress objectives.

The most robust international IP agreement, TRIPS, tends to play an influential role in fostering commerce in creativity and knowledge, addressing trade disputes comprising intellectual property, and enabling WTO members with the autonomy to pursue their domestic policy goals. The agreement also shapes the intellectual property system regarding innovations, technology transmission, and public welfare.

The TRIPS Agreement must be assessed and implemented under the jurisdiction of the TRIPS Council. The TRIPS agreement was outlined at the time of the Uruguay Round that lasted from 1986 to 1994 for GATT. The terminology "Berne and Paris-plus" also refers to the TRIPS Agreement.

What is Intellectual Property Rights?

Intellectual property is an umbrella term for a collection of intangible or non-physical assets and is granted to individuals over their creative creations. Intellectual property is the right to use and control one's intellectual property. It is a legal right for anyone who owns or has an intellectual property right. Therefore, individuals, businesses, and governments must defend their intellectual property. To protect intellectual property rights, individuals and communities must comprehend how they work and the rights they grant to someone. The categories of intellectual property rights include

- industrial designs
- trade secrets
- exclusionary rights over new plant varieties
- geographical indications
- trademarks
- copyright
- patents

Significance of TRIPS

The TRIPS Agreement has its key focus on safeguarding intellectual properties in the World Trade Organization and the international trading system. The TRIPS agreement is often a pillar of the World Trade Organization. The remaining two are trading in services and commodities (the traditional competence of the GATT).

Before TRIPS, there were notable sectoral discrepancies in the range of IP protection and enforcement. However, as IP grew more crucial to commerce, these discrepancies became a focus of contention in international economic relations. To have more order and reliability and to adjudicate controversies in an orderly manner, it was therefore thought advisable to endorse new trade regulatory requirements for IP rights.

Relaxation Favoring TRIPS

The following are the favoring relaxations associated with the TRIPS agreement:

- This would increase access to immunizations for residents of LDCs and developing nations.
- Everybody must be exposed to life-saving medications, medical help, and vaccines. Thus, pharmaceutical enterprises shouldn't strive to make an extraordinary profit from them. It creates moral and ethical dilemmas at stake here.
- Individuals are not safe until everyone is safe, a saying that applies primarily to the COVID-19 outbreak. However, given that it can swiftly spread to all continents, as was the scenario with the first wave, vaccines must be made obtainable to everyone in the affected countries.
- Rules that award monopolies and insert the right to primary healthcare under serious jeopardy must be repealed.

Contradicting Relaxation of TRIPS

Let's discuss some of the contradicting relaxations associated with TRIPS:

- Corporations wouldn't be able to retrieve the funds they invested in R&D unless they were reimbursed for their invention.
- Without the potential to dominate output, there won't be any motivation to encourage innovation.
- They also assert that corporations in impoverished countries are unwilling to generate immunizations or pharmaceuticals on a large scale.

Data Exclusivity and TRIPS

Data exclusivity is the practice of excluding data from use in a research study. It is a form of data mining that allows researchers to access data without handing over the raw data. This can be used to examine the impact of a new research technique or find out what factors affect how well a product performs. Data exclusivity can be used by researchers who want to gain insight into their subject's performance or by those who want to understand the factors that influence how well a product performs.

Data exclusivity can also be used by researchers who want to understand how much data they need to make an informed decision about their research topic. It can also be used by those who want to learn more about the subject under study and how well it performs. Data with high commercial value needs to be well protected. The developed world has put forth the demand for such protection that is not part of the TRIPS Agreement.

TRIPS Agreement - TRIPS Plus

They are not officially associated with TRIPs, despite the moniker "TRIPS-Plus." Instead, the phrase emphasizes that these expectations go above and beyond the fundamental commitments set forth under TRIPs. There is pressure on many FTA-affiliated developing nations to embrace these harsher requirements in their patent legislation.

TRIPS Agreement - Doha Declaration

In 2001, the Doha Declaration was ratified by WTO members. It reiterated TRIPs' adaptability in situations incorporating patent rights. This was done to facilitate access to appropriate medications. According to the Doha Declaration, the TRIPS agreement will not preclude governments from undertaking initiatives to protect public health. This loosened TRIPS guidelines. Everyone has recourse to life-saving medicines and immunizations. Pharmaceutical firms must not benefit from these.

Fears Expressed over the Doha Declaration

Corporate entities contended that they would incur losses relative to the sums invested in their research and development of their inventions were not recognized and compensated. Additionally, removing the right to monopoly in the pharmaceutical industry will lessen the incentives for innovation. Finally, the businesses contended that developing-nation companies could not produce medications and vaccines on a significant scale.

TRIMS

The agreement on Trade-Related Investment Measures or “TRIMs” acknowledges that investment measures can control and twist trade. This states World Trade Organisation members might not apply any measure that differentiates or leads to quantitative restrictions, both of which disobey any fundamental WTO rules.

A list of restricted TRIMS is a part of this agreement. The TRIMS Committee looks after the operation and execution of this agreement and allows members the chance to consult on any appropriate matters.

Agreement on TRIMs resulting from Uruguay recognises that certain investment measures may cause restrictive effects on international trade in goods.

Objectives of Trade-Related Investment Measures

TRIMs believe that there is a strong connection between trade and investment. The goal of trade-related investments measures is to give fair treatment to all investing members across the world.

As the TRIMs deal says, members have to inform the World Trade Organization (WTO) council to buy and sell various services and goods of their current TRIMs that are incompatible with the agreement.

Main Features of TRIMs

- It only applies to investment measures related to goods trade.
- This doesn't apply to service trade.
- It doesn't regulate the entry of foreign industry or investment.
- It is about the discriminatory treatment of imported/exported products.
- Concern measures were applied to both foreign domestic firms.
- A transition period of 2 years in the case of developed countries, 5 years in the case of developing countries and 7 years in the case of LDCs, from the date this agreement came into effect, which is 1st January 1995.

The TRIMs agreement has directly restricted the following.

1. Local Content Requirement

Local content requirement is the measure that if a developed country wants to trade their products in a developing country. Then the developing country wishes to agree on that deal if

and only if it uses one of the domestic items on their product. It means that the growth of domestic products will increase.

For example, suppose a mobile brand wants to sell its products in India. However, India agrees if only the brand uses the battery made in India. If that mobile brand agrees to this agreement, the value of 'made in India' batteries will increase gradually, which is India's benefit.

However, that mobile brand doesn't get proper profit because of purchasing those domestic products or using those products. In other words, a developing country is restricting developed countries from trading their products.

2. Trade Balancing Requirements

Trade balancing measures require that an enterprise's purchase of imported products is equivalent to the quantity or worth of exported products. In simple words, someone from India has a business of desi ghee, and they export it in foreign.

However, they want to expand their business and import foreign cheese. Then, Indian government gave them the condition that they could import the same amount of foreign cheese as the amount of desi ghee they export. Basically, balancing the import and export amount of product or trade is called Trade Balancing.

However, the government is restricting the individual as per the TRIMs agreement.

1. Foreign Exchange Restrictions

These include measures restricting importation by limiting access to foreign exchange.

2. Domestic Sales Requirements

By using domestic sales requirements, many nations restrict the export of domestic products and distort trade. Because of this, the value of those products gradually decreases. As a result, the production of those products is highly available in the market.

Developing countries are permitted to retain Trade-Related Investment Measures by virtue of the economic development needs of developing countries. In TRIMs, some restrictions are overlooked for the developing countries' economic needs.

Industrial Design

What is an Industrial Design?

The owner of a design patent or an industrial design that's registered has the rights to stop third parties from the following:

- Selling articles that have a copy of the design
- Making articles with a similar design
- Importing articles that bear a design that's a copy of the original one

Copies of any form are not allowed when the purpose is for commercial use. There is a wide range of products of handicraft and industry items that these designs apply to. This includes containers, packages, household goods, furnishing, jewelry, lighting equipment, textiles, and electronic devices. Industrial designs also apply to graphical user interfaces, logos, and graphic symbols. Most countries require an industrial design to be registered so it can be protected under industrial design and considered a registered design. Some countries allow unregistered industrial designs and scope limited protection.

Depending on what the design is and what the national law is, industrial designs can be protected under copyright law as works of art. Most countries grant industrial design rights for at least 10 years, although the duration varies in each country. Many countries allow the total duration the rights are protected for to be divided into renewable periods that are successive. These rights are enforced in court and the penalties and remedies vary in each country. The industrial design right shields the aesthetic or appearance features of a product, while an invention is protected by a patent with a new solution.

How Can One Protect Their Designs?

Industrial design protection covers a color, shape, surface pattern, line, or configuration of an article that increases or produces aesthetics and improves the appearance of a design, whether it's two-dimensional or three-dimensional. Protection rights last for a period of 10 years and can be renewed one time for an extra period of five years. In order for a design to qualify for protection, it must be non-obvious, visible on the finished article, applicable to the functional article, and have no prior disclosure of publication of the design.

Applying for Industrial Design Protection

Depending on what the laws are, industrial designs that are independently created must be novel or original. This will vary in each country and is usually considered to be novel or new if it hasn't been disclosed to the public previously. It's considered original if the design is different from other known designs. The office of the intellectual property grants industrial design patents or registrations based on the country where the application gets filed. An application to register the industrial design may be filed or one may be able to appoint an agent in some cases.

The filing cost for protection differs in each country, so it's best to talk to the national intellectual property office to find out more details on the exact fee structure. If filing for protection abroad, it's important to note that there will be more costs for using a local agent and for translation. Industrial design rights are considered territorial, meaning the rights are limited to the region where protection has been granted. No international or world industrial

design right exists. In order to get protection from another country, an application must be submitted for the industrial design for each country.

In order to avoid submitting multiple applications, WIPO's Hague System allows companies to register a maximum of 100 designs in a wide number of territories through one application. It's essential to file the application for registration before the patent is publicly destroyed. This way it retains its originality and novelty. Once the design gets disclosed to the public, it might become part of the public domain and is not considered original or new anymore.

Some countries do allow a grace period of six to twelve months to file after the disclosure of the design. This creates a safeguard for those who disclosed their design before filing for protection.

Integrated Circuits

The conventional method of making circuits was to select components like R, C, L, diode and semiconductors. There are so many factors stopping to build off big circuits like:

1. Bulky in size.
2. Not entirely shockproof
3. Reliability
4. More power consumption
5. Less durability

The concept of fabricating an entire circuit on a single small block (or chip) of a semiconductor has revolutionised electronics technology. Such a circuit is known as Integrated Circuit (IC).

What is an Integrated Circuit (IC)?

Before the discovery of ICs, the basic method of making circuits was to select the components like diodes, transistors, resistors, inductors and capacitors and connect them by shouldering. But due to size and power consumption issues, it was necessary to develop a small size circuit with less power consumption, reliability and shockproof.

After the invention of the semiconductors and transistors, things were quite simplified to a particular extent, but the development of integrated circuits changed electronics technology's face. Jack Kilby from Texas Instruments and Bob Noyce from Intel are the official creators of integrated circuits, and they did it independently.

The integrated circuit is a fundamental concept of electronics that builds on other basic concepts previously discussed in our syllabus. Therefore, for a quick reference, go through the articles listed below:

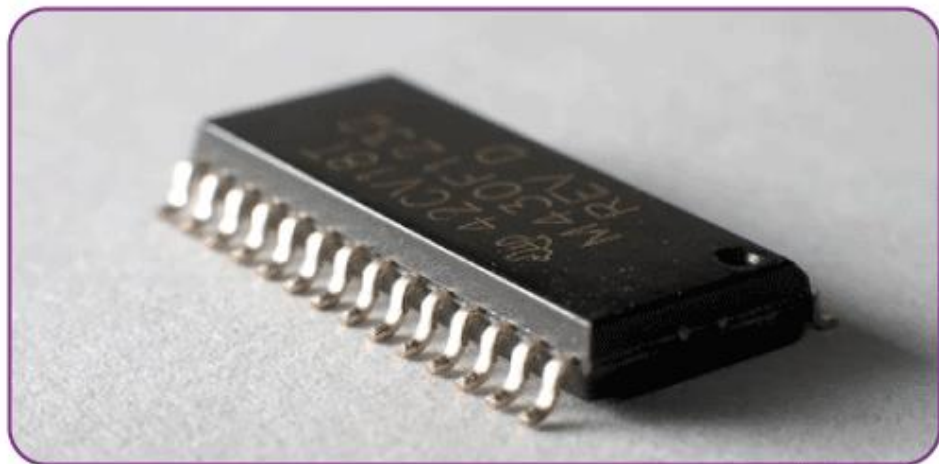
- Electric Circuits
- Resistors

- Transistors
- Diodes
- Capacitors

Definition of Integrated Chip

Integrated circuits are made up of several components such as R, C, L, diodes and transistors. They are built on a small single block or chip of a semiconductor known as an integrated circuit (IC). All of them work together to perform a particular task. The IC is easily breakable, so to be attached to a circuit board, it is often housed in a plastic package with metal pins.

INTEGRATED CIRCUIT



Integrated circuits can function as an oscillator, amplifiers, microprocessors or even as computer memory.

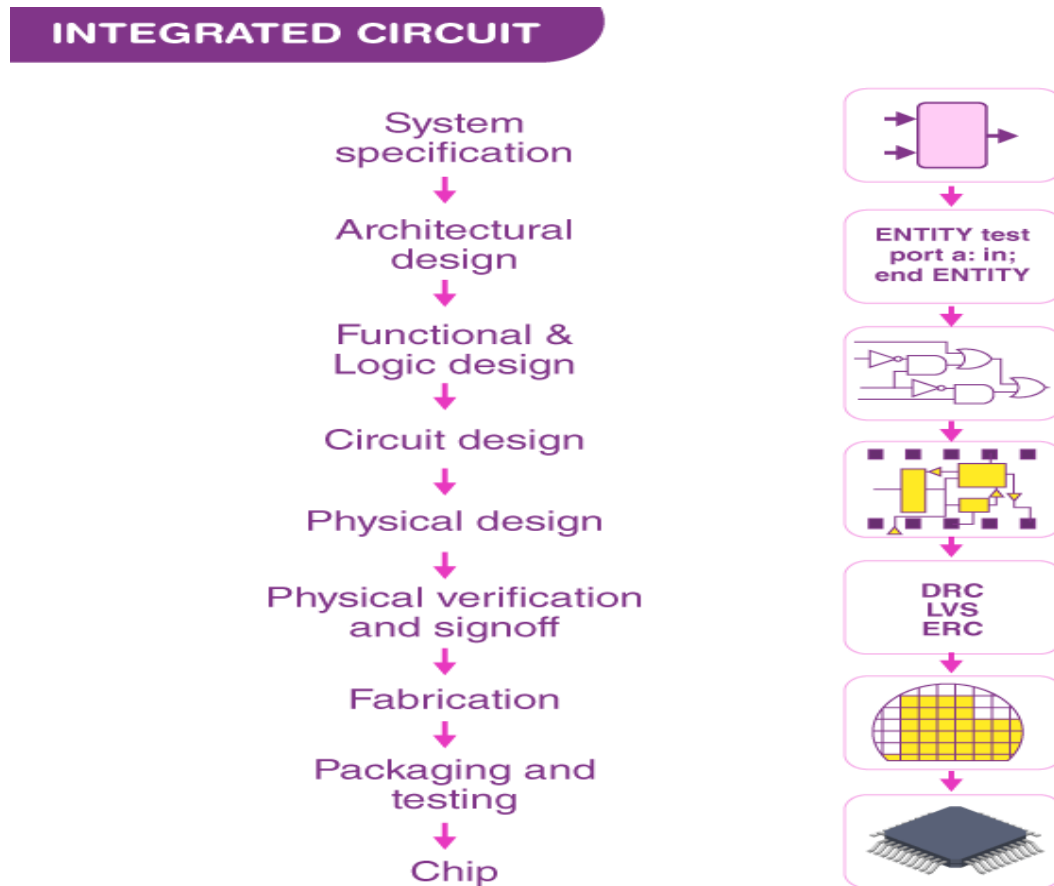
Integrated Circuit Design

An integrated circuit is created using certain logic methods and circuit layouts. The two categories of IC design are as follows:

- **Analog Design**
- **Digital Design**
- **Mixed Design**

Digital Design

The digital design approach is used to create integrated circuits (ICs), which are utilised as computer memories (such as RAM and ROM) and microprocessors. With this approach to design, the circuit density and overall efficiency are both maximised. The ICs created with this technique operate with binary input data like 0 and 1. The process for designing digital integrated circuits is depicted in the diagram below.



Analog Design

IC chip is created by using the analogue design process when:

- ICs are utilised as regulators, filters and oscillators.
- Optimal power dissipation, gain and resistance are required.

Mixed Design

The analog and digital design ideas are used in mixed designs. The mixed ICs perform either Analog to Digital or Digital to Analog conversions.

Integrated Circuit Construction

A complicated stacking of semiconductors, copper, and other related elements to create resistors, transistors, and other components is an integrated circuit. A die is a combination of these wafers that have been sliced and moulded.

The ICs' semiconductor wafers are delicate, and the connections between the layers are extremely complex. The ICs are packaged because an IC die is too small to solder and connect to. The delicate and tiny die is transformed into the familiar black chip by the IC packaging.

The connections between the layers are exceedingly complicated, and the semiconductor wafers used to make the ICs are delicate. Because an IC die is too small to solder, the ICs are packed.

All integrated circuits (ICs) are polarised, and each pin has a specific location and functionality. As seen in the illustration below, integrated chips employ a notch or a dot to denote the first pin.

The subsequent PINs rise consecutively in a counterclockwise way around the chip after the first pin is identified.

Integrated Circuit Features

Construction & Packaging

ICs are built with semiconducting components such as silicon. Because of the small size and delicate nature of IC, a series of tiny gold and aluminium wires are joined together and moulded into a flat block of plastic or ceramic. Metal pins on the block's exterior link to cables inside. The solid block stops the chip from overheating and keeps it cool.

Size of an IC

The size of the integrated chip varies between 1 square mm to more than 200 mm.

Integration of an IC

Because they combine various devices on one chip, integrated chips get their name. A microcontroller is an integrated circuit (IC) that combines a microprocessor, memory, and interface into a single unit.

Commonly Used ICs

Logic Gate ICs

The combinational circuit generates logical outputs based on a variety of input signals. It may only have two to three inputs but one output.

Timer ICs

A Timer IC is produced with accurate timing cycles with a 100 % or 50 % duty cycle.

Operational Amplifiers

An OpAmp or an Operational Amplifier is a high gain voltage amplifier with a differential input and a single-ended output.

Voltage Regulators

A voltage regulator IC provides a constant DC output irrespective of the changes in DC input.

Geographical Indications

What is meant by ‘Geographical Indications’?

Article 22 of the TRIPS Agreements defines Geographical Indication as "indications which identify a good as originating in the territory of a Member, or a region or locality in that territory, where a given quality, reputation or other characteristic of the good is essentially attributable to its geographical origin".

In simpler words, a GI tag is a proof of where the product is born or produced. Such a tag acts as a signaling device that helps producers to differentiate their products from competing products in the market on the basis of place of origin and enables them to build reputation and goodwill around their products, as a result of which the goods fetch a premium price.

A comparative analysis between two types of Intellectual Property – trademarks and GIs, shall help understanding the basis of GIs. A trademark is a sign used by an enterprise in order to distinguish its goods and services from those of others and in turn gives its owner the right to exclude others from using the same. Similarly, a geographical indication acts as a source of origin and has certain characteristics that exist due to the geographical location. However, it is pertinent to note that a geographical indication is not an individual property for use only by the owner but instead a public property, and allows every producer in the specified region to use the said GI as long as the superior quality of the GI goods is attributable to the said region.

Apart from acting as a source identifier, a GI performs other functions as well. Some of the said functions are as follows:

- i) Provides an indication of quality of the goods: A GI tag symbolizes that the specified goods have special qualities as a result of their origin and the said qualities will not be present in other goods of the same category. A GI guarantees that the quality of the goods meets the expectations of the end consumer.
- ii) Culture protecting function: A GI protects local culture by preserving traditional productions methods, habits of consumption and cultural identity.

iv) Boosting the economy as a result of increase in exports: GIs promote economic prosperity by boosting exports as there is a higher demand for goods with GI tags since they let consumers know that the goods come from an area where a given quality, reputation or other characteristic of the goods is essentially attributable to their geographic origin.

Geographical Indications and India

India has a diverse landscape, an extensive rural network and several indigenous goods in the form of handicrafts (Kani Shawl, Bastar Iron Craft) as well as local produce (Nashik Grapes, Mizo Chilly). As a result of being richly endowed with natural and agricultural products, effective protection for GIs was of extreme importance for India.

Thus, in view of the above and in compliance with the TRIPS Agreement of the WTO, 'The Geographical Indications of Goods (Registration and Protection) Act, 1999 (GI Act, 1999) came into force to provide protection to the goods registered under the Act.

The object of the Geographical Indications of Goods (Registration and Protection) Act, 1999, is threefold,

1. By specific law governing the geographical indication of goods in the country which could adequately protect the interest of producers of such goods,
2. To exclude unauthorized persons from misusing geographical indications and to protect consumers from deception and,
3. To promote goods bearing Indian Geographical Indication in the export market.

Further, as per Section 2(1)(e) of the GI Act, 1999^[3], a geographical indication is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. Since the qualities depend on the geographical place of production, there is a clear link between the product and its original place of production.

A striking feature of Indian GIs is the broad variety of product categories which they cater to. The list ranges from agricultural and horticultural products to textiles, handicrafts, paintings and beverages, among other things.

As per Section 11(a) of the GI Act, 1999^[4], an application for registration can be made in writing with the Registrar by an association of persons or producers or any organization or authority established by or under any law, provided the same is in the interest of the producers of the concerned goods.

In order to strengthen GI protection, India submitted two proposals at WTO on Intellectual Property Rights issues covering geographical indications. Additionally, India along with Cuba, Egypt, Indonesia and Pakistan also submitted a joint proposal requesting the additional protection given to wines and spirits, to be extended to other products as well. In the Indian context, this shall help export of agricultural and indigenous products like Basmati rice, Darjeeling tea, alphonso mangoes, which are in abundance in India.

As per the available data, India holds more than 85% share of the global 'Basmati' exports. Further, the European Union had a share of about 8% in India's total basmati rice exports during April-May 2020. Thus, European Union is a major market for Indian products and produce specially basmati rice. Recently, the European Commission has also published India's GI application for 'Basmati' rice in the European Union.

Further, in another recent matter, the mark 'KASHMIR', applied for in respect of furniture, has been accepted by the Romanian IP Office (OSIM). Not only is KASHMIR a commonly known place, but also is closely associated and well-known with various handicrafts, including wooden furniture.

Out of 370 registered GI names on the Indian GI Register, approximately 60% are for handicrafts and in view of the same, one of the issues pertinent for India is the protection of its handicrafts in the European Union, as the European Union has no laws to protect non-agricultural goods under its GI laws.

Establishment of WIPO

Introduction

WIPO, which stands for the World Intellectual Property Organisation, was formed on 14th July 1967 and it is headquartered in Geneva, Switzerland. This article will discuss the history of WIPO, how it took the shape into what it is now, what are the missions, functions, goals of WIPO and how it is working for the betterment of Intellectual Property rights around the world.

World Intellectual Property Organisation

World Intellectual Property Organisation is an organisation based in Geneva and it works with the vision of encouraging creative activity and for promoting the protection of Intellectual Property throughout the world. WIPO is one of the 15 specialised agencies of the United Nations. Currently, there are 193 members in the World Intellectual Property Organisation. WIPO, at the time when it started, was originally about promoting the protection of intellectual property, but when it joined the United Nations in 1974, the objective was redefined as public interest or humanitarian goal. Article 1 of the key agreement establishing WIPO's relationship to the UN restates WIPO's purpose as: "for promoting creative intellectual activity and for facilitating the transfer of technology related to the industrial property to the developing countries in order to accelerate economic, social and cultural development..."

WIPO is a unique organisation among the UN organisations, its activities largely are self-funded, according to the Program and Budget of WIPO for the year 2020-2021, the expected income for the biennium will be over 880 million Swiss francs and the expected expenditure is 768 million Swiss francs. WIPO has expected that nearly 95% of the expected income is going to be generated from the fees that the organisation will receive for their services.

The World Intellectual Property Organisation is the oldest organization in the field of IP protection. Actually, it was created at the diplomatic conference in 1893.

Strategic Goals of WIPO

WIPO works on a 5-year based planned format, the current 5-year plan which is running is from 2017-2021. WIPO makes strategic goals, Implementing Activities and Milestones for a 5-year term and they envision their growth according to that.

The strategic goal number 1 is to use IP assets to advance research and development(R&D) for neglected tropical diseases(NTDs), malaria, and TB through collaboration. The implementing activities for the same goal include establishing and maintaining the Special Programme for Research and Training in Tropical Diseases and others and this work shall be done in collaboration. Establishing new collaborations that can take care of the priority needs and which can provide assistance and support is a major part of the implementing activities, this will also require expansion in the number of company providers who can offer assets and support for the operations of WIPO. The milestones that are expected to be achieved are establishing at least eight new collaborations on an annual basis, preparing comprehensive IP management strategies for two or three partnerships per year.

The strategic goal number 2 is to accelerate the advancement of promising compounds or leads. The implementing activities for this include assisting collaborators for obtaining grants for donor agencies, this will help to accelerate the progress. The projects with high potential for the compound or lead will move rapidly to later stages of evaluation and they will be given the highest priority. The milestones to be achieved by this are connecting two to three partnerships or collaborations per year to relevant funding.

The strategic goal number 3 is to enhance global capacity for IP management and biomedical R&D. The implementing activities include supporting the capacity development of member organisations, providing them research fellowships at leading research centres for the same. Milestone for this goal is providing two to four training sabbaticals annually.

The strategic goal number 4 is to communicate the beneficial role of IP in innovation for NTDs, malaria, and TB. The implementing activities include increasing dissemination of information about the role of IP for R&D in NTDs, malaria, and TB. This is the technology-driven goal to combat the issues in hand. Milestones for this goal are to provide a user-friendly web page consisting of all the available resources which would help facilitate the formation of partnerships and update annually based on users' feedback. This will also publish at least four articles in journals and magazines, it will also make at least six presentations in the subject matter meetings.

Mission

The mission of WIPO is to establish a comprehensive, well balanced and effective system for the protection and enforcement of Intellectual Property Rights. Article 1 of the agreement between WIPO and the UN in 1974 says that “to accelerate economic, social and cultural development”; this is a redefined version from what it earlier used to be and this happened because of the WIPO becoming a part of the UN and the UN works for a larger set of people and is dedicated towards the development of the global society.

Work towards gender equality- WIPO wishes to create an atmosphere where gender equality exists, where men and women are respected in equal measure and they hold important and decision making positions equally. As a lead agency within the UN, WIPO has launched some initiatives which are targeted for awareness-raising, capacity building and leadership.

Leadership– International gender Champions Network is the name of the initiative taken towards leadership from women of the world. Launched in 2015 and at that time the Director-General of WIPO, Francis Gurry was one of the first people to join this initiative.

Raising gender awareness– WIPO has taken a lot of initiatives under the idea of raising gender awareness, example- Celebrating the achievements of women, in 2018 the campaign for World Intellectual Property Day was ‘Powering change: Women in innovation and creativity’.

Another example of initiatives taken by WIPO for gender equality is strengthening data gathering on the use of the IP system by women, the annual data showing Intellectual Property Indicators and Patent Cooperation Treaty Yearly Review will provide the data on the gender dimension of international patenting activity.

Empowering women through IP knowledge is another initiative by WIPO in the direction of gender equality, under this initiative since 1998 WIPO has been empowering women through its IP training and capacity building programs. Out of the estimated 50,000 students that have enrolled with WIPO for such programs, half of them are women.

A growing number of member states present more participation of women in WIPO as delegates and these women provide leadership even though in an unofficial capacity but they are remarkably successful. Women delegates from developing countries, especially from the likes of India and Argentina have proven to be instrumental in building consensus and promoting the developing agenda at WIPO.

Functions

The main functions of WIPO include:

- Assisting campaigns development to improve IP protection all over the world and to harmonize national legislation in this field,
- Signing the international agreements on IP protection,
- Applying the administrative functions of the Paris and Berne Unions,
- Rendering technical and legal assistance in the field of IP,
- collecting and disseminating the information, conducting researches and publishing their results,
- ensuring the work of the services facilitating the international IP protection,
- applying any other appropriate actions.

Administering multilateral international conventions stands to be the most important function done by WIPO, that includes depositing treaties, states' instruments of conflict settlement, ensuring treaties review etc.

WIPO Worldwide Academy has been preparing human resources in the field of IP protection since 1998. This academy provides a distance learning centre where you can gain knowledge with the help of the internet. WIPO Arbitration and Mediation centre were created in 1994, it helps in settling/resolving conflicts.

History of formation

Paris Convention– It all started with the Paris Convention of 1883, the Paris Convention for the Protection of Industrial Property is considered to be the first major step taken towards helping creators to ensure that their intellectual works are protected around the world.

Berne Convention– It started with a campaign by a French writer Victor Hugo, the aim behind it was to give the creators their right to control and to receive payments for the creative done by them on an international level.

Madrid Agreement– The first international IP filing serving was launched in the form of the Madrid Agreement. This agreement was also the first leap towards WIPO.

BIRPI- The United International Bureaux for the Protection of Intellectual Property, 1893 was known better by its French acronym BIRPI. Based in Switzerland, this organisation was the predecessor of WIPO.

WIPO- Headquartered in Geneva, Switzerland, WIPO is what came out of BIRPI and is a product of all those important Conventions and Agreements held before it. WIPO joined the United Nations in 1974.

World Intellectual Property Indicator

The World Intellectual Property Indicator is an annual report published by the World Intellectual Property Organisation, the wide range of indicators that it provides, they cover the areas of intellectual property. The data required for the indicators is provided by national and regional IP offices, the WIPO, World Bank and UNESCO. This annual report has been published by WIPO since 2009.

China dominates the charts when it comes to the data published by WIPO, out of the total patent applications filed worldwide in 2018(3,326,300), 1,542,002 were filed in China which is 46.4% share of the world and the numbers saw a rise by 11.6% when compared to the data published in 2017. Not just for the patent applications but in other fields too, China proves the absolute domination by a large number, like in Utility Models China's share of the world is 96.6%, in Trademarks, it is 51.4%, Industrial Designs, it is 54%, Plant Varieties: 28.5%. With each passing year, a significant rise in the number of applications filed worldwide in all these categories is seen.

Geneva Declaration on the Future of WIPO

In September of 2004, the Geneva declaration on the future of WIPO was published. The people involved in that declaration included prominent legal scholars, public interest NGOs, activists, a former French Prime Minister, a 2002 Nobel prize winner for physiology, scientists and several other concerned citizens of the global society. The declaration was for WIPO to reform its "culture of creating and expanding monopoly privileges, often without regard to the consequences." The declaration said that WIPO's "continuous expansion of these privileges and their enforcement mechanisms has led to grave social and economic costs, and has hampered and threatened other important systems of creativity and innovation."

The declaration was about the excessive protection of intellectual property being put up by the WIPO, the declaration was to make WIPO understand the real economic and social consequences of it. It demanded them to strike a balance between public domain and competition on one side and realm of property rights on the other. It highlighted that the steps being taken can prove to be hazardous for the much-needed factor of competition in the market. The declaration pointed out important factors of countries which are struggling to meet the basic needs of their citizens and it said that a "one size fits all" approach which the WIPO has adopted is causing excessive burden on these countries and if it continues being this way, it will eventually drag them to dust.

This declaration which had no legal significance did a very impressive job in igniting that spark which fuelled many debates on the practices of WIPO, this statement which was made by the declaration was heard all over the world and it garnered a lot of attention from around the world. That was a collectively powered strong word by some of the most prominent people of the global society and it did what it was supposed to do; bring attention to these practices.

The 2004 WIPO General Assembly adopted the resolution which establishes a development agenda to reform the practices of WIPO of blindly increasing intellectual property rights.

IPRs Application and Procedures.

Patents

- **Law:** Patents Act, 1970; Patents (Amendment) Act, 2005; Patents Rules, 2003; Patent (Amendment) Rules, 2020
- **Relevant Ministry:** Office of the Controller General of Patents, Designs and Trade Marks (Indian Patent Office), Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and industry

A patent is a legal right that confers monopoly to a person for their invention.

According to the Patents Act, 1970, an 'invention' must fulfill three criteria:

- It must be a new product or a process that did not previously exist
- It must offer a new technical improvement as simple changes to a previously known technique cannot be patented
- The proposed invention must be useful

Once a product or process is patented, it cannot be commercially produced, distributed, used, or sold without the consent of the patent owner.

India grants patent rights on a first-to-apply basis. The application for a patent can be made by either:

- The inventor
- The assignee
- The legal representative of the inventor

Patent application may also be made jointly by two or more corporations as assignees. A foreign national resident can also apply and obtain a patent in India.

Patent registration procedure

- **Patentability search:** The patentability search is the first step in the patent registration process. The uniqueness, ingenuity, industrial applicability, and enablement of the invention are evaluated through a patentable search.
- **Drafting the patent application:** If an invention is found to be patentable following a search, the applicant must write a patent application that details the invention. The first step after drafting a patent application is to file the patent application.
- **Filing a patent application:** The next step is filing the patent application. Applications must include the invention's specifications as well as any other necessary or pertinent information. To obtain a patent registration for their innovation, an applicant may submit both a provisional and a full application.
- **Publication of application:** The Indian patent Office (IPO) publishes the application in the official patent journal, making it available in the public domain after 18 months from the date of filing or priority. An early publication request is possible.
- **Examination of application:** After publication, the patent office examines the application. Unlike publication, examination doesn't occur automatically upon filing.
- **Objection by the Examiner:** The Examiner scrutinizes the application and issues a report stating all objections related to the application, based on the information mentioned in the application.
- **Office action response:** The applicant must respond to the Examiner's objections to receive a patent grant. They have a maximum of six months from the first examination report's issuing date to respond. The duration can be extended by three months with a request to the IPO.
- **Grant of patent registration:** Once the application addresses all the objections raised by the Examiner, the patent office grants the patent registration and publishes it in the patent gazette.

Essential ingredients of complete specification under patent registration

- Title and preamble
- Prior art
- Loopholes in the prior art
- Solution to drawbacks
- Summary of the invention
- Statement of invention
- Particulars of invention
- Particulars regarding drawings
- Examples
- Claims

Documents required for patent registration in India

For Patent registration, the below-mentioned forms and documents are required:

Form 1: Application for grant of patent	• Identity proof of the inventor and assignee • Address proof of the inventor and assignee • Contact details • Information corresponding to prior patent applications relating to the current invention, which the applicant or any authorized entity has filed • Some declarations, among other information
Form 2: Provisional/complete specification	Form 2 is used to furnish the applicant's patent specification. It may be provisional or a complete patent specification, depending on the type of patent application being filed.
Form 3: Statement and undertaking under Section 8	• Particulars of assignments • Details of foreign filing
Form 5: Declaration as to inventorship	This application is used to declare the inventor(s) of the current patent application.
Technical information	• Background of the invention • Disclosure of need of invention • Summary of the invention

	• Description of how the invention works
Form 18	Request for Examination
Form 19	Request for Early Publication
Form 26	Disclosure of Power of Authority (If assigned to the Patent Agent)
Form 9	Request for Publication. This is optional if the express publication is required.

Trademarks

- **Law:** Trademark Act, 1999; The Trade Marks (Amendment) Act, 2010; Trade Marks Rules, 2003; Trade Marks Rules, 2017
- **Relevant Ministry:** DPIIT, Ministry of Commerce and Industry

For a business, its trademark is among its most valuable assets. The company's trademark protects its distinctive sign or indicator. This may be a symbol, logo, word, phrase, name, sound, design or an image that symbolizes the brand value or goodwill associated with the business or its product and services.

In India, trademarks are registered under the federal registration system with the support of five regional offices: Ahmedabad, Chennai, Delhi, Kolkata, and Mumbai. The Trade Mark Act of 1999 provides trademark rules for the registration, regulation, and protection of trademarks in India. DIPP recently revamped old trademark rules for registering trademarks in India.

Trademark registration procedure

- **Conduct the trademark search**
- **Filing trademark application:** The first and foremost step is to file the prescribed application at the Trademark Registry India. The applicant can also file the application

online. After filing the said application, an acknowledgement receipt is generated by the official portal for future reference.

- **Verification of application and documents:** After filing an application with India's Trademark Registry, the scrutiny process begins and is conducted by a designated official known as an Examiner. The verification process can take around 12-18 months. After inspection, the Examiner may accept the mark unconditionally, conditionally, or object to it.
- If the mark is accepted unconditionally, it will be published in the Trademark Journal.
- If it is not accepted unconditionally, the Examiner will stipulate conditions to be met or objections that must be addressed. The applicant will be given one month to meet the conditions or respond to the objections. If the Examiner finds the applicant's response satisfactory, the trademark in question will be published in the Trademark Journal.
- If the response is not accepted, the applicant will have the chance to represent their case in a hearing. Upon receiving a satisfactory response from the applicant, the Examiner may render their approval and publish the concerned mark in the Trademark Journal.
- **Publication of trademark in trademark journal:** The publication of the mark in the trademark journal enables the authority to invite objections from third parties if no objections are raised within the specified time frame. The authority can provide approval for the next step. On the contrary, if the mark somehow attracts some objections, the authority would conduct a fair hearing.
- **Grant of trademark registration by registrar:** The applicant is subsequently presented with a registration certificate by the registrar, which has a ten-year validity period. The trademark office seal must be included by the registrar on the certificate.

The whole process takes about 15 to 18 months. The trademark is valid for 10 years starting from the date of issuance of the certificate. It can be renewed for another 10 years on the payment of the prescribed fees.

Documents required for trademark registration in India

- Applicant's identity proof
- PAN
- Aadhar card
- Passport
- Certificate of Incorporation (COI), in case of companies registered under Company Act, 2013
- Logo if it is applicable and available
- Address proof

Copyrights

- **Law:** Copyrights Act 1957; The Copyright (Amendment) Act, 2012; Copyright (Amendment) Rules, 2021
- **Relevant Ministry:** Copyright Office, Ministry of Human Resource Development

India has a very strong and comprehensive copyright law based on the Indian Copyright Act of 1957. The copyright law is fully compatible with international norms like the Berne Convention, Trade Related Aspects of Intellectual Property Rights (TRIPS) Agreement, and the World Intellectual Property Organization (WIPO) Copyright Treaty (WCT).

Find Business Support

Have a question about doing business in India? Speak with our experts on the ground

A copyright grants the exclusive right to the creator to control reproduction or adaptation of their work for a certain period of time. Unlike trademarks and patents, a copyright protects the unique expression of ideas and not the idea itself. Further, copyright registration cannot be obtained for titles or names, short word combinations, slogans, short phrases – as these can only be trademarked.

Types of works eligible for copyright registration

- Artistic works
- Producers of cinematograph films
- Music voice recordings
- Literary works
- Musical works, including any accompanying words
- Dramatic works, including any accompanying music
- Pantomimes (drama) and choreographic works
- Pictorial, graphic, and sculptural works
- Motion pictures and other audio-visual works
- Sound recordings

Registration procedure for filing the copyright registration

- Create a user ID and password at the official site, **copyright.gov.in**.
- **File application form:** An applicant can apply either manually in the copyright office or through an e-filing facility available on the official website. Here, the applicant can be the author of the work, the owner of an exclusive right to the work, or an authorized agent.

- **Examination of application:** After the application is filed, the next step is the examination of the copyright application. A minimum waiting period of 30 days is provided for the copyright Examiner to review the application once the diary number is issued. The copyright registration process is then divided into two segments:
 - **Objections raised:** If an objection is raised by someone against the applicant, a letter is sent to both parties, and they are called for a hearing by the registrar. If the objection is rejected upon hearing, the applicant can ask for scrutiny, and the discrepancy procedure is followed.
 - **No objections raised:** If no objections are raised, the Examiner reviews and scrutinizes the application for any disparities. If no discrepancies arise and all necessary documents are provided, the applicant can proceed to the next step. However, if inconsistencies are found, a letter of discrepancy is sent to the applicant. The applicant must reply, and a hearing is conducted by the registrar. Once the difference is resolved, the applicant can move on to the next step. If the difference cannot be resolved, the application is rejected, and a rejection letter is sent to the applicant.
- **Issuance of registration certificate:** The certificate of copyright registration is issued at the final stage. The Registrar could ask for more details or documents at this point. The information regarding the copyright application will be recorded in the register of copyrights, and a certificate of registration will be granted, if the Registrar is fully satisfied with the application made by the applicant.

Documents required for copyright registration in India

Basic documents	• Particulars of the applicant (ID and Address proof of the applicant along with the nationality). • Name, address, and nationality of the author of the work. • Disclosure of the applicant's interest in the copyright – whether they are the author of the work or the representative of the author. • Copies of the original work. • In case for business purpose – incorporation certificate is required. • Details of the nature of the work. • Class, title and description of the work. • Language of the work.
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	• Date of publication – Publication in internal magazines or a research paper submitted to a professor does not count as publication.
Artistic work	• Two copies of the work. • Demand draft of INR (as applicable) according to artistic work. • Author's No Objection Certificate (NOC) if the applicant is different from the author. • Publisher's NOC if the work published and publisher is different from the applicant. • Search certificate from Trade Mark Office in TM-60 if the work is being used on goods or items capable of being used on the goods. • NOC from the individual whose photograph appears on the work. • When filing an application by an attorney, an original copy of a "Power of Attorney" signed by the applicant and also accepted by the attorney.
Cinematograph film	• Two copies of the work. • Demand draft of INR (as applicable) according to cinematographic work. • NOC from different copyright holders or copy of the agreement (deed of assignment). • NOC from the publisher if the work published and publisher is different from the applicant. • When filing an application through an attorney, an original copy of a "Power of

	Attorney” signed by the applicant & also accepted by the attorney.
Music	• Two copies of the work (graphical notes). • Demand draft of INR (as applicable) according to work. • NOC from the publisher if the work published and publisher is different from the applicant. • Author’s NOC if the applicant is other than the author. • When filing an application by an attorney, an original copy of a “Power of Attorney” duly signed by the applicant & also accepted by the attorney.
Software	• Two copies of the work (graphical notes). • Demand draft of INR (as applicable) according to work. • NOC from the publisher if the work published and publisher is different from the applicant. • Author’s NOC if the applicant is other than the author. • When filing an application through an attorney, an original copy of a “Power of Attorney” signed by the applicant & also accepted by the attorney. • The source code and the object code of the work for due verification.

Industrial designs

- **Law:** Designs Act, 2000; Designs Rules, 2001; The Designs (Amendment), Rules, 2021
- **Relevant Ministry:** DPIIT, Ministry of Commerce and industry

A design refers to the distinctive visual characteristics of an article, such as its shape, figure, blueprints, decorations, or arrangement of lines and colors, or a combination of these, created through any manufacturing method, whether manual, mechanical, automated, chemical, or a combination thereof, in either two-dimensional or three-dimensional form. The purpose of obtaining design registration under the Designs Act is to protect a unique or innovative design that is intended to be applied to a particular article during the manufacturing process using an industrial process or method.

Industrial design registration procedure

The industrial design registration procedure involves the following steps:

- **Examination:** The Controller of Designs refers the application to an Examiner of Designs who checks if the application and documents meet formal requirements and if the design is registrable under the Designs Act and Rules.
- **Formality check:** The Examiner ensures that the application is in the prescribed format, fee is paid, applicant's details are provided, and declaration of proprietorship is given. The representation sheet is checked to ensure it complies with Rule 14, and a power of authority is filed if applicable.
- **Substantive examination:** The Examiner checks if the design is new, original, and not prejudicial to public order, morality, or security of India.
- **Consideration of Examiner's report:** If the design is eligible for registration, the Controller registers it and issues a certificate. If objections arise, the applicant is informed, and a statement of objections is communicated. The applicant must comply with the objections or apply for a hearing within three months. If the applicant fails to comply, the application is deemed withdrawn.
- **Design registration and publication:** Registered designs are published in the Patent Office Journal within one month of registration, and the registration number is the same as the application number.
- **Register of Designs:** All registered designs are entered in the Register of Designs maintained at the Patent Office in Kolkata, which is available for public inspection. An e-register is also available on the IPO official portal.

Once the application is accepted, it is notified in the patent office journal and is valid for a period of 10 years from the date of registration. The period may be extended by five years by filing an application for renewal along with the prescribed fee.

Documents required for design registration in India

To file a design application, certain documents and forms need to be prepared and submitted as per the Design Rules. These include:

- Form 1, which should be filled out in the prescribed format as mentioned in Schedule II of the Designs Rules.
- Representations must be prepared in accordance with Rules 12, 13, and 14 of the Design Rules and submitted in duplicate.
- If the application is filed through a patent agent or an advocate, Form 21, the Power of Authority/General Power of Authority (GPA), should be submitted in the original. This form is also mentioned in Schedule II.
- If a copy of the GPA is filed, it should be endorsed with the design application number with which the original GPA was filed.
- Form 24 in the prescribed format should be submitted if the applicant is claiming small entity status, as mentioned in Schedule-II of the Designs Rules. If the applicant is an Indian entity, evidence of registration under the MSME Act, 2006, should also be provided. In the case of foreign entities, an affidavit should be deposed by the applicant or an authorized signatory as per Rule 42 of the Designs Rules.
- The original Priority document should be submitted under Rule 15 of the Designs Rules.
- An authenticated English-translated copy of the priority document should be submitted if the original priority document is in a language other than English.
- If the applicant for the priority application in the convention country is different from the Indian applicant, an assignment in the original format should be submitted.

Geographical indications (GI)

- **Law:** The Geographical Indications of Goods (Registration and protection) Act, 1999; The Geographical Indications of Goods (Registration and Protection) (Amendment) Rules, 2020
- **Relevant Ministry:** DPIIT, Ministry of Commerce and industry

A geographical indication (GI) is a tag used on products based on their geographical uniqueness. The uniqueness of a product is defined either by its origin, process, or availability. GI tags usually help businesses enhance their marketability.

Requirements for registering GI in India: Essential qualifications and prerequisites

- To be eligible for the registration of a geographical indication in India, certain prerequisites and qualifications must be fulfilled, which include:
- The product should originate from a specific geographical location.
- The product should have distinctive qualities, characteristics, or a reputation that are primarily associated with its geographical origin.
- The product should not fall under any prohibited categories, such as generic names or indications that are against public order or morality.

- The applicant must be either an association of persons, producers, or any organization that represents the interests of the producers.
- The applicant should have direct involvement in the production, marketing, or sale of the product.

GI registration procedure in India: To register a geographical indication in India, the following steps need to be taken:

- Conduct a search to confirm that the proposed geographical indication is not already registered or pending registration.
- Prepare the application for registration of the geographical indication in the prescribed form, along with the required documents.
- File the application with the Geographical Indications Registry, along with the prescribed fee.
- The Geographical Indications Registry examines the application to ensure that it meets the eligibility criteria and prerequisites for registration.
- If the application is accepted, the geographical indication is published in the Geographical Indications Journal.
- Any individual may oppose the registration of the geographical indication within three months of its publication in the Geographical Indications Journal.
- If there is no opposition or the opposition is dismissed, the geographical indication is registered, and the registered proprietor has the right to use the registered GI tag on the product.

The registration of a geographical indication is valid for a period of 10 years. It can be renewed for another 10 years at any time.

Documents required for GI registration in India

The following documents are required for the registration of a geographical indication in India:

- Application for registration in the prescribed form.
- Statement of case.
- Details of the product, including its unique features, method of production, and the link between the product and its geographical origin.
- Evidence to establish the link between the product and its geographical origin, such as maps, photographs, and historical records.
- Authorization letter, if the application is filed by an agent.
- Power of attorney, if the application is filed by a legal representative.
- If applicable, a certified copy of the priority document.

*****ALL THE BEST*****